$15\langle 1\rangle + 12\langle -1\rangle$ Lines on a Cubic Surface: On Quadratically Enriched Enumerative Geometry.

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To my grandparents, Margaret and Geoffrey.

Declaration

The work in this thesis is my own except where otherwise stated. This thesis was produced without the assistance of large language models or artificial intelligence.

Liam Murray

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They say it takes a village to raise a child. In this sense, children and theses are isomorphic. I'd like to thank some of the people who've been part of the village that raised this thesis.

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Abstract

In this thesis, we present a complete development of the $\mathbb{A}^1$–Euler number, an invariant of relatively oriented vector bundles valued in the Grothendieck–Witt ring of a field, which allows us to perform enriched enumerative counts over non-algebraically closed fields.

We begin by introducing Grothendieck–Witt theory and establishing the necessary foundations for defining the $\mathbb{A}^1$–Euler number. In particular, we prove the self-duality theorem of Scheja and Storch for complete intersections, and use it to construct local indices of sections of relatively oriented vector bundles. We then establish an invariance theorem using coherent duality which shows the resulting $\mathbb{A}^1$–Euler number is independent of the choice of section.

We apply this theory to obtain an arithmetic count of the lines on a smooth cubic surface, by computing the $\mathbb{A}^1$–Euler number of the bundle $\mathrm{Sym}^3(\mathcal{S}^\vee)$ on the Grassmannian $\mathrm{Gr}_k(4,2)$, where $\mathcal{S}$ is the tautological sub-bundle. We show that

$$e^{\mathbb{A}^1}(\mathrm{Sym}^3(\mathcal{S}^\vee)) = 15\langle 1\rangle + 12\langle -1\rangle,$$

and interpret this as a sum of bilinear forms associated to the lines on the cubic surface. These forms are described in terms of the $\mathbb{A}^1$–degree of a distinguished involution on each line.

We conclude by computing the local indices of all sixteen lines on a singular cubic surface with two A_1 singularities. We find that the eight lines passing through exactly one singularity have hyperbolic indices, while the line passing through both singularities has index $2\mathbb{H}$.

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Notation and terminology

In the following, all rings are taken to be commutative with unity. Unless otherwise specified, all fields are by default of characteristic not equal to 2. We follow the usual conventions of algebraic geometry, with some notation and terminology specific to this thesis collected below.

Notation

k, $\bar{k}$	A field not of characteristic 2 and its algebraic closure.
$\mathrm{GW}_0(k)$	The Grothendieck–Witt ring of a field.
$W_0(k)$	The Witt group of a field.
$\langle\alpha\rangle$	The rank-one bilinear form defined by $x \otimes y \mapsto \alpha xy$.
$\langle a_1, \ldots, a_n\rangle$	A diagonal bilinear form with entries $a_1, \ldots, a_n$.
$\mathbb{H}$	The hyperbolic form $\langle 1\rangle + \langle -1\rangle$.
X	A smooth n-dimensional scheme over k.
V	A rank n vector bundle on X.
σ	A section of V. Usually assumed to have isolated zeroes.
Z	The vanishing subscheme of a section σ of V.
$\kappa(p)$	The residue field of a point p of a scheme.
Φ	Nisnevich coordinates $U \to \mathbb{A}^n_k$.
$\mathbf{r}_U^{\otimes 2}$	The square of a local section $\mathbf{r}_U$ of the line bundle defining the relative orientation, arising from compatible coordinates and trivialisations.

$\mathrm{Ind}_p(\sigma)$	The local index of a section σ at p.
$M^\vee$	The A-linear dual of an A-module M.
$\mathrm{Tr}_{L/k}$	The field trace or the Scharlau transfer.
$\mathrm{Ind}_p(\sigma)$	The index of σ at p.
$e^{\mathbb{A}^1}(V,\sigma)$	The $\mathbb{A}^1$–Euler number of a relatively oriented vector bundle V with respect to a section σ.
$e^{\mathbb{A}^1}(V)$	On a proper scheme, the section-independent $\mathbb{A}^1$–Euler number.
$\mathrm{Ind}_p^{CD}(\sigma)$	The coherent duality index of σ at p.
$e_{CD}^{\mathbb{A}^1}(V,\sigma)$	The coherent duality $\mathbb{A}^1$–Euler number with respect to a section σ.
$e_{CD}^{\mathbb{A}^1}(V)$	The coherent duality $\mathbb{A}^1$–Euler number.
Rf_*	The derived push-forward.
$f^!$	The exceptional inverse image functor.
$K(\sigma)^\bullet$	The Koszul complex arising from the evaluation at σ map $V^* \to \mathcal{O}_X$.
$D^{perf}(X)$	The derived category of perfect complexes
L_f	The cotangent complex of a morphism f
$\widetilde{\det}$	The graded determinant.
$\mathrm{Gr}_k(4,2)$	The Grassmannian of copies of $\mathbb{A}^2_k$ in $\mathbb{A}^4_k$.
$\mathcal{S}$	The tautological sub-bundle on the Grassmannian.
$\mathcal{Q}$	The tautological quotient bundle on the Grassmannian.
$\mathbb{P}(V)$	The GIT quotient $V/k^\times$.
ℓ	A line.
$\mathrm{Type}(\ell)$	The type of the line ℓ.
$\mathrm{Disc}_{x,y}(f,g)$	The discriminant of f and g as polynomials in x and y.

$\mathrm{Res}(f, g)$	The resultant of two polynomials f and g.

Terminology

Bilinear form	Unless otherwise specified, a non-degenerate symmetric bilinear form.
Isolated zero	An isolated point of the vanishing subscheme of a section.
Line	A closed point of $\mathrm{Gr}(4, 2)$.

Introduction

Classically, problems in enumerative geometry have been resolved in two steps. First, one establishes that a count is finite and proves an invariance principle, showing that it is independent of the choice of object within a particular class. One then computes the count for a particularly tractable example.

For example, the classical Cayley–Salmon theorem [Cay49] states that every smooth cubic surface in $\mathbb{P}^3_{\bar{k}}$ contains exactly 27 lines. In the proof presented in [Vak25, Chapter 27], deformation theory and results about the moduli space of cubic surfaces are used to establish an invariance principle, before a computation for a nice example, such as the Fermat cubic, proves the theorem.

However, over a non-algebraically closed field, these invariance principles often fail. For instance, a smooth cubic surface over $\mathbb{R}$ can contain $3, 7, 15$ or 27 real lines, while one over $\mathbb{Q}$ can contain $0, 1, 2, 3, 5, 7, 9, 15$ or 27 rational lines. The issue is that lines defined over $\bar{k}$ need not be defined over k. This dependence on the base field is a general feature of enumerative geometry over non-algebraically closed fields.

Classically, one can recover an invariance principle by counting non-rational solutions with multiplicity, measured by the lengths of their associated stalks. However, while this restores invariance, it loses the arithmetic information concerning the fields over which the solutions are defined.

Another way to recover an invariance principle is to perform an *enriched count*. For example, following classical work of Segre, Okonek and Teleman explain in [OT14] how to obtain an invariant signed count of the lines on a smooth real cubic surface; each line ℓ carries a distinguished involution, which is called hyperbolic if it has real fixed points and elliptic otherwise. They show there are always 3 more hyperbolic lines than elliptic lines. Assigning a contribution of $+1$ to each hyperbolic line and -1 to each elliptic line therefore gives an invariant count valued in $\mathbb{Z}$.

Another example is Gromov–Witten theory, in which virtual counts can take

rational values. Such theories suggest a three-stage approach to an enumerative problem. Before proving invariance and computing the count, one must first ask what kind of quantity the count should take values in.

Quadratically enriched enumerative geometry takes this idea further. Rather than producing numerical contributions, contributions are valued in isometry classes of non-degenerate symmetric bilinear forms over the base field. Specifically, the Grothendieck–Witt ring $\mathrm{GW}_0(k)$ allows us to perform arithmetic on bilinear forms. So, in the case of lines on a cubic surface, each Galois orbit of lines contributes a bilinear form over its field of definition. After transferring these forms to the base field, we add them to obtain the enriched count.

The idea that the degree of an algebraic map should take values in isometry classes of bilinear forms goes back as early as [Eis78]. Earlier work of Eisenbud–Levine [EL77] and Khimshiashvili [Khi77] had constructed a bilinear form on the stalk of an isolated zero whose signature computed its local degree. Eisenbud asked whether the corresponding construction over an arbitrary field could be interpreted as a degree in a suitable algebraic analogue of topology.

Morel–Voevodsky's $\mathbb{A}^1$–homotopy theory provides such a setting. By [Mor07][p. 1037], for $n \geq 2$ Morel's degree map gives an isomorphism

$$\deg^{\mathbb{A}^1} \colon [\mathbb{P}^n_k/\mathbb{P}^{n-1}_k, \mathbb{P}^n_k/\mathbb{P}^{n-1}_k]^{\mathbb{A}^1}_* \to \mathrm{GW}_0(k),$$

where the left hand side consists of $\mathbb{A}^1$–homotopy classes. Over $\mathbb{R}$ and $\mathbb{C}$, there is a homotopy equivalence between $\mathbb{P}^n/\mathbb{P}^{n-1}$ with the Euclidean topology and S^n and S^{2n} respectively. This suggests the Grothendieck–Witt ring can play a role in $\mathbb{A}^1$–homotopy theory analogous to that of the integers in classical topology. This gives motivation for counts valued in $\mathrm{GW}_0(k)$. This provides a target to value our counts in that varies with the base field.

In [KW21], Kass and Wickelgren develop a theory of $\mathbb{A}^1$–Euler numbers. This provides a systematic way to construct quadratically enriched counts for problems that can be expressed as counting the zeros of a section of a vector bundle. Bachmann and Wickelgren provide in [BW23] an invariance principle whenever one is counting the zeroes of a vector bundle over a proper base scheme.

The problem of counting lines on a cubic surface fits into this framework. A cubic polynomial f determines a section σ_f of a bundle on the Grassmannian of lines in $\mathbb{P}^3_k$. The zeros of σ_f correspond to the lines contained in the cubic surface. Counting these zeros by their local indices therefore gives an $\mathbb{A}^1$–Euler number.

Furthermore, Kass and Wickelgren give a geometric interpretation of these indices in terms of the distinguished involutions of the lines. A geometric inter-

pretation of the indices is not in general known for the $\mathbb{A}^1$–Euler number. For a line ℓ on a smooth cubic defined over L, its involution determines a rank 1 form $\text{type}(\ell) \in \text{GW}_0(L)$. Their main theorem, which we prove as Theorem 3.34, states that for a smooth cubic surface over a field k of characteristic different from 2,

$$\sum_{\text{Lines}} \text{Tr}_{k(\ell)/k}(\text{type}(\ell)) = 15\langle 1\rangle + 12\langle -1\rangle.$$

Over $\mathbb{R}$, taking the signature recovers the signed count, since the contributions from non-real lines have signature zero. Taking the rank recovers the count of 27 geometric lines. Over finite fields of odd characteristic, the determinant gives parity restrictions on the numbers of lines with particular types of involution.

This framework also assigns indices to isolated zeros that are not simple. In the quadratic setting, the local index can instead be represented by a form of rank greater than 1. Consequently, for a singular cubic surface with isolated lines, Theorem 2.58 shows that the sum of the indices agrees with the smooth count, while the individual indices can retain arithmetic information about the non-simple zeros. As we discuss, the structure theorems of [QSW22] constrain the forms that can occur, and hence the information these indices can encode.

Outline of the Thesis

The goal of this thesis is to give a detailed exposition of the algebraic construction and invariance of the $\mathbb{A}^1$–Euler number, and to apply this theory to counting lines on cubic surfaces. The expository part of this thesis aims to reorganise several papers in the field into a uniform exposition, taking into account newer developments that streamline existing arguments. Towards the end we compute a novel example that we have not found elsewhere in the literature.

In Chapter 1, we define the Grothendieck–Witt ring as the group completion of the semi-ring of isometry classes of non-degenerate symmetric bilinear forms. We first work over a commutative ring before specialising to fields k of characteristic different from 2. Following the first two chapters of [Lam05], we prove Theorem 1.29, which describes the structure of the Grothendieck–Witt ring over a field. We reorder some results to emphasise this structure and adjust their proofs accordingly. We then compute several examples and discuss invariants used to distinguish isometry classes. Finally, we briefly introduce bilinear forms in an abstract category, providing background for Chapter 2.

Chapter 2 is the technical heart of the thesis. Following Section 3 of [KW21],

we begin with relative orientations, Nisnevich coordinates and compatible trivialisations of vector bundles. We then show that the local algebras at isolated zeros of a section admit complete intersection presentations. Following the outline of [Kuh+22], we prove Scheja–Storch duality, which allows us to construct bilinear forms from these presentations. These forms define the local indices, whose sum is the $\mathbb{A}^1$–Euler number. We then discuss how to compute this number. Finally, following Section 2 of [BW23], we give a cohomological interpretation of the $\mathbb{A}^1$–Euler number and prove that, for a fixed relatively oriented vector bundle on a smooth proper base, it is independent of the section and well-defined.

In Chapter 3, we compute the arithmetic count of lines on a smooth cubic surface. We first specify the bundle, relative orientation, coordinates and trivialisations needed to define the $\mathbb{A}^1$–Euler number. We then interpret the local indices in terms of the $\mathbb{A}^1$–degrees of the distinguished involutions. After performing the computation, we deduce parity restrictions on the numbers of lines with particular types of involution over finite fields. Finally, we use Macaulay2 to compute the local indices and Euler number for a singular cubic surface. We discuss the structure of indices at zeros of multiplicity greater than 1 and consider a possible geometric interpretation.

For the hurried reader, we suggest first reading the statement of the main theorem. A route to understanding its proof begins with the definition of the Grothendieck–Witt ring and the statement of the structure theorem over a field, followed by the complete intersection theorem and the statement of Scheja–Storch duality. We then recommend reading the definition of the $\mathbb{A}^1$–Euler number and section 2.4 presenting an algorithm for its computation, including Proposition 2.53. The statement of Theorem 2.58 can be read without its proof. Finally, the definition of the distinguished involution, Corollary 3.32 and the result that lines on a smooth cubic surface are simple provide the remaining background for the proof of the main theorem. This route should suffice to understand the main computation and its consequences.

We assume familiarity with scheme theory at the level of Chapters II and III of [Har13]. Some further commutative algebra, particularly Koszul complexes and the length of modules, as discussed in [Eis95], would be helpful for the proof of Scheja–Storch duality. The proof of Theorem 2.58 is substantially more technical than the rest of the thesis and requires familiarity with spectral sequences, derived categories and coherent duality. Readers may accept the statement of this theorem without following its proof and still understand the main computation and its consequences.

Chapter 1

Grothendieck–Witt Theory

In this chapter we will investigate the Grothendieck–Witt ring. First we define it over an arbitrary ring and its connection to K-theory. Then, we will focus on the case of a field of characteristic not equal to 2 and produce a presentation of the Grothendieck–Witt ring. Next, we will cover some invariants associated to non-degenerate symmetric bilinear forms, and use them to compute some examples. We finish the chapter with a brief note on non-degenerate symmetric bilinear forms in categories with duality. This part is only required to understand part of the proof of Theorem 2.58.

Since the only bilinear forms we are interested in are non-degenerate and symmetric, we will often simply use "bilinear form" to refer to non-degenerate and symmetric forms.

1.1 The Grothendieck–Witt Ring of a Ring

Definition 1.1 ([Wei13, p. 63])**.** The **Grothendieck completion** of an abelian monoid M is the abelian group $K^0(M)$, equipped with a monoid morphism $M \to K^0(M)$, such that any monoid map from M to an abelian group G factors uniquely through $K^0(M)$.

Remark 1.2. By standard universal property arguments, $K^0(M)$ is unique up to unique isomorphism.

Lemma 1.3 ([Wei13, p. 64])**.** *The Grothendieck completion $K^0(M)$ exists for every abelian monoid M.*

Proof. To construct $K^0(M)$, consider the set $M \times M$. We quotient by the equivalence relation $(a, b) \sim (c, d)$ whenever there exists an $e \in M$ such that

$a + d + e = b + c + e$. Call this set A. It has a group structure under the operation $[(a, b)] + [(c, d)] = [(a + c, b + d)]$. First, observe that A has an identity as $[(a, b)] + [(0, 0)] = [(a + 0, b + 0)] = [(a, b)]$. Next, we have the existence of inverses as $[(a, b)] + [(b, a)] = [(a + b, a + b)]$ and $(a + b, a + b) \sim (0, 0)$.

Now we have to prove this group satisfies the universal property. Observe we have a monoid map $M \to A$ via $a \mapsto [(a, 0)]$. Likewise, suppose we have $\varphi\colon M \to G$. Consider the factoring $\bar{\varphi}\colon A \to G$ via $[(a, b)] \mapsto \varphi(a)\varphi(b)^{-1}$. First, we claim this is well defined. Suppose $(a, b) \sim (c, d)$. Then $a + d + e = b + c + e$. Hence, $\varphi(a + d + e) = \varphi(b + c + e)$ so $\varphi(a)\varphi(d)\varphi(e) = \varphi(b)\varphi(c)\varphi(e)$. Then, $\varphi(a)\varphi(b)^{-1} = \varphi(c)\varphi(d)^{-1}$, so the map $\bar{\varphi}$ is well defined.

Finally, we claim the factoring is unique. Note that any factoring $M \to K^0(M) \to G$ must satisfy $a \mapsto [(a, 0)] \mapsto \varphi(a)$. Next we notice that since $[(a, 0)]^{-1} = [(0, a)]$ that we must also have $[(0, a)] \mapsto \varphi(a)^{-1}$. Finally, since $[(a, b)] = [(a, 0)] + [(0, b)]$ we must have $[(a, b)] \mapsto \varphi(a)\varphi(b)^{-1}$. So this is the only possible factoring. Hence A is a $K^0(M)$. □

Remark 1.4. We say a monoid M is cancellative if $a + c = b + c$ implies $a = b$. For M cancellative we have $a + d + e = b + c + e$ implies $a + d = b + c$. So the equivalences in the construction of $K^0(M)$ are equivalent to setting $(a, b) \sim (c, d)$ if $a + d = b + c$. Then $K^0(M)$ may be thought of as M with formal additive inverses of each element adjoined.

Lemma 1.5 ([Wei13, Corollary 1.2])**.** *The canonical map $M \to K^0(M)$ given by $a \mapsto [(a, 0)]$ is injective if and only if M is cancellative.*

Proof. Suppose M is cancellative. We have $[(m, 0)] = [(n, 0)]$ implies $m+0 = n+0$, so $m = n$. Hence the canonical map is injective.

Now suppose the map is injective. Say we had $a+c = b+c$ in M. Applying the canonical map, we have $[(a + c, 0)] = [(b + c, 0)]$ which implies $[(a, 0)] + [(c, 0)] = [(b, 0)] + [(c, 0)]$. This then gives $[(a, 0)] = [(b, 0)]$. Taking pre-images under the injective canonical map, we conclude $a = b$. So M is cancellative. □

Example 1.6. Consider the monoid of natural numbers $\mathbb{N}$. Its Grothendieck completion $K^0(\mathbb{N})$ is isomorphic to $\mathbb{Z}$ via the map $[(m, n)] \mapsto m - n$.

Lemma 1.7. *Suppose that a monoid M is additionally a semi-ring. Then there is an induced ring structure on the Grothendieck completion $K^0(M)$.*

Proof. This follows from our construction. We set $[(a, b)] \cdot [(c, d)] = [(ac + bd, bc + ad)]$. It can then be verified this satisfies all the axioms of a ring, with multiplicative identity $[(1, 0)]$. □

Definition 1.8. Let A be a ring and P a projective A-module. A **symmetric unimodular form** on P is an A-linear map $\beta\colon P \otimes_A P \to A$ satisfying:

- $\beta(a \otimes b) = \beta(b \otimes a)$.
- The map $\beta^*\colon P \to \mathrm{Hom}(P, A)$ given by $p \mapsto (q \mapsto \beta(p, q))$ is an isomorphism.

We observe that the set of pairs (P, β) may be endowed with the structure of a semi-ring. This follows by setting $(P, \beta) + (Q, \beta') = (P \oplus Q, \beta \perp \beta')$. Here, the form $\beta \perp \beta'$ is defined by setting $(\beta \perp \beta')(p \oplus q, p' \oplus q') = \beta(p, p') + \beta'(q, q')$. Similarly, we set $(P, \beta) \times (Q, \beta') = (P \otimes_A Q, \beta \otimes \beta')$, where $\beta \otimes \beta'$ is defined by setting $(\beta \otimes \beta')(p \otimes q, p' \otimes q') = \beta(p, p') \cdot \beta(q, q')$.

Notation 1.9. Denote by M_A the semi-ring of isometry classes of finitely generated projective A modules equipped with symmetric unimodular forms.

Definition 1.10. The **Grothendieck–Witt ring** $\mathrm{GW}_0(A)$ is the ring $K^0(M_A)$.

Remark 1.11. To explain the finite hypothesis on M_A, we observe that there exists an isomorphism $\bigoplus_{i=1}^{\infty} P \cong P \oplus \bigoplus_{i=1}^{\infty} P$ by associativity of the direct sum. Hence, if we permitted even countably generated modules, applying the Eilenberg-Mazur swindle, we would have in the Grothendieck completion that $[(\bigoplus_{i=1}^{\infty} P, 0)] = [(P, 0)] + [(\bigoplus_{i=1}^{\infty} P, 0)]$ so $[(P, 0)] = 0$ for all P equipped with a form, so the Grothendieck completion would be trivial.

One can regard the Grothendieck–Witt ring as an enrichment of the zeroth K-group of A; there is a natural projection map $\mathrm{GW}_0(A) \to K^0(A)$ via $[(P, \beta)] \mapsto P$. Similarly, there is a natural map $K^0(A) \to \mathrm{GW}_0(A)$ given by $P \mapsto P \oplus P^\vee$ where $P^\vee = \mathrm{Hom}_A(P, A)$. This space carries a natural bilinear form given by $\beta((v, w), (v', w')) \coloneqq w(v') + w'(v)$.

The cokernel of the map $K^0(A) \to \mathrm{GW}_0(A)$ is called the Witt group, $W_0(A)$. This group may alternatively be described as the quotient of $\mathrm{GW}_0(A)$ by the subgroup of metabolic spaces, i.e. projective A modules with a unimodular form β such that there exists a subspace $N = N^\perp$.

Since finitely generated projective modules are reflexive, dualisation induces an action of the cyclic group C_2 on the set of finitely generated projective A-modules. Let $K_0(A)_{C_2}$ denote the set of co-invariants of this action.

Proposition 1.12 ([Cal+23, p. 3])**.** *There is an exact sequence:*

$$K_0(A)_{C_2} \to \mathrm{GW}_0(A) \to W_0(A) \to 0$$

Proof. This is a direct corollary of the definition of $W_0(A)$ as the cokernel of the map $K^0(A) \to \mathrm{GW}_0(A)$ along with the observation the map $K^0(A) \to \mathrm{GW}_0(A)$ is C_2-equivariant by reflexivity of projective modules. □

1.2 The Structure of $\mathrm{GW}_0(k)$ over a Field

From this section onwards, let k be a field of characteristic not equal to 2. In this case, the structure of $\mathrm{GW}_0(k)$ is quite easy to understand. Indeed, we will provide a simple presentation theorem as Theorem 1.29. We note that many of the results stated here hold for a bilinear form that is not assumed to be non-degenerate, or have analogous statements. However, as these forms are not relevant to our enumerative purpose we do not give proofs of these more general facts.

First, we have the four perspectives on bilinear forms over a field not of characteristic 2. We can associate to a symmetric bilinear form the following objects:

- The linear map $\beta\colon V \otimes V \to k$.
- The quadratic form, $Q_\beta\colon V \to k$ defined as $Q_\beta(v) = \beta(v, v)$

If we fix a basis $B = \{b_1, ..., b_n\}$ of V;

- The matrix M_β defined via $v^\intercal M_\beta w = \beta(v, w)$.
- The homogeneous quadratic polynomial $f_\beta(x_1, ..., x_n) = \sum \beta(b_i, b_j)x_i x_j$.

It turns out that we can re-derive from the last three pictures facts about the linear map.

- A quadratic form $Q\colon V \to k$ is a degree 2 homogeneous function, or a function satisfying $Q(\alpha v) = \alpha^2 Q(v)$ for all $\alpha \in k$. Two quadratic forms Q, Q' are equivalent if there exists a map $A \in \mathrm{GL}_n(V)$ such that $Q(Av) = Q'(v)$ for all v. Given a quadratic form Q_β arising from a bilinear form, we can return the bilinear form β by the polarisation identity, i.e. setting
$$\beta(u, v) = \frac{1}{2}(Q(u + v) - Q(u) - Q(v)).$$
- A matrix M with respect to a basis B defines a bilinear form via $(v, w) \mapsto v^\intercal M w$. We have this form is non-degenerate if and only if $\det(M) \neq 0$, and symmetric if and only if $M^\intercal = M$. Likewise, we have M and M' are equivalent if there exists an invertible matrix P such that $P^\intercal M P = M$.

- A homogeneous quadratic polynomial in n-variables is one of the forms $\sum a_{ij}x_ix_j$. It defines a bilinear form via:

$$\left(\sum \alpha_i v_i, \sum \beta_i v_i\right) \mapsto \sum a_{ij}\alpha_i\beta_i.$$

  Two homogeneous quadratic polynomials $f(X)$ and $g(X)$ are equivalent if there exists a matrix A such that $f(AX) = g(X)$ for all X. The form defined by f is symmetric if and only if $a_{ij} = a_{ji}$.

Definition 1.13. A vector space V equipped with a non-degenerate symmetric bilinear form β is called a **quadratic space**.

Remark 1.14. Many of the pathologies of the characteristic 2 case arise from the observation that alternating and symplectic forms now satisfy the definition of a symmetric form. Indeed, the failure of quadratic forms to correspond to symmetric bilinear forms in this case arises from alternating forms being non-isometric, but having the zero quadratic form associated to them. Likewise, Theorem 1.16 and Theorem 1.18 fail in this case, as the information associated to a diagonalisation fails to capture any off diagonal data in characteristic 2.

Notation 1.15. Denote by $\langle\alpha\rangle$ the map $k \otimes k \to k$ given by $x \otimes y \mapsto \alpha xy$. Then, set $\langle a_1, \ldots, a_n\rangle = \langle a_1\rangle \perp \cdots \perp \langle a_n\rangle$. This corresponds to a diagonal matrix.

Theorem 1.16 ([Lam05, Corollary I.2.4.])**.** *Let β be a non-degenerate symmetric bilinear form on an n-dimensional k-vector space. Then there exist not necessarily distinct $\alpha_1, \alpha_2, ..., \alpha_n \in k^\times$ such that β is isometric to $\langle\alpha_1, \ldots, \alpha_n\rangle$.*

Proof. We proceed by induction. The argument is effectively by the Gram–Schmidt process. If $n = 1$, then β is equal to $\langle\beta(e_1, e_1)\rangle$, since $\beta(e_1, e_1)$ is non-zero by non-degeneracy. Now, suppose the theorem is true for $n = m$. Let β be a form on an $(m + 1)$-dimensional vector space V.

Next, we argue there exists at least one vector e_0 such that $\beta(e_0, e_0) \neq 0$. To see this, consider the associated quadratic form $Q_\beta(v)$. By the polarisation identity, $\beta(u, v) = \frac{1}{2}(Q(u + v) - Q(u) - Q(v))$. Hence if $Q_\beta(v) = 0$ for all v, we get β is identically 0 for all pairs of vectors and hence degenerate. Ergo, there exists some vector e_0 such that $\beta(e_0, e_0) = Q_\beta(e_0) \neq 0$.

Now, consider the space $e_0^\perp = V'$. We observe by the rank-nullity theorem, that V' has dimension m. Furthermore, we have $e_0 \notin e_0^\perp$, so as a vector space $V \cong ke_0 \oplus V'$. Furthermore, by definition of V' we have as quadratic spaces $(V, \beta) \cong (k \cdot e_0, \langle\beta(e_0, e_0)\rangle) \perp (V', \beta|_{V'})$.

To conclude, we want to argue that $\beta|_{V'}$ is non-degenerate. To see this, let's pick a basis of V, $e_0, e_1, ..., e_m$ such that $e_1, ..., e_m \in V'$. Then, consider the associated matrix to β in this basis, M_β. We have by construction

$$M_\beta = \begin{bmatrix} \beta(e_0, e_0) & 0 \\ 0 & M_{\beta|_{V'}} \end{bmatrix}$$

So $\det(M_\beta) = \det(M_{\beta|_{V'}}) \cdot \beta(e_0, e_0)$. Since $\det(M_\beta) \neq 0$ by non degeneracy of β and $\beta(e_0, e_0) \neq 0$ we conclude $\det(M_{\beta|_{V'}}) \neq 0$. Thus, $\beta|_{V'}$ is non-degenerate. So applying the theorem, $\beta|_{V'} = \langle \alpha_1, \cdots, \alpha_m \rangle$. Taking $\beta(e_0, e_0) = \alpha_0$ we conclude $\beta \cong \langle \alpha_0, \cdots, \alpha_m \rangle$ as desired. □

Remark 1.17. We observe that our characteristic assumption was essential here, as we utilised the polarisation identity. Indeed, consider the vector space $\mathbb{F}_2^2$, with coordinate basis e_1, e_2. The symmetric bilinear form β defined by $\beta(e_i, e_i) = 0$ and $\beta(e_1, e_2) = 1$ is non-degenerate yet satisfies $\beta(v, v) = 0$ for all v. Hence it cannot be expressed as a diagonal form.

So as a first step towards the structure of the Grothendieck–Witt ring, we have demonstrated that M_k is generated by symbols of the form $\langle \alpha \rangle$ for each $\alpha \in k^\times$. Next we will prove M_k is cancellative.

Theorem 1.18 ([Lam05, Witt Cancellation Theorem, Theorem I.4.2.])**.** *Suppose that $q_1 \perp q \cong q_2 \perp q$. Then $q_1 \cong q_2$.*

Proof. Let $\langle a_1, \ldots, a_n \rangle$ be a diagonalisation of q. By associativity of direct sums and induction on m, we reduce to the case $n = 1$. So we want to prove $q_1 \perp \langle a_1 \rangle \cong q_2 \perp \langle a_1 \rangle$ implies $q_1 \cong q_2$.

Then, consider the isometry $T \colon (W, q_1 \perp \langle a_1 \rangle) \xrightarrow{\sim} (V, q_2 \perp \langle a_1 \rangle)$. From the orthogonal decomposition of the two forms, we have an orthogonal decomposition of the underlying vector spaces, $W = E \oplus E^\perp$ and $V = F \oplus F^\perp$, where E and F are the 1-dimensional subspaces corresponding to the $\langle a_1 \rangle$ components.

Now, consider the image of $T|_{E^\perp}$. If it is $F^\perp$, then we are done, as the restriction of an isometry is an isometry, so $T|_{E^\perp}$ induces an equivalence $q_1 \cong q_2$. Then, suppose that $T(E^\perp)$ does not lie in $F^\perp$. Let $v \in E$ be a vector such that $(q_1 \perp \langle a_1 \rangle)(v, v) = a_1$. We then know that $T(v) \in V$ is some vector satisfying $(q_2 \perp \langle a_1 \rangle)(T(v), T(v)) = a_1$. Furthermore, $T(v) \notin F$, since isometries preserve orthogonality, so if it were then $T(E^\perp) \subseteq F^\perp$. Hence it suffices to find some isometry τ of $(V, q_2 \perp \langle a_1 \rangle)$ such that $\tau(T(v)) \in F$, since if the image of E lies in F, then the image of $E^\perp$ will lie in $F^\perp$, as isometries preserve orthogonality.

To furnish such a choice of orthogonal transformation, we prove the following result:

Proposition 1.19 ([Lam05, Proposition I.4.7.])**.** *Let V be a space endowed with a bilinear form β. Let x, y be vectors such that $\beta(x,x) = \beta(y,y) \neq 0$. Then there exists $\tau \in O(V)$ such that $\tau(x) = y$.*

Proof. First, we recall the notion of a reflection. Given a vector v, the reflection with respect to the orthogonal hyperplane to v is classically defined as $\tau_v(w) = w - \frac{2\beta(v,w)}{\beta(v,v)}v$. This is easily shown to be both an orthogonal transformation and an involution. However, this map only makes sense if v is anisotropic, i.e., $\beta(v,v) \neq 0$.

τ_{x-y} is our candidate reflection; however, we must first demonstrate that $x-y$ is anisotropic. First, note that $\beta(x+y,x+y) + \beta(x-y,x-y) = 2\beta(x,x) + 2\beta(y,y) = 4\beta(x,x) \neq 0$. So we cannot have both $\beta(x-y,x-y) = 0$ and $\beta(x+y,x+y) = 0$. We may then assume $\beta(x-y,x-y) \neq 0$, since if it is zero, then $\beta(x+y,x+y) \neq 0$, in which case we will find a transformation such that $\tau(x) = -y$. But then $-\tau$ is our desired orthogonal transformation. So, assuming otherwise, we compute that $\beta(x-y,x-y) = \beta(x,x) + \beta(y,y) - 2\beta(x,y)$, which by our assumption on x and y reduces to $2[\beta(x,x) - \beta(x,y)] = 2\beta(x,x-y)$. Hence $\tau_{x-y}(x) = x - \frac{2\beta(x,x-y)}{2\beta(x,x-y)}(x-y) = y$. So we are done. □

Applying then the appropriate map τ such that it interchanges $T(v)$ with a generator of F, we are done proving the cancellation theorem. □

Remark 1.20. The Witt cancellation theorem does not require non-degeneracy. However, since we're focussed on Grothendieck–Witt theory, we do not prove it in this level of generality. We direct the interested reader to [Lam05, Chapter 1] for the proofs in the degenerate cases.

Corollary 1.21. *The monoid of non-degenerate symmetric bilinear forms on k-vector spaces is cancellative.*

Corollary 1.22. *The map $M \to \mathrm{GW}_0(k)$ where M is the monoid of non-degenerate symmetric bilinear forms on k-vector spaces is an injection when* $\mathrm{char}(k) \neq 2$.

Next we probe the relations on M_k and $\mathrm{GW}_0(k)$.

As a first relation, we have $\langle\alpha\rangle = \langle\beta^2\alpha\rangle$ for all $\alpha, \beta \in k^\times$. This follows from the isometry $(k, \langle\alpha\rangle) \to (k, \langle\beta^2\alpha\rangle)$ given by $1 \mapsto \beta$. As a second relation, we observe that $\langle\alpha\rangle \cdot \langle\beta\rangle \cong \langle\alpha\beta\rangle$.

For finding further relations, we introduce the following notion.

Definition 1.23 ([Lam05, Definition I.2.1.])**.** Let β be a quadratic form on V. We say $\alpha \in k^\times$ is **represented** by β if there exists $v \in V$ such that $\beta(v, v) = \alpha$. The set of values represented by β is denoted $D(\beta)$.

Proposition 1.24 ([Lam05, Representation Criterion, I.2.3])**.** *Let (V, β) be a space with a bilinear form. Then $\alpha \in D(\beta)$ if and only if there exists a space (V', β') such that $\beta \cong \langle \alpha \rangle \perp \beta'$.*

Proof. If $\beta \cong \langle \alpha \rangle \perp \beta'$, then we obviously have $\alpha \in D(\beta)$, by just taking an appropriately scaled vector from the subspace of V corresponding to $\langle \alpha \rangle$.

For the reverse direction, we observe this is a direct corollary of our proof of the diagonalisation theorem, by taking a v such that $\beta(v, v) = \alpha$ as the e_0 in that proof. □

To better understand how to relate a diagonalisation of β to another diagonalisation we introduce the determinant.

Definition 1.25. The **determinant map**, $\det\colon GW_0(k) \to k^\times/(k^\times)^2$ is the map induced via the monoid map $M_k \to k^\times/(k^\times)^2$ which sends any bilinear form β to the class of the determinant of some matrix representation of it.

Remark 1.26. This map is well defined, since if two bilinear forms are isometric, we have a relation between their associated matrices, $M = P^\intercal M' P$. Since the determinant is invariant under transposition, we have $\det(M) = \det(M')\det(P)^2$.

Lemma 1.27. *Suppose we have $\langle a_1, a_2 \rangle \cong \langle c_1, c_2 \rangle$. Then $c_2 = a_1 a_2 c_1$ in $k^\times/(k^\times)^2$.*

Proof. Equating determinants, we have $a_1 a_2 = c_1 c_2$. Since c_1, c_2 are units, we get $c_2 = \frac{a_1 a_2}{c_1}$. Multiplying by c_1^2 leaves the square classes invariant, and yields $c_2 = a_1 a_2 c_1$. □

Corollary 1.28. *If $\alpha + \beta \neq 0$ we have $\langle \alpha, \beta \rangle \cong \langle \alpha + \beta, (\alpha + \beta)\alpha\beta \rangle$.*

Proof. We observe that $\alpha + \beta \in D(\langle \alpha, \beta \rangle)$ by considering the vector $e_1 + e_2$. Hence applying the Representation Criterion, we have $\langle \alpha, \beta \rangle \cong \langle \alpha + \beta, c \rangle$ for some $c \in k^\times$. Then, by the Lemma 1.27, since the isometry class of $\langle c \rangle$ is only determined by the square class of c, we can take $c = \alpha\beta(\alpha + \beta)$. □

We claim we've found all relations on the Grothendieck–Witt ring. Specifically, our goal is to prove the following presentation for $\mathrm{GW}_0(k)$.

Theorem 1.29 ([Lam05, Theorem II.4.1.])**.** $\mathrm{GW}_0(k)$ *is the ring generated by symbols of the form $\langle \alpha \rangle$ where $\alpha \in k^\times$ subject to the relations:*

1. $\langle\alpha\rangle = \langle\beta^2\alpha\rangle$
2. $\langle\alpha\rangle\langle\beta\rangle = \langle\alpha\beta\rangle$
3. *If* $\alpha+\beta \neq 0$ *then* $\langle\alpha\rangle + \langle\beta\rangle = \langle\alpha+\beta\rangle + \langle\alpha\beta(\alpha+\beta)\rangle$

Where $\alpha, \beta \in k^\times$.

To prove this theorem, we have a preliminary required notion and result.

Definition 1.30 ([Lam05, p. 15])**.** Let $p = \langle a_1, \ldots, a_n\rangle$ and $q = \langle b_1, \ldots, b_n\rangle$. p and q are **simple equivalent**, denoted $p \sim_S q$ if there exist two indices i and j such that $\langle a_i, a_j\rangle \cong \langle b_i, b_j\rangle$ and $a_k = b_k$ for $k \neq i, j$.

Then, we say two forms p and q are **chain equivalent**, denoted $p \cong_C q$ if there exists a chain of simple equivalences $p = f_0 \sim_S f_1 \sim_S \cdots \sim_S f_m = q$

Theorem 1.31 ([Lam05, Chain Equivalence Theorem, Theorem I.5.2.])**.** *For any two diagonalised bilinear forms,* p *and* q *we have* $p \cong_C q$ *if and only if* $p \cong q$.

Proof. That chain equivalence implies isometry is obvious, since each simple equivalence is itself an isometry. The non-trivial part is showing isometry implies chain equivalence.

Now suppose $p = \langle a_1, \ldots, a_n\rangle$ and $q = \langle b_1, \ldots, b_n\rangle$. First, observe that since the symmetric group is generated by transpositions we have that for any permutation $\sigma \in S_n$ setting $p^\sigma := \langle a_{\sigma(1)}, ..., a_{\sigma(n)}\rangle$ that $p^\sigma \cong_C p$. This holds, as for the transposition (ij) we have $\langle a_i, a_j\rangle \cong \langle a_j, a_i\rangle$. Hence any expression of σ as a product of transpositions yields a chain equivalence.

We now proceed by induction on n. For $n = 1$ and $n = 2$ an isometry is a simple equivalence so we are done. Otherwise, consider the chain equivalence class of p. Then, choose a form $\langle c_1, ..., c_n\rangle$ such that for some value of m we have $\langle c_1, ..., c_m\rangle$ represents b_1. Such a form exists, as p is chain equivalent to itself, and as $p \cong q$, we know p must represent everything q does.

We claim the minimal such value of m is 1. Given any form with diagonalisation $\langle c_1, ..., c_p\rangle$ representing b_1, we have $b_1 = c_1e_1^2 + \cdots + c_me_m^2$, by applying the form to some vector in k^p. By minimality of m, no sub-sum can be 0. So $d = c_1e_1^2 + c_2e_2^2 \neq 0$. By the representation criterion and the lemma, we have $\langle c_1, c_2\rangle = \langle d, dc_1c_2\rangle$. Then, we note that $\langle c_1, c_2, ..., c_m\rangle$ is chain equivalent to $\langle d, c_3, ..., c_m, dc_1c_2\rangle$, and that $d \cdot e_1^2 + c_3 \cdot e_3^2 + \cdots + c_me_m^2 = b_1$, contradicting minimality of m unless we take $m = 1$.

Hence, $\langle c_1\rangle \cong \langle b_1\rangle$. So $q \cong \langle b_1, c_2, ..., c_n\rangle$ for some c_i. Then applying the cancellation theorem, we conclude that $\langle b_1, c_2, ..., c_n\rangle \cong \langle b_1, b_2, ..., b_n\rangle$ implies

$\langle c_2, ..., c_n\rangle \cong \langle b_2, ..., b_n\rangle$. Finally, applying the inductive hypothesis, $\langle c_2, ..., c_n\rangle \cong_C \langle b_2, ..., b_n\rangle$, and hence $p \cong_C q$ as desired. □

Remark 1.32. This theorem explains why the basic relations we have on $\mathrm{GW}_0(k)$ only involve two–dimensional forms; it states that any isometry of forms can be reduced to the two–dimensional case.

We are now positioned to prove the structure theorem.

Proof of Theorem 1.29. Denote by $R1, R2, R3$ the 3 relations in the statement of the theorem. Consider the free ring $\mathcal{F}$ generated by $\langle\alpha\rangle$ for $\alpha \in k^\times$, and the quotient $R = \mathcal{F}/\langle R1, R2, R3\rangle$. Since these 3 relations hold in the Grothendieck–Witt ring, as previously computed, we have a map induced by the universal properties of the free ring and of quotients, mapping $R \twoheadrightarrow \mathrm{GW}_0(k)$. To show that this is an isomorphism, it suffices to produce an inverse. We will begin by defining a map $M_k \to R$.

Take $(V, \beta) \in M_k$. Suppose $\langle a_1, a_2, ..., a_n\rangle$ is a diagonalisation of β. We want to map $q \mapsto \langle a_1\rangle + \cdots + \langle a_n\rangle \in R$. A priori, it is non obvious that this map is well defined. Suppose we have two diagonalisations of β, given by $\langle a_1, ..., a_n\rangle$ and $\langle b_1, ..., b_n\rangle$. By the Witt chain equivalence theorem, we may assume they're simple equivalent, since any isometry is a composition of simple equivalences, so if any 2 simple equivalent forms are equal in R, by transitivity of equality, so will any two isometric forms. Without loss of generality, assume $a_i = b_i$ for $i \geq 3$, and $\langle a_1, a_2\rangle \cong \langle b_1, b_2\rangle$. We need to show this implies $\langle a_1\rangle + \langle a_2\rangle = \langle b_1\rangle + \langle b_2\rangle$ in R.

We observe by $R1$ that $\langle\alpha^2\rangle = \langle 1\rangle$, which by $R2$ is the multiplicative identity. Then, by the representation criterion, we set $b_1 = a_1x^2 + a_2y^2$ and equating determinants yields $a_1a_2 = b_1b_2c^2$ for a unit c. If x or y are 0, we note since $b_1 \in k^\times$ they can't be both 0. Assume without loss of generality that $x \neq 0$. For our first case, we assume $y = 0$. We obtain $b_1 = x^2a_1$. Then by $R1$ we obtain $\langle b_1\rangle = \langle a_1\rangle$. Likewise, $\langle a_2\rangle = \langle\frac{b_1b_2c^2}{a_1}\rangle$, which applying R_2 and the previous result is just $\langle b_2\rangle$.

For our next case, if $x, y \neq 0$ we have

$$\begin{aligned}
\langle a_1\rangle + \langle a_2\rangle &= \langle a_1x^2\rangle + \langle a_2y^2\rangle && \text{(R1)}\\
&= \langle a_1x^2 + a_2y^2\rangle + \langle a_1a_2(xy)^2(a_1x^2 + a_2y^2)\rangle && \text{(R3)}\\
&= \langle b_1\rangle + \langle b_1b_2c^2(xy)^2(b_1)\rangle && \text{(Prop 1.24)}\\
&= \langle b_1\rangle + \langle b_2\rangle && \text{(R1)}
\end{aligned}$$

So this is a well defined map. By the universal property of the Grothendieck completion, it extends to a unique map $\mathrm{GW}_0(k) \to R$. Then, we observe by construction that the map is an inverse for the canonical map $R \to \mathrm{GW}_0(k)$, and we're done. □

Corollary 1.33 (Sylvester's Law of Inertia)**.** *The signature of a non-degenerate symmetric quadratic form on a real vector space is independent of choice of diagonalisation.*

Proof. We can observe that the statement is equivalent to saying that the assignment $\langle a_1, ..., a_n\rangle \mapsto \sum_{i=1}^{n} \mathrm{sgn}(a_i)$ is a well defined map $\varphi\colon \mathrm{GW}_0(\mathbb{R}) \to \mathbb{Z}$. By Theorem 1.29, it suffices to check applying the 3 relations preserves the signature.

First, we have $\varphi(\langle\alpha\beta^2\rangle) = \mathrm{sgn}(\alpha\beta^2) = \mathrm{sgn}(\alpha)$ since $\beta^2 > 0$.

Next, we have:

$$\begin{aligned}\varphi(\langle\alpha\beta\rangle) &= \mathrm{sgn}(\alpha\beta)\\ &= \mathrm{sgn}(\alpha)\,\mathrm{sgn}(\beta)\\ &= \varphi(\langle\alpha\rangle)\varphi(\langle\beta\rangle)\end{aligned}$$

The second line follows from a simple exhaustion of the four cases of signs of α and β.

For the third relation, if both $\alpha, \beta > 0$, we have $\alpha + \beta > 0$ and $\alpha\beta(\alpha + \beta) > 0$ so both sides have the same signature.

Meanwhile, if α and β are negative, their sum is negative so $\alpha\beta(\alpha + \beta)$ has 3 negative factors and is itself negative.

Finally, if α and β have opposite signs, the sign of $\alpha + \beta$ will be opposite to $\alpha\beta(\alpha + \beta)$ since $\alpha\beta < 0$. So both sides of the third relation have a signature of 0.

Hence as all 3 relations preserve signature, we conclude the signature defines a well defined map on the Grothendieck–Witt ring and hence is invariant under choice of basis. □

Corollary 1.34. *For all $\alpha \in k^\times$ we have $\langle 1\rangle + \langle -1\rangle = \langle\alpha\rangle + \langle -\alpha\rangle$.*

Proof. Set $\alpha \neq 0$. Then by Relation three we obtain

$$\begin{aligned}\langle\alpha\rangle + \langle 1 - \alpha\rangle &= \langle 1\rangle + \langle\alpha(1 - \alpha)\rangle\\ \langle\alpha\rangle + \langle -1\rangle &= \langle\alpha - 1\rangle + \langle -\alpha(\alpha - 1)\rangle\\ &= \langle\alpha - 1\rangle + \langle\alpha(1 - \alpha)\rangle\end{aligned}$$

From these equations we derive,

$$\begin{aligned} \langle 1\rangle &= \langle \alpha\rangle + \langle 1-\alpha\rangle - \langle \alpha(1-\alpha)\rangle \\ \langle -1\rangle &= \langle \alpha-1\rangle + \langle \alpha(1-\alpha)\rangle - \langle \alpha\rangle \\ \implies \langle 1\rangle + \langle -1\rangle &= \langle \alpha-1\rangle + \langle -(\alpha-1)\rangle \end{aligned}$$

Letting $\alpha - 1$ vary over all values of $\alpha \neq 1$ we obtain all units of k. □

Notation 1.35. Denote by $\mathbb{H}$ the hyperbolic form, $\langle 1\rangle + \langle -1\rangle$.

We have $W_0(k) \cong \frac{\mathrm{GW}_0(k)}{\mathbb{Z}\mathbb{H}}$, as $\mathbb{Z}\mathbb{H}$ is an ideal by Corollary 1.34.

1.3 Invariants and Computations

In this section we survey some invariants of bilinear forms, and the corresponding maps on $\mathrm{GW}_0(k)$. We also compute examples of $\mathrm{GW}_0(k)$.

We can associate the following classes of invariants to non-degenerate symmetric bilinear forms over a field not of characteristic 2:

- The rank map $\mathrm{rk}\colon \mathrm{GW}_0(k) \to \mathbb{Z}$ is generated by $\langle\alpha\rangle \to 1$. It is the extension of the monoid map $M_k \to \mathbb{Z}$ sending each bilinear form to the dimension of the underlying vector space.
- The signature map, $\mathrm{sgn}\colon \mathrm{GW}_0(\mathbb{R}) \to \mathbb{Z}$ defined by sending $\sum n_i\langle\alpha_i\rangle$ to $\sum n_i \operatorname{sgn}(\alpha_i)$, or to the signature of the associated matrix.
- The determinant map $\det\colon \mathrm{GW}_0(k) \to k^\times/(k^\times)^2$ is generated by the map sending every element of M_k to the class of the determinant of an associated matrix. It sends $\sum n_i\langle\alpha_i\rangle \mapsto \prod \alpha_i^{n_i}$.

We will now use these invariants to compute $\mathrm{GW}_0(k)$

Example 1.36. Let k be algebraically closed. Then for any $\alpha \in k^\times$ the equation $\alpha = x^2$ has a solution. So α has a square root. Then $\langle\alpha\rangle = \langle 1\cdot(\sqrt{\alpha}^2\rangle = \langle 1\rangle$. Hence, every element of $\mathrm{GW}_0(k)$ is of the form $n\cdot\langle 1\rangle$. So the map $\mathrm{rk}\colon \mathrm{GW}_0(k) \to \mathbb{Z}$ is an isomorphism.

Remark 1.37. The above argument holds more generally for 'quadratically closed' fields, i.e. fields where every non-zero element has a square root and the characteristic is not 2. Some examples include the field of constructible numbers, and for any finite field, after fixing an algebraic closure, the field $\bigcup_{n=1}^{\infty} \mathbb{F}_{p^{2^n}}$.

Example 1.38. Fix $k = \mathbb{R}$. We will show $\mathrm{GW}_0(\mathbb{R}) \cong \mathbb{Z}[C_2]$. First, observe that $\alpha \in \mathbb{R}^*_{>0}$ we have $\sqrt{\alpha} \in \mathbb{R}$. Hence by $R1$ $\langle\alpha\rangle = \langle 1\rangle$. By the same argument, if $\alpha < 0$, we have $\langle\alpha\rangle = \langle -1\rangle$. Hence, every element $\beta \in \mathrm{GW}_0(\mathbb{R})$ is equivalent to something of the form $m\langle 1\rangle + n\langle -1\rangle$. Then, taking the rank of a form gives us $m + n = \mathrm{rk}(\beta)$ and Corollary 1.33 tells us the signature $m - n = \mathrm{sgn}(\beta)$ is an invariant of the equivalence class. Together the rank and signature equations completely determine m, n for any $\beta \in \mathrm{GW}_0(k)$, so their values are unique. Then define the map $m\langle 1\rangle + n\langle -1\rangle \mapsto m(1) + n(-1) \in \mathbb{Z}[C_2]$, taking the presentation $C_2 = \{-1, 1\}$. This map is easily checked to be a ring homomorphism, and is clearly seen to be a bijection, so we're done.

Example 1.39. Fix $k = \mathbb{F}_p$ for $p \neq 2$. We will show as additive groups $\mathrm{GW}_0(\mathbb{F}_p) \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

First, since squaring is a 2 to one map on $\mathbb{F}_p^*$, we have $\mathbb{F}_p^*/(\mathbb{F}_p^*)^2$ has exactly 2 elements. Then we have if $\alpha \neq d^2$ for any d, that $\langle\alpha\rangle \cdot \langle\alpha\rangle = \langle\alpha^2\rangle = \langle 1\rangle$.

Lemma 1.40. *There exist $a, b \in \mathbb{F}_p^*$ and $\alpha \notin \mathbb{F}_p^2$ such that $\alpha = a^2 + b^2$.*

Proof. Consider the ordering on $\mathbb{F}_p^*$ defined by $1 < 2 < \cdots < p - 1$. We split into two cases.

- Suppose -1 is a square. Then since 1 is a square, there exists a minimal α such that α is a non square. Then by minimality $\alpha - 1 = b^2$ for some b as $\alpha \neq 1$. Hence, $\alpha = b^2 + 1$ so we're done.

- Suppose -1 is not a square. Then there is some maximal square a^2. So $a^2 + 1$ is a non-square unit α, since $a^2 + 1 \neq 0$, as $a^2 \neq -1$. □

Then our third relation yields $\langle a^2\rangle + \langle b^2\rangle = \langle\alpha\rangle + \langle a^2b^2(\alpha)\rangle$ so $2\langle 1\rangle = 2\langle\alpha\rangle$. So every element of $\mathrm{GW}_0(\mathbb{F}_p)$ can be written in the form $n\langle 1\rangle$ or $n\langle 1\rangle + \langle\alpha\rangle$. Then, the map $B \mapsto (\mathrm{rk}(B), \det(B))$ provides our desired isomorphism.

Remark 1.41. Note that the ring structure on $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ doesn't agree with $\mathrm{GW}_0(\mathbb{F}_p)$ since $(\langle 1\rangle - \langle\alpha\rangle)^2 = 0$ while the direct sum is a reduced ring.

Next, we ask how many distinct invariants we can associate to a non-degenerate symmetric bilinear form. The Milnor conjecture intuitively says the number of useful invariants is equivalent to the étale cohomological dimension of a certain sheaf over our field.

Theorem 1.42 ([Voe97, Theorem 5.1.])**.** *Denote by I the kernel of the map* $\mathrm{rk}\colon \mathrm{W}(k) \to \mathbb{Z}/2\mathbb{Z}$. *We have $I^n/I^{n+1} \cong H^n_{et}(k, \mathbb{Z}/2\mathbb{Z})$.*

Proposition 1.43. *Suppose* $\mathrm{rk}(\beta) = \mathrm{rk'}(\beta)$ *and* $[\beta]_W = [\beta']_W$ *in* $W_0(k)$*. Then in* $\mathrm{GW}_0(k)$ *we have* $[\beta] = [\beta']$

Proof. Since $[\beta]_W = [\beta']_W$ we have in $\mathrm{GW}_0(\mathrm{k})$ that $[\beta] - [\beta]' \in \mathbb{Z}\mathbb{H}$. Hence, $[\beta] - [\beta'] = n\mathbb{H}$. Taking ranks we get $0 = 2n$, so $n = 0$. □

Remark 1.44. Applying this lemma, one has a bijection $I \cong \hat{I}$ where $\hat{I}$ is the kernel of the map $\mathrm{rk}\colon \mathrm{GW}_0(k) \to \mathbb{Z}$. This is because we can send any representative of $[A]_W \in I$ to the, by Proposition 1.43, unique representative A with rank 0.

As explained by [PW21, Example 8], the invariant for $n = 0$ is the parity of the rank. In $n = 1$ we have the determinant. For $n = 2$ the associated invariant is the Hasse-Witt invariant, taking values in the Brauer group. For $n = 3$, we obtain what is called the Arason invariant. In general, this theorem says that if k has finite mod 2 étale cohomological dimension, then there is a finite number of obstructions blocking any bilinear form being trivial.

This leads us to $\mathrm{GW}_0(\mathbb{Q})$. In general, this ring is rather complicated. We state without proof two results describing its structure.

Theorem 1.45 ([Lam05, Hasse–Minkowski Principle, VI.3.1.])**.** *Two quadratic forms* β *and* β' *over* $\mathbb{Q}$ *are equivalent if and only if they're equivalent at every place. I.e, the forms induced by* β *and* β' *at each* $\mathbb{Q}_p$ *must be equivalent, along with the forms induced at* $\mathbb{R}$*.*

Corollary 1.46 ([Lam05, Theorem VI.4.1.])**.** *There is a group isomorphism:*

$$\mathrm{GW}_0(\mathbb{Q}) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/2\mathbb{Z} \oplus \bigoplus_{p=1 \bmod 4} (\mathbb{Z}/2\mathbb{Z})^2 \bigoplus_{p=3 \bmod 4} \mathbb{Z}/4\mathbb{Z},$$

where the p *in the indices are prime.*

However, the author noticed the following more elementary result while trying to produce explicit isometries over $\mathbb{Q}$. It allows us to bypass the theory of Brauer invariants when proving Theorem 3.34.

Proposition 1.47. *Let* α *be a non-zero rational number. Then there is an equality in the Grothendieck–Witt ring:*

$$4\langle\alpha\rangle = 4\langle\mathrm{sgn}(\alpha)\rangle.$$

Proof. By the multiplication structure of the Grothendieck–Witt ring, it suffices to prove the result for $\alpha > 0$. Furthermore, since multiplying α by a rational square does not change its square class, we may assume that α is a positive integer.

We first establish an analogous result for $\alpha = x^2 + y^2$, where $x, y \in \mathbb{Q}$ are non-zero. By Theorem 1.29

$$\begin{aligned} 2\langle 1\rangle &= \langle x^2\rangle + \langle y^2\rangle \\ &= \langle x^2 + y^2\rangle + \langle x^2y^2(x^2 + y^2)\rangle \\ &= 2\langle \alpha\rangle. \end{aligned}$$

The same equality trivially holds if exactly one of x, y vanishes.

By [Con97, p.140] only eight odd integers at most 41, together with integers of the form $2 \cdot 4^k$, $6 \cdot 4^k$ or $14 \cdot 4^k$ are not the sum of four non-zero squares. So possibly replacing α with $43^2\alpha$ we may assume that $\alpha = a^2 + b^2 + c^2 + d^2$ for some non-zero integers a, b, c, d. Then we obtain:

$$\begin{aligned} 4\langle 1\rangle &= \langle a^2\rangle + \langle b^2\rangle + \langle c^2\rangle + \langle d^2\rangle \\ &= \langle a^2 + b^2\rangle + \langle a^2b^2(a^2 + b^2)\rangle \\ &\quad + \langle c^2 + d^2\rangle + \langle c^2d^2(c^2 + d^2)\rangle \\ &= 2\left(\langle a^2 + b^2\rangle + \langle c^2 + d^2\rangle\right) \\ &= 2\langle \alpha\rangle + 2\langle (a^2 + b^2)(c^2 + d^2)\alpha\rangle \\ &= 2\langle \alpha\rangle + 2\langle \alpha\rangle\langle (a^2 + b^2)(c^2 + d^2)\rangle. \end{aligned}$$

Finally, by the two-square identity,

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2.$$

Hence, by the first part of the proof,

$$2\langle (a^2 + b^2)(c^2 + d^2)\rangle = 2\langle 1\rangle.$$

Substituting this into the preceding calculation gives

$$4\langle 1\rangle = 4\langle \alpha\rangle,$$

as required. □

Remark 1.48. The essential property of $\mathbb{Q}$ used above is that every positive rational number is a sum of four squares. More generally, the argument shows that if k is a field not of characteristic 2 and $\alpha \in k^\times$ is a sum of 2^n squares, for $n = 0, 1, 2$, then

$$2^n\langle \alpha\rangle = 2^n\langle 1\rangle.$$

1.4 Symmetric Spaces in a Category with Duality

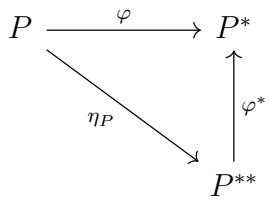
To end this chapter, we note that we can generalise the notion of bilinear forms to other contexts via categories with duality. The material in this section is only necessary to define the cohomology theory used in the proof of Theorem 2.58.

Definition 1.49 ([Bal05, Definition 1.1.5.])**.** A **category with duality** is a triple $(\mathcal{C}, *, \eta)$ where $\mathcal{C}$ is an additive category with a functor $*\colon \mathcal{C}^{op} \to \mathcal{C}$, equipped with a natural transformation $\eta\colon \mathrm{Id}_{\mathcal{C}} \to * \circ *$ satisfying the following condition. Denote for $M \in \mathcal{C}$ by M^* the object $*(M)$, and likewise for morphisms. Then we require

$$(\eta_M)^* \circ \eta_{M^*} = \mathrm{Id}_{M^*}$$

For all objects M.

Definition 1.50 ([Bal05, Definition 1.1.2.])**.** A **symmetric space** in $(\mathcal{C}, *, \eta)$ is an object P equipped with an isomorphism $\varphi\colon P \xrightarrow{\sim} P^*$, called the form, satisfying $\varphi^* \circ \eta_P = \varphi$.



Definition 1.51 ([Bal05, Definition 1.1.4.])**.** An **isometry** of symmetric spaces $(P, \varphi), (Q, \psi)$, is an intertwining map $h\colon P \to Q$, or a map such that $h^*\psi h \cong \varphi$

We will only need to consider 2 examples of such categories.

Example 1.52. Take the derived category, $D(k)$. Then, one can consider the duality functor $\mathrm{Hom}(-, k[0])$. This is equivalent to th category of *graded* vector spaces, equipped with a *graded* bilinear form of degree 0. I.e. a form β on a graded vector space V satisfying $\beta(V_i, V_j) = 0$ if $i + j \neq 0$

Then, more generally:

Example 1.53. Take the category of vector bundles over a scheme, with the duality functor, $*$ being given by $\mathrm{Hom}(-, \mathcal{L})$ for a line bundle $\mathcal{L}$. We will also look at the derived category of quasi-coherent sheaves, and consider $\mathrm{Hom}(-, \mathcal{L}[n])$.

Definition 1.54 ([Bal05, Definition 1.1.5.])**.** A **form preserving functor** F between two categories with duality, $(\mathcal{C}, *_{\mathcal{C}}, \eta)$ and $(\mathcal{D}, *_{\mathcal{D}}, \eta')$ is a functor satisfying $*_{\mathcal{D}}F = F*_{\mathcal{C}}$.

Example 1.55. Taking cohomology is duality preserving on $D(k)$. This holds since viewed as the category of graded vector spaces, dualising is exact and hence preserves cohomology.

Chapter 2

$\mathbb{A}^1$–Euler Numbers

Let k be a field not of characteristic 2. In this chapter, we study the titular $\mathbb{A}^1$–Euler number, a $\mathrm{GW}_0(k)$-valued invariant associated to 'relatively oriented' vector bundles. To motivate it, we recall the Poincaré–Hopf theorem from differential topology.

Theorem 2.1 (Poincaré–Hopf)**.** *Let M be a compact smooth manifold. Let V be a vector field on M with isolated zeroes. Then we have*

$$\chi(M) = \sum_{V(x_i)=0} \mathrm{index}_{x_i}(V).$$

Here the index is the Brouwer degree of the map $\partial D \to S^{n-1}$ defined by $z \mapsto V(z)/\|V(z)\|$, for a spherical neighbourhood D containing only the zero x_i.

In $\mathbb{A}^1$–homotopy theory, the Brouwer degree map $\deg\colon [S^n, S^n] \to \mathbb{Z}$ is replaced by Morel's degree map $\deg^{\mathbb{A}^1}\colon [\mathbb{P}^n/\mathbb{P}^{n-1}, \mathbb{P}^n/\mathbb{P}^{n-1}]^{\mathbb{A}^1} \to \mathrm{GW}_0(k)$. Hence, we're inspired to define the $\mathbb{A}^1$–Euler number $e^{\mathbb{A}^1}(V)$ using a formula of the form

$$e^{\mathbb{A}^1}(V) = \sum_{p \in Z} \mathrm{ind}_p(\sigma).$$

Here Z is the vanishing locus of a section σ of V, and $\mathrm{ind}_p(\sigma)$ is a to-be-defined element of $\mathrm{GW}_0(k)$ associated to the zero p.

In the first section, we define relative orientation and the appropriate coordinates for computing the local indices, Nisnevich coordinates. Then, in the second section, we prove that the stalks at the zeroes of a section of a vector bundle with isolated zeroes admit a presentation as a complete intersection. Next, we provide an overview of the self-duality theorem of Scheja-Storch, and use it to define the indices. Then, we define the $\mathbb{A}^1$–Euler number, compute several examples and

prove a few theorems that are useful for computing it. Finally, we end the chapter with a proof that the $\mathbb{A}^1$–Euler number is well defined and on proper schemes independent of the choice of section.

2.1 Relative Orientation and Nisnevich Coordinates

Let k be a field of characteristic not equal to 2, and X be a smooth k-scheme of dimension n. Fix a rank n vector bundle $\pi\colon V \to X$.

Definition 2.2 ([Mor07, Definition 4.3.])**.** An **orientation** of V is a choice of isomorphism $\det V \xrightarrow{\sim} \mathcal{L}^{\otimes 2}$ for some line bundle $\mathcal{L}$.

A related notion for quadratically enriched enumerative geometry, initially introduced in the real case in [OT14], is a relative orientation.

Definition 2.3 ([KW21, Definition 17])**.** A **relative orientation** of a vector bundle V is an isomorphism $j\colon \mathcal{H}om(\det TX, \det V) \xrightarrow{\sim} \mathcal{L}^{\otimes 2}$ for some line bundle $\mathcal{L}$.

The following alternative definition of relative orientation appears elsewhere in the literature, for instance, [BW23].

Lemma 2.4. *A choice of relative orientation is canonically equivalent to the choice of an isomorphism* $\rho\colon \det(V) \otimes \omega_{X/k} \to \mathcal{L}^{\otimes 2}$.

Proof. This follows from the fact for locally free sheaves, we have a canonical isomorphism $\mathrm{Hom}(\mathcal{G}, \mathcal{F}) \cong \mathcal{F} \otimes \mathcal{G}^\vee$ [Har13, Exercise II.5.1]. So we obtain $\mathrm{Hom}(\det TX, \det V) \cong \det V \otimes (\det TX)^\vee$. Since determinants and duals commute, we're left with $\det V \otimes \det T^*X$, and the right tensor factor by definition is $\omega_{X/k}$. Hence $\mathrm{Hom}(\det TX, \det V) \cong \det V \otimes \omega_{X/k}$, giving us our equivalence. □

Then, our intuition for the relative orientation comes from [BB25, p. 13], where the authors explain it morally provides choice of square root of the Jacobian determinant to quadratically enrich our count with, in the same sense that the sign of the Jacobian determinant is used in more traditional signed counts. We will make this more precise in the proof of Theorem 2.58.

Since the $\mathbb{A}^1$–Euler number is defined as a sum of local contributions, we will work locally using suitable coordinates.

Definition 2.5 ([KW21, Definition 18])**.** Let $p \in X$ be a closed point of X. By **Nisnevich coordinates** around p, we mean an étale map $\Phi\colon U \to \mathbb{A}^n_k$, for some Zariski open U containing p, such that the induced map on residue fields $\Phi^\sharp_{\phi(p)}\colon \kappa(\phi(p)) \to \kappa(p)$ is an isomorphism.

While these coordinates are motivated by the Nisnevich site in $\mathbb{A}^1$–homotopy theory, we can more directly motivate them as follows; the goal of the Nisnevich condition is to ensure that passing to coordinates preserves the arithmetic information associated to the residue field of a particular point. Since our final index will be a bilinear form on the stalk of a point in X, preserving the residue field ensures we're able to define forms on the stalks in a chart.

Standard coordinates on $\mathbb{A}^n_k$ or Plücker coordinates on the Grassmannian provide examples of Nisnevich coordinates since they're locally isomorphisms of schemes. We provide the following non-example and example to further illustrate this point.

Example 2.6. Let X be the curve defined by $y^2 = x^3 + 1$ over $\mathbb{Q}$. There is a natural coordinate system on X given by projection to the x-axis; the ring map $\mathbb{Q}[t] \to \frac{\mathbb{Q}[x,y]}{\langle y^2-x^3-1\rangle}$ defined by $t \mapsto x$. Consider the closed point p defined by the maximal ideal $\langle x-1, y^2-2\rangle$. This map is locally étale since the Jacobian is invertible.

The image of p is the point defined by the ideal $\langle t-1\rangle$. The residue field at this point is $\frac{\mathbb{Q}[t]}{\langle t-1\rangle} \cong \mathbb{Q}$. Meanwhile, the residue field at p is given by $\frac{\mathbb{Q}[x,y]}{\langle x-1, y^2-2, y^2-x^3-1\rangle} \cong \mathbb{Q}(\sqrt{2})$. These two fields are not isomorphic, hence the projection cannot be restricted to Nisnevich coordinates at p.

The essential reason for the failure of this map to be Nisnevich is because it does not preserve the arithmetic information of the degree of the solution $(1, \pm\sqrt{2})$ over $\mathbb{Q}$.

Conversely, we could instead take the projection to the y-axis, or the map $X \to \mathbb{A}^1_{\mathbb{Q}}$ arising from the map $\mathbb{Q}[t] \to \frac{\mathbb{Q}[x,y]}{\langle y^2-x^3-1\rangle}$ defined by $t \mapsto y$. This map provides Nisnevich coordinates around p, since the image of p is now the point corresponding to the ideal $\langle t^2-2\rangle$. Hence, it has residue field $\frac{\mathbb{Q}[t]}{\langle t^2-2\rangle} \cong \frac{\mathbb{Q}[x,y]}{\langle y^2-x^3-1, x-1, y^2-2\rangle}$. This map is locally étale since the Jacobian is non-vanishing.

Remark 2.7 ([KW21, p. 687])**.** It is not obvious that Nisnevich coordinates exist. Consider the k-scheme $X = \mathrm{Spec}(L)$ for L a non trivial yet finite and separable extension of k. The only possible Nisnevich coordinate map would be the canonical map $\mathrm{Spec}(L) \to \mathrm{Spec}(k)$, which is clearly not Nisnevich, as $L \ncong k$, yet it is étale as L is assumed to be separable.

Proposition 2.8 ([KW21, Proposition 20])**.** *There are Nisnevich coordinates around any closed point of a smooth k-scheme of dimension $n \geq 1$.*

To prove existence, we prove as separate results the cases where the residue field is separable and the case where it is an infinite field. Since any finite field is a separable extension of another finite field, this exhausts all cases.

Lemma 2.9 ([KW21, Lemma 19])**.** *Nisnevich coordinates exist around any closed point with separable residue field.*

Proof. First we reduce to the affine case by taking U to be an affine open containing p, and denote by $\mathfrak{p}$ the corresponding prime ideal. By [GR03, II Corollaire 5.9], there exist $x_1, \ldots, x_n$ in $\mathcal{O}_X(U)$ such that $\mathfrak{p} = \langle x_1, \ldots, x_n \rangle$. By abuse of notation, we've identified the x_i's with their image in the stalk. Then, by the primitive element theorem we may choose $x \in \mathcal{O}_X(U)$ such that the image of x in $\kappa(p)$ generates $\kappa(p)$ as an extension of k. If $dx = 0$ in $\Omega^1_X \otimes \kappa(p)$, then $d(x + x_1), dx_2, \ldots, dx_n$ is a $\kappa(p)$ basis of $\Omega^1_X \otimes \kappa(p)$. Therefore, the map $X \to \operatorname{Spec}(k[x+x_1, x_2, \ldots, x_n])$ induced by the obvious map on rings is étale, since it induces an isomorphism on the Kähler differentials, and induces an isomorphism on the residue field by construction. If dx is non-zero we can repeat the exact same argument, but use $dx, dx_1, \ldots, \hat{dx_i}, \ldots, dx_n$ as our basis. □

Proposition 2.10 ([Knu12, Proposition VIII.3.2.1])**.** *Nisnevich coordinates exist around any closed point with an infinite residue field.*

To resolve this case, we will need three lemmata.

Lemma 2.11 ([Knu12, Lemma VIII.3.2.2])**.** *Let k be an infinite field and A, B two finitely generated k-algebras. Next, fix a k-algebra homomorphism $\varphi_0 \colon A \to B$ and assume that B is generated as a k-algebra by elements $y_1, \ldots, y_N$. Further fix distinct maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$ of B.*

Then, there exists a nonempty Zariski open subset $U \subseteq k^N$ such that for every $(a_1, \ldots, a_N) \in U$, the homomorphism $\varphi \colon A[T] \to B$ defined by $\varphi|_A = \varphi_0$ and $\varphi(T) = a_1 y_1 + \cdots + a_N y_N$ satisfies the following:

1. *The ideals $\varphi^{-1}(\mathfrak{m}_1), \ldots, \varphi^{-1}(\mathfrak{m}_r)$ are distinct maximal ideals of $A[T]$.*

2. *If $B/\mathfrak{m}_i$ is separable over $A[T]/\varphi^{-1}(\mathfrak{m}_i)$, then φ induces an isomorphism between them.*

Remark 2.12. On the level of spectra, each maximal ideal $\mathfrak{m}_i$ defines a closed point of $\mathrm{Spec}(B)$. Then φ induces a map from the spectrum of B to the spectrum of A. The result says that if we instead project onto $\mathrm{Spec}(A) \times_k \mathbb{A}^1_k$, a generic choice of such a projection will keep our points distinct, and preserve the residue fields if the extension is separable.

Proof. Fix an algebraic closure $\bar{k}$ of k. Then, note for each $\mathfrak{m}_i$ we have embeddings $\sigma_{ij} \colon B/\mathfrak{m}_i \to \bar{k}$ where $j = 1, \ldots, n_i$ with n_i being the separable degree of $B/\mathfrak{m}_i$ over k. Let $\alpha_1, \ldots, \alpha_s$ be all the distinct k-linear homomorphisms obtained via the compositions $B \twoheadrightarrow B/\mathfrak{m}_i \xrightarrow{\sigma_{ij}} \bar{k}$. Consider the vector space $V = ky_1 + \cdots + ky_N \subseteq B$. Observe that since the y_i are generators of B, the distinct α_i are determined completely by their restriction to the y_i and hence to V. Thus, let $\beta_1, \ldots, \beta_s$ be the distinct maps obtained via the restriction of the α_i to V.

We then set $V' = \bigcup_{i \neq j} \{v \in V \colon \beta_i(v) = \beta_j(v)\}$. Since k is infinite, for any pair $i \neq j$ there must be some points where $\beta_i(v) \neq \beta_j(v)$ (if two linear forms agree at every point over an infinite field, they are equal). Thus, each $\{v \in V \colon \beta_i(v) = \beta_j(v)\}$ is a proper closed subspace of V, making $V' \subseteq V$ a proper Zariski closed set. Furthermore, it is not empty as it contains the origin.

We define U as the set of tuples $(a_1, \ldots, a_N)$ such that $a_1 y_1 + \cdots + a_N y_N \notin V'$. This implies $\varphi(T) \notin V'$. Because U is the pre-image of $V \setminus V'$ under the linear map $k^N \to V$ defined by $(a_1, \ldots, a_N) \mapsto a_1 y_1 + \cdots + a_N y_N$, it is open.

Suppose then that $\varphi^{-1}(\mathfrak{m}_i) = \varphi^{-1}(\mathfrak{m}_j) = \mathfrak{m}$ for some $i \neq j$. We obtain $n_i + n_j$ maps $A[T]/\mathfrak{m} \to \bar{k}$ via the compositions $A[T]/\mathfrak{m} \xrightarrow{\bar{\varphi}} B/\mathfrak{m}_i \xrightarrow{\sigma_{il}} \bar{k}$ and likewise for j. This would yield $n_i + n_j$ distinct embeddings of $A[T]/\mathfrak{m}$ into $\bar{k}$ (since they evaluate distinctly on $\varphi(T)$ by the construction of U). However, the separable degree of $A[T]/\mathfrak{m}$ over k is bounded above by the separable degree of $B/\mathfrak{m}_i$ (which is n_i), since $\bar{\varphi}$ embeds $A[T]/\mathfrak{m}$ as a subfield of $B/\mathfrak{m}_i$. This implies there can be at most n_i such embeddings, which is a contradiction. Hence, the pre-images must be distinct.

Now suppose that $B/\mathfrak{m}_i$ is separable over $A[T]/\varphi^{-1}(\mathfrak{m}_i)$ and has degree n. Then there exist n different embeddings among the σ_{ij} which induce the same map as σ_{i1} on $\varphi(A[T]/\varphi^{-1}(\mathfrak{m}_i))$. However, the σ_{ij} are distinct, so $\sigma_{i1}(\varphi(T) + \mathfrak{m}_i) \neq \sigma_{ij}(\varphi(T) + \mathfrak{m}_i)$ for $j \neq 1$. Since $\varphi(T) + \mathfrak{m}_i$ is in the image of $A[T]$, these embeddings cannot agree on the subfield unless $n = 1$, giving the desired isomorphism. □

Lemma 2.13 ([Knu12, Lemma VIII.3.2.3])**.** *Let k be an arbitrary field, and let B be a finitely generated k-algebra of dimension n with generators $y_1, \ldots, y_N$. Fix a maximal ideal $\mathfrak{m}$ of B and suppose that B is smooth over k at $\mathfrak{m}$. Then there*

exists a Zariski open subset U of the $n \times N$ matrices with entries in k such that for any $(a_{ij}) \in U$, the homomorphism of k-algebras $\varphi\colon k[T_1, \ldots, T_n] \to B$ defined by $\varphi(T_i) = a_{i1}y_1 + \cdots + a_{iN}y_N$ is étale at $\mathfrak{m}$.

Remark 2.14. This lemma morally says that generic linear combinations of ambient coordinates give local coordinates on $\operatorname{Spec}(B)$. It is therefore analogous to the consequence of the inverse function theorem that a generic linear projection of an embedded smooth n-dimensional manifold into $\mathbb{R}^n$ is a local diffeomorphism at any fixed point. This follows from [Wal16, Corollary 1.2.6].

Proof. Suppose that $B = k[Y_1, \ldots, Y_N]/I$, where I is the kernel of the surjection $\pi\colon k[Y_1, \ldots, Y_N] \to B$ defined by $Y_i \mapsto y_i$. As B is smooth over k at $\mathfrak{m}$, the Jacobian criterion for smoothness yields that the Jacobian defining B, when viewed as an affine variety, must have rank $N - n$. Thus, there exist $F_{n+1}, \ldots, F_N \in I$ such that at least one $(N-n) \times (N-n)$ minor of the matrix $\left(\frac{\partial F_i}{\partial Y_j}\right)$ is invertible at $\mathfrak{p} \coloneqq \pi^{-1}(\mathfrak{m})$. Denote the residue field $B/\mathfrak{m}$ by L.

The classes of $F_{n+1}, \ldots, F_N$ then form a linearly independent set in the cotangent space $\mathfrak{p}/\mathfrak{p}^2$ over L. In particular, possibly by relabelling the variables Y_i, the map $F \mapsto \left(\frac{\partial F}{\partial Y_{n+1}}, \ldots, \frac{\partial F}{\partial Y_N}\right)$ sends the F_i to $N - n$ linearly independent vectors.

Since B is smooth and n-dimensional at $\mathfrak{m}$, the localisation $B_{\mathfrak{m}}$ is an n-dimensional regular local ring, giving $\dim_L(\mathfrak{m}/\mathfrak{m}^2) = n$. Note that $\mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{p}/(I + \mathfrak{p}^2)$ since the pre-image of an element $m \in \mathfrak{m}^2$ under π is either an element of $\mathfrak{p}^2$ if $m = ab$ for two non-zero elements $a, b \in \mathfrak{m}$, else, the pre-image of 0 is all of I. Thus, $\dim_L(\mathfrak{p}/(I + \mathfrak{p}^2)) = n$. By the third isomorphism theorem, we have:

$$\frac{\mathfrak{p}}{I + \mathfrak{p}^2} \cong \frac{\mathfrak{p}/\mathfrak{p}^2}{(I + \mathfrak{p}^2)/\mathfrak{p}^2}$$

Since $k[Y_1, \ldots, Y_N]$ is N-dimensional, $\dim_L(\mathfrak{p}/\mathfrak{p}^2) = N$. Hence $\dim_L((I + \mathfrak{p}^2)/\mathfrak{p}^2) = N - n$.

Let $C = k[Y_1, \ldots, Y_N]$ and define the ideal $J = (F_{n+1}, \ldots, F_N)$. The image of J in $\mathfrak{p}/\mathfrak{p}^2$ has L-dimension $N - n$ because the vectors $F_{n+1}, \ldots, F_N$ are linearly independent. Since $J \subseteq I$, equality of dimension forces $J + \mathfrak{p}^2 = I + \mathfrak{p}^2$.

By Nakayama's Lemma, one can show that $I_{\mathfrak{p}} = J_{\mathfrak{p}}$. Recall that a set of elements in a module M over a local ring $(R, \mathfrak{m})$ is a minimal set of generators if and only if it descends to a basis of $M/\mathfrak{m}M$. Applying this to the local ring $C_{\mathfrak{p}}$, its maximal ideal $\mathfrak{p}C_{\mathfrak{p}}$, and the module $I_{\mathfrak{p}}$, we study the vector space $I_{\mathfrak{p}}/\mathfrak{p}I_{\mathfrak{p}}$. Since $B_{\mathfrak{m}} \cong C_{\mathfrak{p}}/I_{\mathfrak{p}}$ is regular of dimension n, and $C_{\mathfrak{p}}$ is regular of dimension N, $I_{\mathfrak{p}}$ must be minimally generated by $N - n$ elements. Therefore, $F_{n+1}, \ldots, F_N$ form a basis for this vector space and generate $I_{\mathfrak{p}}$.

Hence, by localising at some $s \notin \mathfrak{p}$, we can assume that $B = C/J = k[Y_1, \ldots, Y_N]/(F_{n+1}, \ldots, F_N)$. By [Ray06, Théorème 5 (2)], because the Jacobian determinant does not vanish, we recover that B is étale at $\mathfrak{m} = \mathfrak{p}/I$ over $k[Y_1, \ldots, Y_n]$.

Now, we want to study variations of this map and see when it remains étale. Specifically, if we consider maps $\varphi\colon k[T_1, \ldots, T_n] \to B$ defined by $\varphi(T_i) = a_{i1}y_1 + \cdots + a_{iN}y_N$, we want to determine for which matrices (a_{ij}) the map is étale.

Let us consider the linear change of variables $Y_i = \sum_{j=1}^{N} c_{ij}Z_j$ where $\det(c_{ij}) \neq 0$. Applying the multi-variable chain rule, we obtain $\frac{\partial F_i}{\partial Z_j} = \sum_{k=1}^{N} \frac{\partial F_i}{\partial Y_k}\frac{\partial Y_k}{\partial Z_j}$, which simplifies to $\sum_{k=1}^{N} c_{kj}\frac{\partial F_i}{\partial Y_k}$.

The $N \times N$ matrix (c_{ij}) can be decomposed into four blocks:

$$\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix}$$

where c_1 is $n \times n$. Denote by Δ' the $(N-n) \times n$ matrix $\left(\frac{\partial F_i}{\partial Y_k}\right)$ for $n < i \leq N$ and $1 \leq k \leq n$. Let Δ'' be the $(N-n) \times (N-n)$ matrix for $n < i \leq N$ and $n < k \leq N$. Then, the $(N-n) \times (N-n)$ matrix $\left(\frac{\partial F_i}{\partial Z_j}\right)$ for $n < i \leq N$ and $n < j \leq N$ is equal to $\Delta' c_2 + \Delta'' c_4$.

If $\Delta' c_2 + \Delta'' c_4 \in \mathrm{GL}_{N-n}(L)$, we have $B = k[Z_1, \ldots, Z_N]/(F_{n+1}, \ldots, F_N)$ and $\det\left(\frac{\partial F_i}{\partial Z_j}\right) \notin \mathfrak{p}$, meaning B is still étale over $k[Z_1, \ldots, Z_n]$ at $\mathfrak{m}$.

The set of invertible $N \times N$ matrices for which $\Delta' c_2 + \Delta'' c_4$ is also invertible forms an open set U', since it requires two polynomial constraints (the determinants) to be non-zero. The set U' is non-empty because it contains the identity matrix (as established in our first case where Δ'' was shown to be invertible). The set of inverses $U'' = (U')^{-1}$ is likewise open and non-empty.

The projection mapping a matrix $\begin{bmatrix} d_1 & d_2 \\ d_3 & d_4 \end{bmatrix}$ to its top $n \times N$ block maps U'' onto an open set U of $n \times N$ matrices. Therefore, if $\varphi\colon k[T_1, \ldots, T_n] \to B$ is defined by $\varphi(T_i) = a_{i1}y_1 + \cdots + a_{iN}y_N$ with $(a_{ij}) \in U$, we can choose a full matrix $c \in U''$ such that applying this change of variables yields $\varphi(T_i) = Z_i$, ensuring B is étale over $k[Z_1, \ldots, Z_n]$ at $\mathfrak{m}$. This provides our desired open set. □

Lemma 2.15. *[Knu12, Lemma VIII.3.2.4] Let k be an infinite field and let C be a k-algebra generated by $y_1, \ldots, y_N$ such that C is of dimension less than or equal to $n-1$. Then there exists a non-empty Zariski open subset of the $(n-1) \times N$ matrices (a_{ij}) such that the map of k-algebras $\psi\colon k[T_1, \ldots, T_{n-1}] \to C$ defined by $\psi(T_i) = a_{i1}y_1 + \cdots + a_{iN}y_N$ is finite.*

Proof. If $N = n - 1$, any invertible matrix will work, since the fibre will just be the pre-image of the corresponding point under the map. We proceed by induction. Suppose that $N \geq n$ and the lemma is true for algebras generated by $N - 1$ elements. Since $\dim(C) < N$, there is some nontrivial polynomial relation such that $F(y_1, \ldots, y_N) = 0$ for some $F \in k[Y_1, \ldots, Y_N]$. By making a change of variables, $Y_N = Z_N$ and $Y_i = Z_i + c_i Z_N$ for some choice of c_i, we can make F monic and its highest degree term a power of Z_N, for a generic choice of $(c_1, \ldots, c_{N-1}) \in k^{N-1}$ (by generic we mean the collection of such c_i forms an open subset).

Then, for such c_i we have $z_N = y_N$ is integral over $C' = k[z_1, \ldots, z_{N-1}]$, where $z_i = y_i - c_i y_N$. By the inductive hypothesis, there is a nonempty set U' of $(n-1) \times (N-1)$ matrices (a'_{ij}) such that the map $\psi' \colon k[T_1, \ldots, T_{n-1}] \to C'$ defined by $\psi'(T_i) = a'_{i1} z_1 + \cdots + a'_{i(N-1)} z_{N-1}$ is finite. Since C is finite over C' and finiteness is transitive, we can just take for U the set of matrices a_{ij} such that $a_{ij} = a'_{ij}$ when $1 \leq j \leq N - 1$, and $a_{iN} = -(a'_{i1} c_1 + \cdots + a'_{i(N-1)} c_{N-1})$. □

With these three lemmata resolved, we're now in place to prove Proposition 2.10.

Proof of Proposition 2.10. We note that the existence of Nisnevich coordinates in our setup is equivalent, upon passing from spectra to rings, to the following: the existence of a polynomial ring $A = k[T_1, \ldots, T_n]$, such that for a finitely generated k-algebra B with a maximal ideal $\mathfrak{m}$ such that B is smooth over k at $\mathfrak{m}$ and has dimension n at $B_{\mathfrak{m}}$, there exists a morphism of k-algebras $\varphi \colon A \to B$ which is étale at $\mathfrak{m}$ and induces an isomorphism $\bar{\varphi} \colon (A/J)_{\mathfrak{p}} \xrightarrow{\sim} (B/I)_{\mathfrak{m}}$ where $J = \varphi^{-1}(I)$ and $\mathfrak{p} = \varphi^{-1}(\mathfrak{m})$.

Let $\bar{y}_1, \ldots, \bar{y}_N$ be the classes of $y_1, \ldots, y_N$ in B/I. By Lemma 2.15, there is an open set U_4 of $(n-1) \times N$ matrices (a_{ij}) such that the map of k-algebras $\bar{\varphi}_0 \colon k[T_1, \ldots, T_{n-1}] \to B/I$ defined by $\bar{\varphi}_0(T_i) = a_{i1}\bar{y}_1 + \cdots + a_{iN}\bar{y}_N$ is finite. Then, let $\mathfrak{m}_1 = \mathfrak{m}$, and let $\mathfrak{m}_2, \ldots, \mathfrak{m}_r$ be the maximal ideals of B lying over $\bar{\varphi}_0^{-1}(\mathfrak{m})$. By Lemma 2.11, there exists an open set $U_1 \subseteq k^N$ such that for any $(a_{n1}, \ldots, a_{nN}) \in U_1$, the map $\bar{\varphi} \colon k[T_1, \ldots, T_n] \to B/I$ defined by $\bar{\varphi}(T_i) = \bar{\varphi}_0(T_i)$ for $i < n$ and $\bar{\varphi}(T_n) = a_{n1}\bar{y}_1 + \cdots + a_{nN}\bar{y}_N$ satisfies the conditions of that lemma.

Then, by Lemma 2.13, there exists an open set U_3 of $n \times N$ matrices (a_{ij}) so that the map $\varphi \colon k[T_1, \ldots, T_n] \to B$ defined by $\varphi(T_i) = a_{i1} y_1 + \cdots + a_{iN} y_N$ is étale at $\mathfrak{m}$. So if we choose our matrix to be in $U_3 \cap (U_4 \times U_1)$ we get a desired map. Then, since φ is étale at $\mathfrak{m}$, $A/\varphi^{-1}(\mathfrak{m})$ is separable over $B/\mathfrak{m}$ and hence an isomorphism. Since $\mathfrak{m}/I$ is the unique maximal ideal lying over $\varphi^{-1}(\mathfrak{m}) = \mathfrak{p}$, we

get that $k[T_1, \ldots, T_n]_{\mathfrak{p}} \to (B/I)_{\mathfrak{m}}$ is local and finite, and by Nakayama's lemma, surjective. So we are done. □

Tying these results together, we've resolved Proposition 2.8. As we can observe from our proofs of existence, there are usually many choices of Nisnevich coordinates. We further restrict the choice of coordinates by requiring that we have 'compatible' Nisnevich coordinates.

Specifically, since our Nisnevich coordinates Φ are étale, let us consider the pullback of the standard basis of $\mathbb{A}^n_k$ to the tangent bundle TX by Φ. This gives a distinguished ordered basis of TX.

Furthermore, since being a Nisnevich coordinate map is a local property, by perhaps shrinking the neighbourhood U of p, we can assume that V is trivialisable on U. Upon choosing a trivialisation $i\colon V \to \mathcal{O}^n_U$, pulling back the standard basis on $\mathbb{A}^n_k$ gives a distinguished basis of V.

Definition 2.16 ([KW21, Definition 21])**.** Consider the image ψ under the relative orientation $j\colon \underline{\mathrm{Hom}}(\det TX, \det V) \to \mathcal{L}^{\otimes 2}$ of the map $\det TX \to \det V$ induced by mapping the distinguished ordered bases of TX and V to each other. We say the trivialisation and the Nisnevich coordinates are **compatible** if $\psi = \mathbf{r}_U^{\otimes 2}$ for some $\mathbf{r}_U \in \mathcal{L}$.

The choice of this particular compatibility condition is motivated by the invariance of elements of $\mathrm{GW}_0(k)$ under multiplication by squares. Hence, if conversion between a computation on the induced coordinates on the trivialisation to a computation in our Nisnevich coordinates results in a change by a square factor, our Grothendieck–Witt class will not change.

2.2 Complete Intersection

Denote by σ a section of the vector bundle V. We then set Z to be the vanishing scheme of σ, it is naturally a closed subscheme of X. Fix a closed point p of Z.

Definition 2.17. We say a closed subscheme $Z \subseteq X$ is **isolated** if it contains finitely many points.

We say a point $p \in Z$ is **isolated** if there exists a Zariski open neighbourhood $p \in U$ such that $U \cap Z = \{p\}$.

We say σ has **isolated zeroes** if the vanishing subscheme Z is isolated.

When X is assumed to be locally of finite-type, for example, smooth, we can give the following algebraic characterisation:

Proposition 2.18 ([KW21, Proposition 23])**.** *We have p is an isolated point of Z if and only if $\mathcal{O}_{Z,p}$ is a finite k-algebra.*

A subscheme Z is an isolated closed subscheme if and only if $\mathcal{O}_Z$ is a finite k-algebra.

Remark 2.19. To demonstrate that this equivalence relies on our base scheme being locally of finite type, set $L = k(t)$. Consider the k-scheme $X = \mathbb{A}^1_L$. Then, consider the closed subscheme $Z = \operatorname{Spec}(L[x]/\langle x\rangle)$ of X. This is a closed subscheme, and is topologically isolated, consisting of a single point. However, algebraically, the stalk $\mathcal{O}_{X,p}$ is not finite over k, as it is the infinite degree extension L of k.

Proof. Suppose that p is an isolated point. There exists an open subscheme U such that $U \cap Z = p$. Hence, $U \cap Z$ is a finite type k scheme whose underlying topological space is a single point. So its coordinate ring will be a zero-dimensional finitely generated k-algebra, and hence must be finite dimensional. Localising to get the stalk preserves finite dimensionality.

Now suppose that $\mathcal{O}_{Z,p}$ is a finite k-algebra. Then, $\mathcal{O}_{Z,p}$ is dimension 0. Since p is a closed point, $\mathcal{O}_{Z,p}$ has dimension 0. Let Z^0 be the irreducible component of Z containing p. As Z^0 is of finite type over a scheme, $\dim Z^0 = \dim \mathcal{O}_{Z,p} + \dim \mathcal{O}_{Z,p}/\mathfrak{p}$, which is 0. So Z^0 is an irreducible dimension 0 scheme of finite type over k. Hence it will be a single point, which is p. So we can take U to be the complement of the other irreducible components of $\mathfrak{p}$.

Suppose that Z consists of finitely many points. Then Z is a zero-dimensional finite type k-algebra. As a result it is finite.

If $\mathcal{O}_Z$ is a finite k-algebra, it is a zero-dimensional Noetherian ring. Thus it will consist of finitely prime ideals. □

Now fix Nisnevich coordinates $\Phi\colon U \to \mathbb{A}^n_k = \operatorname{Spec}(k[x_1, \ldots, x_n])$ around p, and a compatible trivialisation. The following is the main theorem of this section.

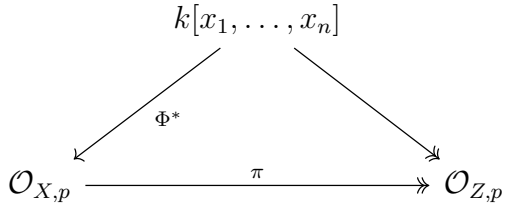
Theorem 2.20 ([KW21, Lemma 27])**.** *Suppose that Z is the vanishing subscheme of a section σ of a relatively oriented rank n vector bundle V on a smooth n-dimensional scheme X, and that σ has isolated zeroes.*

Fix $p \in Z$, and choose Nisnevich coordinates $\Phi\colon U \to \mathbb{A}^n_k$ on an affine neighbourhood U of p, together with a compatible trivialisation of V in which $\sigma = (f_1, \ldots, f_n)$. Let $\mathfrak{p} \subseteq \mathcal{O}_{X,p}$ be the maximal ideal, and let $\mathfrak{q} \subseteq k[x_1, \ldots, x_n]$ be the ideal corresponding to $\Phi(p)$.

Then there exists an m such that $\mathfrak{p}^m = 0$ in $\mathcal{O}_{Z,p}$ and elements $g_1, \ldots, g_n$ of $k[x_1, \ldots, x_n]_{\mathfrak{q}}$ such that;

- *The quotient $\frac{k[x_1,\ldots,x_n]_{\mathfrak{q}}}{\langle g_1,\ldots,g_n\rangle}$ is a presentation of $\mathcal{O}_{Z,p}$ as a complete intersection.*
- *For all i we have $\Phi^* g_i - f_i \in \mathfrak{p}^{2m}$.*

To prove this result, we need some machinery. Our choice of trivialisation gives σ in the form $(f_1, \ldots, f_n)$ for $f_i \in \mathcal{O}_X(U)$. We then have $\mathcal{O}_{Z,p} = \frac{\mathcal{O}_{X,p}}{\langle f_1,\ldots,f_n\rangle}$, where by abuse of notation we're using f_i to mean the restriction $f_i|_{\mathcal{O}_{X,p}}$. We then obtain the following diagram.



Let $\mathfrak{p}$ be the maximal ideal of $\mathcal{O}_{X,p}$. Since $\mathcal{O}_{Z,p}$ is finite by our assumption on σ, there necessarily exists a value of $m \geq 0$ such that $\mathfrak{p}^m = 0$ in $\mathcal{O}_{Z,p}$. This then implies that $\langle f_1, \ldots, f_n\rangle = \langle f_1, \ldots, f_n\rangle + \mathfrak{p}^m$ in $\mathcal{O}_{X,p}$.

Lemma 2.21 ([KW21, Lemma 24])**.** *The stalk $\mathcal{O}_{Z,p}$ is generated as a k-algebra by the images of $\Phi^* x_1, \ldots, \Phi^* x_n$ under the quotient map.*

$\mathcal{O}_{X,p}/\mathfrak{p}^m$ is generated by $\Phi^* x_1, \ldots, \Phi^* x_n$ for all $m > 0$.

Proof. By our previous discussion, there exists an m such that we have an equality of the ideals $\langle f_1, \ldots, f_n\rangle$ and $\langle f_1, \ldots, f_n\rangle + \mathfrak{p}^m$ in $\mathcal{O}_{X,p}$ for large enough m. Note then that a generating set of $\mathcal{O}_{X,p}/\mathfrak{p}^m$ will descend to a generating set of $\mathcal{O}_{Z,p}$. Hence it suffices to prove only the second claim.

By the Nisnevich condition on our coordinate map Φ, they induces an isomorphism between the residue fields $k[x_1, \ldots, x_n]/\mathfrak{q} \to \mathcal{O}_{X,p}/\mathfrak{p}$ where $\mathfrak{q} = \Phi^*(\mathfrak{p})$.

We claim by induction that the induced map $k[x_1, \ldots, x_n] \to \mathcal{O}_{X,p}/\mathfrak{p}^m$ is a surjection. Suppose it is a surjection for $\mathcal{O}_{X,p}/\mathfrak{p}^m$. Then, given $y \in \mathcal{O}_{X,p}/\mathfrak{p}^{m+1}$, since $k[x_1, \ldots, x_n]$ surjects onto $\mathcal{O}_{X,p}/\mathfrak{p}^m$, we can take some $\psi \in k[x_1, \ldots, x_n]$ such that ψ maps onto the same class as y in $\mathcal{O}_{X,p}/\mathfrak{p}^m$. Then, denote by y' the image of ψ in $\mathcal{O}_{X,p}/\mathfrak{p}^{m+1}$. Then, $y - y' \in \mathfrak{p}^m$.

This implies $y - y' = \sum_i a_i b_i$ where $a_i \in \mathfrak{p}^{m-1}$ and $b_i \in \mathfrak{p}$. As Φ is étale, it induces an isomorphism on cotangent spaces, $\mathfrak{p}/\mathfrak{p}^2 \xrightarrow{\sim} \mathfrak{q}/\mathfrak{q}^2$. This implies that $\mathfrak{p}^{m-1}/\mathfrak{p}^m \cong \mathfrak{q}^{m-1}/\mathfrak{q}^m$. Then there exists an $\eta \in \mathfrak{q}^{m-1}$ that gets mapped down to

$a_i \pmod{\mathfrak{p}^m}$. Denote its image in $\mathfrak{p}^{m-1}$ as a_i'. Then, $a_i - a_i' \in \mathfrak{p}^m$. Likewise, there exists a b_i' in the image such that $b_i - b_i' \in \mathfrak{p}^2$.

So we will get modulo $\mathfrak{p}^{m+1}$ that $\sum_i (a_i - a_i')b_i = 0$, and $\sum_i a_i'(b_i - b_i') = 0$, which implies that $\sum_i a_i b_i = \sum_i a_i' b_i = \sum_i a_i' b_i'$. Then, we note by construction that as a_i' and b_i' are in the image of the induced map by Φ onto $\mathcal{O}_{X,p}/\mathfrak{p}^{m+1}$, it follows y is also in the image of the map. □

Hence, the map $k[x_1, \ldots, x_n] \to \mathcal{O}_{X,p}/\mathfrak{p}^{2m}$ is surjective. Thus, by picking an element of $k[x_1, \ldots, x_n]$ that maps onto the image of $-f_i$ under the projection to $\mathcal{O}_{X,p}/\mathfrak{p}^{2m}$, there exists $g_i \in k[x_1, \ldots, x_n]$ for $i = 1, \ldots, n$ such that $\Phi^* g_i - f_i \in \mathfrak{p}^{2m}$. Then note that $\mathfrak{p}^{2m} \subseteq \mathfrak{p}^{m+1}$.

Lemma 2.22 ([KW21, Lemma 25])**.** *We have an equality of ideals* $\langle \Phi^* g_1, \ldots, \Phi^* g_n \rangle^j = \langle f_1, \ldots, f_n \rangle^j$ *for all* $j > 0$.

Proof. It is sufficient to prove for $j = 1$. Since $\Phi^* g_i = (\Phi^* g_i - f_i) + f_i \in \mathfrak{p}^{m+1} + \langle f_1, \ldots, f_n \rangle$, we then obtain $\langle g_1, \ldots, g_n \rangle \subseteq \langle f_1, \ldots, f_n \rangle + \mathfrak{p}^m = \langle f_1, \ldots, f_n \rangle$ by our assumption on m. Hence $\langle g_1, \ldots, g_n \rangle \subseteq \langle f_1, \ldots, f_n \rangle$.

In the other direction, we have the g_i generate $\langle g_1, \ldots, g_n \rangle + \mathfrak{p}^m$ modulo $\mathfrak{p}^{m+1}$. That is because modulo $\mathfrak{p}^{m+1}$ we have $g_i = f_i \bmod \mathfrak{p}^{m+1}$. Hence $\mathfrak{p}^m \subseteq \langle g_1, \ldots, g_n \rangle$, and $\langle f_1, \ldots, f_n \rangle \subseteq \langle g_1, \ldots, g_n \rangle$. □

Lemma 2.23 ([KW21, Lemma 26])**.** *There is an equality of ideals* $\langle g_1, \ldots, g_n \rangle^j = (\Phi^*)^{-1}(\langle f_1, \ldots, f_n \rangle^j)$ *in* $k[x_1, \ldots, x_n]_{\mathfrak{q}}$ *for all* $j > 0$.

Proof. We have an obvious inclusion. Hence it suffices to show the quotient map $a\colon k[x_1, \ldots, x_n]_{\mathfrak{q}}/\langle g_1, \ldots, g_n \rangle^j \to k[x_1, \ldots, x_n]_{\mathfrak{q}}/(\Phi^*)^{-1}(\langle f_1, \ldots, f_n \rangle^j)$ is injective. As Φ is flat, the map $\Phi^*\colon k[x_1, \ldots, x_n]_{\mathfrak{q}} \to \mathcal{O}_{X,p}$ is faithfully flat. Then after tensoring, it is sufficient to show $\mathcal{O}_{X,p}/\langle g_1, \ldots, g_n \rangle^j \to \mathcal{O}_{X,p}/\mathcal{O}_{X,p}(\Phi^*)^{-1}(\langle f_1, \ldots, f_n \rangle^j)$ is surjective. However, as we just proved those ideals are equal in $\mathcal{O}_{X,p}$ we're done. □

Then, we have proved our theorem, as the map $k[x_1, \ldots, x_n]_{\mathfrak{q}} \to \mathcal{O}_{Z,p}$ is surjective with kernel $(\Phi^*)^{-1}\langle f_1, \ldots, f_n \rangle = \langle g_1, \ldots, g_n \rangle$. The presentation $\mathcal{O}_{Z,p} \cong \frac{k[x_1, \ldots, x_n]_{\mathfrak{q}}}{\langle g_1, \ldots, g_n \rangle}$ presents the left hand side as a local complete intersection.

2.3 Scheja-Storch Duality

In this section we give an overview of a duality theorem of Scheja–Storch for complete intersections.

Definition 2.24 ([Stacks, Tag 00SP])**.** A ring map $A \to B$ is a **relative global complete intersection** if there exists a presentation $A[x_1, \ldots, x_n]/(f_1, \ldots, f_c) \cong B$ and every non-empty geometric fibre of the induced map $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ has dimension $n - c$.

Remark 2.25. In the case where A is a field k, a relative global complete intersection recovers the usual meaning of just requiring the variety be of dimension $n - c$, as $\mathrm{Spec}(B) \to \mathrm{Spec}(k)$ has a single fibre over k.

Then, in [SS75] the authors proved the following result;

Theorem 2.26. *A choice of presentation of an A-algebra B as a finite, flat, relative global complete intersection over a Noetherian ring A determines a distinguished isomorphism $\mathrm{Hom}_A(B, A) \xrightarrow{\sim} B$ of B-modules.*

To see the relevance of this theorem, recall by Theorem 2.20 at each isolated zero p of a section of a vector bundle, the stalk $\mathcal{O}_{Z,p}$ admits a compatible presentation as a complete intersection. Then Theorem 2.26 yields an isomorphism of $\mathcal{O}_{Z,p}$-modules, $\mathrm{Hom}_k(\mathcal{O}_{Z,p}, k) \xrightarrow{\sim} \mathcal{O}_{Z,p}$. This isomorphism will be used in the following section to define the $\mathbb{A}^1$–index of a zero of σ.

The proof of Theorem 2.26 may be skipped by the hurried reader. However, the construction of the isomorphism is important for understanding the proofs of some later results and our example computations of the $\mathbb{A}^1$–Euler number.

To construct said isomorphism, we begin by noting a presentation of a finitely generated A-algebra B arises from a surjection $\pi\colon A[x_1, \ldots, x_n] \to B$. Then we obtain the following commutative diagram

$$\begin{array}{ccc} A[x_1, \ldots, x_n] \otimes_A A[x_1, \ldots, x_n] & \xrightarrow[m_1]{} & A[x_1, \ldots, x_n] \\ \downarrow{\scriptstyle \pi\otimes\pi} & & \downarrow{\scriptstyle \pi} \\ B \otimes_A B & \xrightarrow{\qquad m_2 \qquad} & B \end{array}$$

Here, m_1 and m_2 are the multiplication maps $a \otimes b \mapsto ab$. Next, suppose the kernel of π is generated by the elements $g_1, \ldots, g_n \in A[x_1, \ldots, x_n]$. We then have $g_j \otimes 1 - 1 \otimes g_j \in \ker(m_1)$. However, we can observe $\ker(m_1)$ is generated by the set of elements of the form $x_i \otimes 1 - 1 \otimes x_i$. Hence, we can find a_{ij} such that

$$g_j \otimes 1 - 1 \otimes g_j = \sum_{i=1}^{n} a_{ij} \cdot (x_i \otimes 1 - 1 \otimes x_i).$$

Then, we fix $\Delta \coloneqq (\pi \otimes \pi)(\det(a_{ij})) \in B \otimes_A B$.

Next, we define the A-module map $\chi\colon B \otimes_A B \to \mathrm{Hom}_A(\mathrm{Hom}_A(B, A), B)$ as the map generated by setting $\chi(b \otimes c) = [\varphi \mapsto \varphi(b)c]$. Then, Theorem 2.26 is more precisely stated as:

Theorem 2.27 ([Kuh+22, Theorem 3.4.])**.** *The map of B-modules*

$$\chi(\Delta)\colon \mathrm{Hom}_A(B, A) \to B$$

is an isomorphism. Here, $\mathrm{Hom}_A(B, A)$ *is a B-module under pre-multiplication.*

In Corollary 2.41, we will see that Δ is independent of the choice of a_{ij}, and only depends on our choice of presentation. This motivates the following definition.

Definition 2.28. The element $\det(a_{ij})$ is called the **distinguished socle element**.

We now dedicate the rest of this section to proving Theorem 2.26. While the original result was shown in [SS75], we follow the presentation of [Kuh+22]. This paper has the advantage of not being written in German. First, we have some preliminary results and definitions from commutative algebra.

Definition 2.29. Let M be an A module. Consider a 2 term resolution of M, $A^{\oplus N} \to A^{\oplus n} \to M$. Consider the $n \times N$ matrix T representing $A^{\oplus N} \to A^{\oplus n}$. The i-th **Fitting ideal** $\mathrm{Fit}^i_A(M)$ is the ideal of A generated by the determinants of the $(n - i) \times (n - i)$ sub-matrices of T.

Lemma 2.30 ([Stacks, Tag 07Z8])**.** *The Fitting ideals,* $\mathrm{Fit}^i_A(M)$*, are invariant of the choice of free resolution $A^N \to A^n \to M$.*

Definition 2.31. Let $\varphi\colon E \to A$ be a map of A-modules. Then the Koszul complex is the complex:

$$K_\bullet(\varphi) \coloneqq \cdots \to \wedge^3 E \to \wedge^2 E \to E \xrightarrow{\varphi} A \to 0$$

with the differential $d_m\colon \wedge^m E \to \wedge^{m-1} E$ defined by setting

$$d_m(e_1 \wedge \cdots \wedge e_n) \coloneqq \sum_{i=1}^{n} (-1)^{i+1} \varphi(e_i) \cdot e_1 \wedge \cdots \wedge \hat{e_i} \wedge \cdots \wedge e_n.$$

Lemma 2.32 ([Stacks, Tag 0621])**.** *Suppose that we have 2 maps of A-modules, $\varphi\colon E \to A$ and $\varphi'\colon E' \to A$, and a map $\psi\colon E \to E'$ such that $\varphi'\psi = \varphi$. Then ψ induces a map of Koszul complexes $K_\bullet(\varphi) \to K_\bullet(\varphi')$.*

Proof. The desired vertical maps are given by $\wedge^j \psi \colon \wedge^j E \to \wedge^j E'$. That they commute with the differential follows from our requirement $\varphi' \psi = \varphi$. □

Notation 2.33. Let $f_1, \ldots, f_n \in A$. Consider the map $\varphi \colon A^{\oplus n} \to A$ given by multiplication by f_i on the i-th coordinate. We set $K_\bullet(f_1, \ldots, f_n) = K_\bullet(\varphi)$.

Lemma 2.34 ([Stacks, Tag 062F])**.** *Suppose $f_1, \ldots, f_n$ is a regular sequence in A. Then $H_i(K_\bullet(f_1, \ldots, f_n)) = 0$ for $i \neq 0$.*

Remark 2.35. The converse of this statement does not hold. We say a sequence is Koszul-regular if $H_i(K_\bullet(f_1, \ldots, f_n)) = 0$ for $i \neq 0$. We observe being Koszul-regular is invariant under permutation. However, being regular is in general not. Following [Stacks, Tag 00LG], take $A = k[x, y, z]$ and consider the sequence $x(y-1), y, z(y-1)$. This is a regular sequence, however $x(y-1), z(y-1), y$ is not as $z(y-1)$ is a zero divisor in $k[x, y, z]/(x(y-1))$. Both are Koszul-regular.

Remark 2.36. The main consequence of Lemma 2.34 we will employ is that the corresponding Koszul complex yields an acyclic resolution of $A/\langle f_1, ..., f_n \rangle$. Specifically, the projection map $K_\bullet(f_1, \ldots, f_n) \to A/\langle f_1, ..., f_n \rangle$ is a quasi-isomorphism.

Remark 2.37. The necessity of Koszul complexes can be understood via derived algebraic geometry. In this setting, the 'correct structure sheaf' for $\mathcal{Z}$ is the Koszul complex of the map $V^* \to \mathcal{O}_X$ given by evaluation at σ. Hence, if we're doing homological algebra, we should expect to work with it.

Lemma 2.38 ([Stacks, Tag 00SC])**.** *A presentation $\frac{A[x_1, \ldots, x_n]}{\langle f_1, \ldots, f_n \rangle}$ of B is a relative global complete intersection if and only if the sequence $f_1, \ldots, f_n$ is regular.*

Lemma 2.39 ([Stacks, Tag 00SW])**.** *Relative global complete intersection maps are flat.*

We can now begin to prove the theorem.

Proposition 2.40 ([Kuh+22, Lemma 2.1.])**.** *Suppose that A is a Noetherian ring, $f_1, \ldots, f_n$ and $g_1, \ldots, g_n$ two sequences of elements of A such that*

1. *We have an inclusion of ideals $\mathfrak{b} \coloneqq \langle g_1, \ldots, g_n \rangle \subseteq \mathfrak{a} \coloneqq \langle f_1, \ldots, f_n \rangle$.*
2. *If $\mathfrak{p}$ is a prime ideal such that $\mathfrak{a} \subseteq \mathfrak{p}$, both sequences are regular in $A_\mathfrak{p}$.*

Then, write $g_i = \sum_{j=1}^n a_{ij} f_j$ where (a_{ij}) is the arising matrix of coefficients. Then, set $\Lambda \coloneqq \det(a_{ij})$, and $\bar{\Lambda}$ the image of Λ under the projection $A \to A/\mathfrak{b}$. We then have

1. The element $\bar{\Lambda}$ is independent of the choice of (a_{ij}).

2. We have the equality of ideals $\langle \bar{\Lambda} \rangle = \mathrm{Fit}_{A/\mathfrak{b}}(\mathfrak{a}/\mathfrak{b})$.

3. We have the equalities of ideals $\langle \bar{\Lambda} \rangle = \mathrm{Ann}_{A/\mathfrak{b}}(\mathfrak{a}/\mathfrak{b})$ and $\mathfrak{a}/\mathfrak{b} = \mathrm{Ann}_{A/\mathfrak{b}}(\bar{\Lambda})$.

Proof. We begin by reducing to the case where A is a local ring. We note that two ideals of a commutative ring A are equal if and only if they're equal in every localisation of A at a maximal ideal. Likewise, two elements are equal if and only if they're equal in all localisations of A at a maximal ideal. So the lemma is true if it is true in every $A_{\mathfrak{m}}$, and localising at a maximal ideal yields a local ring. So, checking the result on all local rings suffices.

Then, we may further assume that $\mathfrak{a} \subseteq \mathfrak{m}$. This follows as for a set of generators $f_1, \ldots, f_n$ if we have $\mathfrak{a} \not\subseteq \mathfrak{m}$ then at least one of the f_j must not lie in $\mathfrak{m}$. However, as A is local, this then implies that f_j is invertible. So we first obtain $\mathfrak{a} = A$. We also have in $A/\mathfrak{b}$, the equation $g_i = \sum_{i=1}^n a_{ij} f_j$ becomes $0 = \sum_{i=1}^n a_{ij} f_j$. This would imply $-f_j^{-1} \sum_{i=1}^n a_{ij} f_j = a_{ij}$. This yields a linear dependence among the a_{ij}, which would imply the determinant of (a_{ij}) in $A/\mathfrak{b}$ will be 0, so $\bar{\Lambda}$ is zero. We further have $\mathrm{Fit}^0_{A/\mathfrak{b}}(A/\mathfrak{b}) = 0$, since $\mathrm{Fit}^0_C(C) = 0$ for all rings, as $0 \to C \xrightarrow{Id} C$ is our desired 2-term free resolution. Finally, as $A/\mathfrak{b}$ has an identity, $\mathrm{Ann}_{A/\mathfrak{b}}(A/\mathfrak{b}) = 0 = \langle \bar{\Lambda} \rangle$, and $A/\mathfrak{b} = \mathrm{Ann}_{A/\mathfrak{b}}(0)$. Hence, we can assume that $\mathfrak{a} \neq A$.

In this setting, we then have assumption 2 implies (f_i) and (g_i) are both regular sequences. This follows as for a local ring $(A, \mathfrak{m})$ the canonical map induces an isomorphism $A_{\mathfrak{m}} \cong A$, and $\mathfrak{m}$ is a prime containing $\mathfrak{a}$.

To prove the first assertion, suppose that we have an alternative presentation $g_i = \sum b_{ij} f_j$. Then statement 1 reduces to showing that $\det(a_{ij}) - \det(b_{ij}) \in \mathfrak{b}$. It is sufficient to prove this under the assumption $a_{ij} = b_{ij}$ for $i \neq n$. In this case, we're just altering the presentation of g_n in terms of the a_{nj}. However, by A being local and Noetherian, we may permute the order of (g_i), and preserve regularity [Stacks, Tag 00LJ]. Hence it is sufficient to only alter the presentation of one g_i at a time.

Then, we set $c_{ij} = a_{ij} = b_{ij}$ for $i \neq n$, and $c_{nj} = a_{nj} - b_{nj}$. Then cofactor expansion along the n-th row yields $\det(c_{ij}) = \det(a_{ij}) - \det(b_{ij})$. We also have

the matrix equation

$$[c_{ij}] \cdot \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_{n-1} \\ f_n \end{bmatrix} = \begin{bmatrix} \sum a_{1j} f_j \\ \sum a_{2j} f_j \\ \vdots \\ \sum a_{(n-1)j} f_j \\ \sum (a_{nj} - b_{nj}) f_j \end{bmatrix} = \begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_{n-1} \\ 0 \end{bmatrix}.$$

Then, we recall Cramer's rule states that if we have the matrix equation $Ax = b$, the solution is given by setting $x_i = \frac{\det(A_i)}{\det(A)}$, where A_i is the matrix obtained by replacing the i-th column of A by a copy of b. This implies $\det(A)x_i = \det(A_i)$. Then, applying this to our matrix equation, we have $x_i = f_i$, but we know that $\det(A_i) \in \langle g_1, \dots, g_{n-1} \rangle$ by co-factor expansion along the i-th column. So we obtain $\det(c_{ij}) f_k \in \langle g_1, \dots, g_{n-1} \rangle$ for all k. However, as $\mathfrak{a}$ is generated by the f_k, we have $\det(c_{ij}) \cdot \mathfrak{a} \subseteq \langle g_1, \dots, g_{n-1} \rangle$. But as $g_n \in \mathfrak{a}$ this would imply $\det(c_{ij}) g_n \in \langle g_1, \dots, g_{n-1} \rangle$. However, passing to $\frac{A}{\langle g_1, \dots, g_{n-1} \rangle}$ this implies we have $\det(c_{ij}) \cdot g_n = 0$. Then since (g_i) is a regular sequence, g_n cannot be a zero-divisor in this ring. Hence we conclude $\det(c_{ij}) \in \langle g_1, \dots, g_{n-1} \rangle \subseteq \mathfrak{b}$ and we've proved the first assertion.

To prove the second assertion, we begin by noting that by right exactness of the tensor product, a resolution $A^N \to A^n \to \mathfrak{a}/\mathfrak{b} \to 0$ yields a resolution $(A/\mathfrak{b})^N \to (A/\mathfrak{b})^n \to \mathfrak{a}/\mathfrak{b} \to 0$. Then, the entries of the matrix representing $(A/\mathfrak{b})^N \to (A/\mathfrak{b})^n$ are, by computing the Kronecker product with the identity on $A/\mathfrak{b}$, the residues of the entries modulo $\mathfrak{b}$. This then tells us that the minors of this map are the minors of our original map $A^N \to A^n$ reduced modulo $\mathfrak{b}$. So we get the equality $\mathrm{Fit}^0_A(\mathfrak{a}/\mathfrak{b}) \bmod \mathfrak{b} = \mathrm{Fit}^0_{A/\mathfrak{b}}(\mathfrak{a}/\mathfrak{b})$.

Hence it is sufficient to show that $\mathrm{Fit}^0_A(\mathfrak{a}/\mathfrak{b}) = \langle \Lambda \rangle + I$ for some ideal $I \subseteq \mathfrak{b}$. Then by Lemma 2.30 we can choose to compute the Fitting ideal of $\mathfrak{a}/\mathfrak{b}$ via the presentation $A^n \oplus A^{\binom{n}{2}} \xrightarrow{T} A^n \to \mathfrak{a}/\mathfrak{b} \to 0$, where $T = (a_{ij}) \times d_2^{Kosz}$, where the Koszul differential d_2^{Kosz} becomes the matrix whose column corresponding to the transposition (ij) for $i < j$ is defined by having f_i in the j'th row, and $-f_j$ in the i'th row. This is exact by acyclicity of the Koszul complex for a regular sequence.

Then, the 0-th Fitting ideal is given by the $n \times n$ minors of T. The first minor is Λ. Then let Λ' be another $n \times n$ minor. It will be the determinant of a matrix T' which contains r columns of (a_{ij}) and $n - r$ columns of d_2^{Kosz}. We can assume T' contains the first r columns of (a_{ij}) by reordering the g_i, as regularity is preserved under permutation when A is local and Noetherian [Stacks, Tag 00LJ].

Then, we get the matrix equation;

$$(T')\begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix} = \begin{bmatrix} g_1 \\ \vdots \\ g_r \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

This holds as the Koszul differential, essentially by construction, vanishes on this vector. Then, applying Cramer's rule, we'll obtain $\det(T')f_i \in \mathfrak{b}$. Hence, $\Lambda' \cdot \mathfrak{a} \in \langle g_1, \dots, g_r \rangle \subseteq \langle g_1, \dots, g_{n-1} \rangle$. Hence, $\Lambda' \cdot g_n \in \langle g_1, \dots, g_{n-1} \rangle$. Then regularity of the g_i like before implies that $\Lambda' \in \mathfrak{b}$, so taking I to be the ideal generated by all minors $\Lambda' \neq \Lambda$, we obtain $I \subseteq \mathfrak{b}$ and $\mathrm{Fit}^0_A(\mathfrak{a}/\mathfrak{b}) = \Lambda + I$ and we're done.

Now we prove the third claim. First, we will show that $\mathrm{Ann}_{A/\mathfrak{b}}(\mathfrak{a}/\mathfrak{b}) \cong \mathrm{Tor}^A_n(A/\mathfrak{b}, A/\mathfrak{b})$. As a regular sequence is Koszul regular, we have a quasi-isomorphism $K_\bullet(f_1, \dots, f_n) \cong A/\mathfrak{a}$. Then, the Koszul complex for a length n sequence is of length n itself. So in degrees $n+1$ to $n-1$, we obtain $K_\bullet(f_1, \dots, f_n) \otimes A/\mathfrak{b}$ is given by

$$0 \to A/\mathfrak{b} \xrightarrow{(f_1, \dots, f_n)} (A/\mathfrak{b})^{\oplus n} \to$$

So $\mathrm{Tor}^A_n(A/\mathfrak{b}, A/\mathfrak{b})$ is the kernel of the map $(f_1 \dots, f_n)$. However, the kernel of this is the annihilator of $\mathfrak{a}/\mathfrak{b}$, as any element x in the kernel has to multiply with all of the residues of the f_i to yield 0, so $x \cdot \mathfrak{a}/\mathfrak{b} = 0$.

Next, we claim that $\mathrm{Tor}^A_n(A/\mathfrak{a}, A/\mathfrak{b}) \cong \Lambda \cdot A/\mathfrak{b}$. This holds because we have a short exact sequence of A-modules $0 \to \mathfrak{a}/\mathfrak{b} \to A/\mathfrak{b} \to A/\mathfrak{a} \to 0$. This induces a long exact sequence on torsion. Then we claim that for $j \geq 1$, this exact sequence splits into sub-sequences

$$0 \to \mathrm{Tor}^A_j(A/\mathfrak{b}, \mathfrak{a}/\mathfrak{b}) \to \mathrm{Tor}^A_j(A/\mathfrak{b}, A/\mathfrak{b}) \to \mathrm{Tor}^A_j(A/\mathfrak{b}, A/\mathfrak{a}) \to 0.$$

To see this, we note that the Koszul complex yields isomorphisms $\mathrm{Tor}^A_j(A/\mathfrak{b}, A/\mathfrak{b}) \cong (A/\mathfrak{b})^{\binom{n}{j}}$ and $\mathrm{Tor}^A_j(A/\mathfrak{b}, \mathfrak{a}/\mathfrak{b}) \cong (\mathfrak{a}/\mathfrak{b})^{\binom{n}{j}}$. This is because the Koszul differentials vanish on tensoring with $A/\mathfrak{b}$, as multiplication by g_i acts as 0 on $A/\mathfrak{b}$. Then, the inclusion $\mathfrak{a}/\mathfrak{b} \to A/\mathfrak{b}$ causes the induced map $\mathrm{Tor}^A_j(A/\mathfrak{b}, \mathfrak{a}/\mathfrak{b}) \to \mathrm{Tor}^A_j(A/\mathfrak{b}, A/\mathfrak{b})$ to just be $\binom{n}{j}$ copies of the inclusion $\mathfrak{a}/\mathfrak{b} \to A/\mathfrak{b}$. Then, as that map is injective, the map $\mathrm{Tor}^A_{j+1}(A/\mathfrak{b}, A/\mathfrak{a}) \to \mathrm{Tor}^A_j(A/\mathfrak{b}, \mathfrak{a}/\mathfrak{b})$ is forced to be the zero map as

it lands in the kernel of $\mathrm{Tor}_j^A(A/\mathfrak{b}, \mathfrak{a}/\mathfrak{b}) \to \mathrm{Tor}_j^A(A/\mathfrak{b}, A/\mathfrak{b})$. Hence we get the desired splitting.

Then, to conclude, we recall the Koszul complex is functorial. So the inclusion $(a_{ij})\colon A^n \to A^n$ induces a map of complexes $A/\mathfrak{b} \otimes K_\bullet(g_1, \dots, g_n) \to A/\mathfrak{b} \otimes K_\bullet(f_1, \dots, f_n)$. Since the vertical maps are given by the wedges of (a_{ij}) the left end is then given by:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A/\mathfrak{b} & \longrightarrow & (A/\mathfrak{b})^n & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle \bar{\Lambda}} & & \downarrow{\scriptstyle \wedge^{n-1}(a_{ij})} & & \\ 0 & \longrightarrow & A/\mathfrak{b} & \longrightarrow & (A/\mathfrak{b})^n & \longrightarrow & \cdots \end{array}$$

As $\wedge^{top}(a_{ij}) = \det(a_{ij})$. Since by our splitting we know that $\mathrm{Tor}_n^A(A/\mathfrak{b}, A/\mathfrak{b}) \to \mathrm{Tor}_n(A/\mathfrak{b}, A/\mathfrak{a})$ is a surjection, we get $\mathrm{Tor}_n(A/\mathfrak{b}, A/\mathfrak{a}) \cong \mathrm{Im}(\bar{\Lambda}) \cong \Lambda \cdot A/\mathfrak{b}$ as desired.

Then, as $\mathrm{Ann}_{A/\mathfrak{b}}(\bar{\Lambda})$ is the kernel of the left vertical map in the above, it is isomorphic to $\mathrm{Tor}_n^A(A/\mathfrak{b}, \mathfrak{a}/\mathfrak{b})$ which we computed is $\mathfrak{a}/\mathfrak{b}$. □

Denote by I the kernel of the multiplication map $B \otimes_A B \to B$. Then, as a corollary we have the following.

Corollary 2.41 ([Kuh+22, Proposition 3.2.])**.** *Let $\pi\colon A[x_1, \dots, x_n] \to B$ be a presentation of B as a finite, flat, relative global complete intersection.*

- *The element $\Delta \coloneqq (\pi \otimes \pi)(\det(a_{ij}))$ is independent of the choice of a_{ij}.*
- *There is an equality $\langle \Delta \rangle = \mathrm{Fit}_{B \otimes_A B}(I)$ of ideals in $B \otimes_A B$.*
- *We have the equalities of ideals, $\langle \Delta \rangle = \mathrm{Ann}_{B \otimes_A B}(I)$ and $\mathrm{Ann}_{B \otimes_A B}(\Delta) = I$.*

Proof. We reduce to the set up of Proposition 2.40. Specifically, consider the map $\pi \otimes 1\colon A[x_1, \dots, x_n] \otimes_A A[x_1, \dots, x_n] \to B \otimes_A A[x_1, \dots, x_n]$. Our equality $f_i \otimes 1 - 1 \otimes f_i = \sum a_{ij}(x_i \otimes 1 - 1 \otimes x_i)$ in $A[x_1, \dots, x_n] \otimes A[x_1, \dots, x_n]$ then induces the equality $-1 \otimes f_i = \sum_{i=1}^n (\pi \otimes 1)(a_{ij}) \cdot (\pi(x_i) \otimes 1 - 1 \otimes x_i)$.

Now, we notice that Δ is the image of $\det((\pi \otimes 1)(a_{ij}))$ under the map $1 \otimes \pi\colon B \otimes_A A[x_1, \dots, x_n] \to B \otimes_A B$. Furthermore, $\pi \otimes 1$ is the quotient map $B \otimes_A A[x_1, \dots, x_n] \to B \otimes_A B$. This inspires us to take $\mathfrak{b}$ to be the ideal generated by $(-1 \otimes f_i)$. Then, we take $\mathfrak{a} = \langle \pi(x_i) \otimes 1 - 1 \otimes x_i \rangle$. Finally, $I = \mathfrak{a}/\mathfrak{b}$ since we know that the $\pi(x_i)$ generated B, so elements of the form $\pi(x_i) \otimes 1 - 1 \otimes \pi(x_i)$

will generate the kernel of the multiplication map. Then all the desired properties follow from Proposition 2.40 after we show the regularity hypothesis holds.

Relative global complete intersections are flat [Stacks, Tag 00SW]. Regular sequences are preserved under flat morphisms. So the regularity of the f_i in $A[x_1, \ldots, x_n]$ (as the generators of the kernel in a relative global complete intersection are regular), the flatness of $A \to B$ gives us the desired regularity. Then, the collection of elements of the form $\pi(x_i) - x_i$ gives a regular sequence in $B[x_1, \ldots, x_n] \cong B \otimes_A A[x_1, \ldots, x_n]$. □

Next, we remark that $B \otimes_A B$ and $\mathrm{Hom}_A(\mathrm{Hom}_A(B, A), B)$ are both B-bimodules. The former under left and right multiplication by B. The latter by pre- and post-multiplication by elements of B.

Lemma 2.42 ([Kuh+22, Lemma 3.3.])**.** *The map χ induces a B-module isomorphism* $\mathrm{Ann}_{B\otimes_A B} I \xrightarrow{\sim} \mathrm{Hom}_B(\mathrm{Hom}_A(B, A), B)$.

Proof. As relative global intersections are flat, and B is finite, we have B is a projective A-module. Then, every finitely generated projective module is reflexive, since Hom commutes with direct sums. So we have $B^{\vee\vee} \cong B$. This gives us the isomorphism of A-modules

$$B \otimes_A B \cong (B^\vee)^\vee \otimes_A B \cong \mathrm{Hom}_A(B^\vee, B) = \mathrm{Hom}_A(\mathrm{Hom}_A(B, A), B).$$

The map χ is the composition of these isomorphisms as the composition is the map generated by

$$b \otimes b' \mapsto \mathrm{ev}_b \otimes b' \mapsto [f \mapsto \mathrm{ev}_b(f)b'] = [f' \mapsto f(b)b'],$$

since the canonical map $\psi\colon M^\vee \otimes_A N \to \mathrm{Hom}_A(M, N)$ is defined by $\psi(f \otimes n)(m) = f(m)n$. However, this composition is exactly χ, so it yields an isomorphism of A-modules.

Then we observe that χ is a B-bimodule homomorphism. Then, we observe the largest sub-module of $B \otimes_A B$ where the two B-module structures agree is $\mathrm{Ann}_{B\otimes_A B} I$. Suppose $r \in B \otimes_A B$. It is annihilated by all $a \otimes 1 - 1 \otimes a$ when $(a \otimes 1)r = (1 \otimes a)r$. This occurs when the two B-module structures agree.

Then, the largest sub-modules of $\mathrm{Hom}_A(\mathrm{Hom}_A(B, A), B)$ where pre- and post-composing with multiplication agree is when we have $b\varphi(\psi) = \varphi(b\psi)$. This is precisely B-linearity, which is the condition to be a map of B-modules.

Hence we have our desired claims, as χ must map the two largest sub-modules where the two structures agree onto one another. □

Now we have three preliminary commutative algebra results to prove before Theorem 2.27. The following Lemma appeared in [Kuh+22] as part of the proof of Lemma 2.44. We have isolated it as we apply it twice to bypass a step in their proof we found unclear.

Lemma 2.43 ([Kuh+22, Lemma 2.3.])**.** *Let A be a Noetherian local ring, B be a finite flat A-algebra, and N be a finite B-module. If* $\operatorname{depth}(A) \geq 2$*, then* $\operatorname{depth}_A(\operatorname{Hom}_B(N, B)) \geq 2$*.*

Proof. By flatness, an A-regular sequence a_1, a_2's images b_1, b_2 are a regular sequence on B. Observe that b_1 is not a zero divisor on $\operatorname{Hom}_B(N, B)$, as for $f \in \operatorname{Hom}(N, B)$ if $b_1 \cdot f = 0$, then $b_1 f(n) = 0$ for all $n \in N$. But $f(n) \in B$, so $b_1 f(n) = 0$ implies $f(n) = 0$ so f must be the zero homomorphism. Then we note that the cokernel of multiplication by b_1 can be identified as a sub-module of $\operatorname{Hom}(N, B/b_1B)$ as the map $\operatorname{Hom}(N, B) \to \operatorname{Hom}(N, B/b_1B)$ induced by the quotient makes the image of the multiplication by b_1 map to zero. But, as b_1, b_2 is regular, we conclude b_2 is a not a zero divisor on this module either. So $\operatorname{Hom}(N, B)$ has depth greater than or equal to 2. □

Lemma 2.44 ([Kuh+22, Lemma 2.4.])**.** *Suppose A is Noetherian and B is a finite flat A algebra. Then a finite B-module M is reflexive if and only if the following hold:*

1. *If $\mathfrak{p}$ is a prime ideal of A with* $\operatorname{depth}(A_\mathfrak{p}) \leq 1$*, then $M_\mathfrak{p}$ is reflexive;*
2. *If $\mathfrak{p}$ is a prime ideal of A with* $\operatorname{depth}(A_\mathfrak{p}) \geq 2$*, then* $\operatorname{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) \geq 2$*.*

Proof. (Reflexive $\implies$ 2 conditions): Suppose M is reflexive. Reflexivity is a local condition. Hence condition 1 is automatically satisfied. For condition 2, let us suppose that $\mathfrak{p}$ is a prime ideal with the depth of $A_\mathfrak{p}$ greater than or equal to 2. Then there must exist a regular sequence a_1, a_2 on $A_\mathfrak{p}$. By flatness of B over A, these are mapped to a regular sequence b_1, b_2 in $B_\mathfrak{p}$. Then, applying Lemma 2.43 to $M_\mathfrak{p}^\vee$, we have $M_\mathfrak{p}^{\vee\vee}$ has depth greater than or equal to 2, and since $M_\mathfrak{p} \cong M_\mathfrak{p}^{\vee\vee}$ we're done.

(Not reflexive $\implies$ not 2 conditions): Suppose that M isn't reflexive. By way of contradiction, let us assume the 2 conditions hold. Then, by locality of reflexivity, there exist some primes $\mathfrak{p}$ such that $M_\mathfrak{p}$ is not reflexive. Let us assume that $\mathfrak{p}$ is a minimal such prime. Then, since $M_\mathfrak{p}$ isn't reflexive by condition 1 we must have $\operatorname{depth}(\mathrm{A}_\mathfrak{p}) \geq 2$. Now let us consider the exact sequence

$$0 \to \ker(\varphi) \to M_\mathfrak{p} \to M_\mathfrak{p}^{\vee\vee} \to \operatorname{coker}(\varphi) \to 0.$$

Here φ is the canonical map. By minimality of $\mathfrak{p}$ we know upon passing to any further localisation, both $\ker(\varphi)$ and $\operatorname{coker}(\varphi)$ must vanish, as $M_\mathfrak{p}$ becomes reflexive upon localising to a smaller prime. However, a finite module over a local ring with support at only the maximal ideal of said local ring is necessarily of finite length [Stacks, Tag 00L5]. So both the kernel and cokernel of φ have finite length. However, note that as $\operatorname{depth}(M_\mathfrak{p}) \geq 2$ by the second assumption, we have some a_1 such that a_1 is not a zero divisor on $M_\mathfrak{p}$. But then a_1 must also be a non zero-divisor on the submodule, $\ker(\varphi)$. However, as $\ker(\varphi)$ is of finite length this implies it must be zero. So the kernel vanishes.

Likewise, by Lemma 2.43, since $\operatorname{depth}(A_\mathfrak{p}) \geq 2$, the dual of any $B_\mathfrak{p}$-module has depth greater than or equal to 2. So we conclude $M^{\vee\vee}$ has depth at least 2. Then by [Stacks, Tag 00LX], $\operatorname{coker}(\varphi)$ has depth at least 1, which by the same argument as the kernel implies due to finiteness it is 0. Hence, we conclude φ is an isomorphism, so $M_\mathfrak{p}$ is reflexive. This is a contradiction, so one of the two conditions must not hold. □

Lemma 2.45. *Suppose A is Noetherian and B is a finite flat A-algebra. Let M be a finite B-module which is projective as an A-module. If $\operatorname{Hom}_B(M, B)$ is projective as a B-module, then M is projective as a B-module. Furthermore, if $\operatorname{Hom}_B(M, B)$ is free, then M is free.*

Proof. As the dual of a finitely generated free or projective module is itself free or projective, it suffices to show that M is reflexive. Then, we simply have to prove the conditions of Lemma 2.44. Condition 2 is immediate, since by the Auslander-Buchsbaum formula, we have $\operatorname{pd}_\mathfrak{p}(M_\mathfrak{p}) + \operatorname{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \operatorname{depth}(A_\mathfrak{p})$ and since M was assumed to be projective, it has projective dimension 0, yielding $\operatorname{depth}(A_\mathfrak{p}) = \operatorname{depth}_{A_\mathfrak{p}}(M_\mathfrak{p})$. Hence condition 2 is satisfied.

So we just have to prove Condition 1. We can then assume that A is Noetherian local with $\operatorname{depth} A \leq 1$. We will show M is reflexive by showing it is projective as a B-module. Then, denote by $\varphi\colon M \to M^{\vee\vee}$ the canonical map. Let $C \coloneqq \operatorname{coker}(\varphi)$. A presentation of M yields the exact sequence

$$0 \to U \to F \to M \to 0$$

where F is free. Dualising, we obtain by left exactness of the Hom functor,

$$0 \to M^\vee \to F^\vee \to U^\vee \tag{2.1}$$

Set $Q \coloneqq \operatorname{Im}(F^\vee \to U^\vee)$. As $M^\vee$ is projective by assumption, 2.1 yields a projective resolution of Q. This implies that $\operatorname{pd}_B(Q)$ is zero or one. We then have

the following commutative diagram:

$$\begin{array}{ccccccc} F & \twoheadrightarrow & M & & & & \\ \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \varphi} & & & & \\ F^{\vee\vee} & \longrightarrow & M^{\vee\vee} & \longrightarrow & \mathrm{Ext}^1_B(Q,B) & \longrightarrow & 0 \end{array}$$

The lower row is exact, being the extension of the dual of the exact sequence $0 \to M^\vee \to F^\vee \to Q^\vee \to 0$.

Since $F \to M$ is a surjection, we have that the image of $F \to M \to M^{\vee\vee}$ will just be the image of $M \to M^{\vee\vee}$. But by commutativity, this is the image of the map $F \to F^{\vee\vee} \to M^{\vee\vee}$, which by surjectivity of $F \to F^{\vee\vee}$, will be the image of $F^{\vee\vee} \to M^{\vee\vee}$. So, $M \to M^{\vee\vee}$ and $F^{\vee\vee} \to M^{\vee\vee}$ have the same cokernel, so we obtain $C = \mathrm{coker}(\varphi) \cong \mathrm{Ext}^1(Q, B)$.

Since B is finite flat over A, by [Stacks, Tag 0337] we have $\mathrm{depth}(B_\mathfrak{m}) = \mathrm{depth}(A)$ for all maximal ideals $\mathfrak{m}$ of B. Now suppose $\mathrm{depth}(A) = 0$. By Auslander-Buchsbaum, we have $\mathrm{depth}(B_\mathfrak{m}) = \mathrm{depth}(Q_\mathfrak{m}) + \mathrm{pd}(Q_\mathfrak{m})$. But the left hand side is 0, so we conclude both terms on the right hand side must be 0 for every maximal ideal $\mathfrak{m}$ of B. As $\mathrm{pd}(Q) = \sup_\mathfrak{m} \mathrm{pd}(Q_\mathfrak{m})$ we finish by concluding the projective dimension of $Q_\mathfrak{m}$ is 0.

Likewise, suppose $\mathrm{depth}(A) = 1$. By [Stacks, Tag 00LX], as $0 \to U \to F \to M \to 0$ is short exact, the depth of U is at least 1. So by [Stacks, Tag 0AV5], $\mathrm{depth}_{B_\mathfrak{m}} U^\vee_\mathfrak{m} \geq 1$. Hence, the regular element on $U^\vee_\mathfrak{m}$ descends to a regular element on the submodule $Q_\mathfrak{m}$, so $\mathrm{depth}_{B_\mathfrak{m}}(Q_\mathfrak{m}) \geq 1$. Then, applying Auslander-Buchsbaum, $\mathrm{depth}(B_\mathfrak{m}) = \mathrm{pd}(Q_\mathfrak{m}) + \mathrm{depth}_{B_\mathfrak{m}}(Q_\mathfrak{m})$. The left hand side is 1, and the depth of $Q_\mathfrak{m}$ is at least 1, so $\mathrm{pd}(B_\mathfrak{m}) = 0$. Hence we obtain $C \cong \mathrm{Ext}^1(Q, B) \cong= 0$.

Hence, $M \to M^{\vee\vee}$ is surjective. As $M^{\vee\vee}$ is projective, this gives us a splitting $M \cong M^{\vee\vee} \oplus N$ for some B-module N. We obtain by commutative of dualising and direct sums from the reflexivity of $M^\vee$, that $N^{vee} = 0$. Since we reduced to A local, we further obtain N is free. By assumption, both M and $M^{\vee\vee}$ are free over the local ring A. A surjection of finite free A-modules is an isomorphism if they have the same rank.

As our modules are finite, the rank is preserved under localisation. So we can check it at a convenient localisation. We then take a minimal prime ideal $\mathfrak{q}$ of B. Then the localisation $B_\mathfrak{q}$ is zero-dimensional ring. Since we're working over A Noetherian, we further have $B_\mathfrak{q}$ is Artinian. Now, potentially by taking an intermediary localisation, we can assume B is local with maximal ideal. We want

to show $\mathrm{Hom}_{B_\mathfrak{q}}(N_\mathfrak{q}, B_\mathfrak{q}) = 0$ implies $N_\mathfrak{q} = 0$. To see this, suppose $N_\mathfrak{q} \neq 0$. We consider it as a B module. Then by Nakayama's Lemma, $N_\mathfrak{q}/\mathfrak{m}N_\mathfrak{q}$ is non-zero and is a vector space over $B/\mathfrak{m}B$, hence there exists a non-zero map $N_\mathfrak{q} \to N_\mathfrak{q}/\mathfrak{m}N_\mathfrak{q}$. As $B_\mathfrak{q}$ has finite length as it is Artinian, there is a non-zero element annihilated by $\mathfrak{m}$. This gives us a B-homomorphism $B/\mathfrak{m} \to B_\mathfrak{q}$, which composes to yield the desired non-zero homomorphism $N_\mathfrak{q} \to B_\mathfrak{q}$. Thus $N_\mathfrak{q}$ vanishes, and $M_\mathfrak{q}$ and $M_\mathfrak{q}^{\vee\vee}$ have the same rank and the canonical map is an isomorphism. □

We are now able to prove Theorem 2.27.

Proof of Theorem 2.27. By Corollary 2.41 we know $\mathrm{Ann}_{B\otimes_A B} I = \Delta(B \otimes_A B)$ and $\mathrm{Ann}_{B\otimes_A B} \Delta(B \otimes_A B) = I$. Recall $m_2 \colon B \otimes_A B \to B$ is the canonical multiplication map. This gives us the following chain of equalities;

$$\mathrm{Ann}_{B\otimes_A B} I = \Delta(B\otimes_A B) = \frac{\Delta(B \otimes_A B)}{\mathrm{Ann}_{B\otimes_A B} \Delta} = \frac{\Delta(B \otimes_A B)}{I} = \frac{\Delta(B \otimes_A B)}{\ker(m_2)} = m_2(\Delta)B$$

The last equality follows from the first isomorphism theorem. However, since χ induces an isomorphism between $\mathrm{Hom}_B(\mathrm{Hom}_A(B, A), B)$ and $\mathrm{Ann}_{B\otimes_A B} I$, this gives us $\mathrm{Hom}_B(\mathrm{Hom}_A(B, A), B)$ is a free B-module of rank 1. Applying Lemma 2.45 we acquire $\mathrm{Hom}_A(B, A)$ is a free rank one B-module. So since $\chi(\Delta)$ is not the zero map, it is an isomorphism. □

2.4 Computing $\mathbb{A}^1$–Euler Numbers

Armed with Scheja and Storch's duality theorem, we are now in position to define the $\mathbb{A}^1$–Euler number.

Definition 2.46 ([KW21, Definition 30])**.** Let V be a relatively oriented rank n vector bundle on a smooth scheme X over k of dimension n, with compatible Nisnevich coordinates and trivialisations. Suppose σ is a section of V with isolated zeroes. Denote by Z its vanishing subscheme. Let $p \in Z$ be an isolated zero. By Theorem 2.20, select a compatible presentation $\mathcal{O}_{Z,p} \cong \frac{k[x_1,\ldots,x_n]_\mathfrak{q}}{\langle g_1,\ldots,g_n\rangle}$.

By Theorem 2.27, this presentation induces a distinguished isomorphism of $\mathcal{O}_{Z,p}$-modules $\chi(\Delta) \colon \mathrm{Hom}_k(\mathcal{O}_{Z,p}, k) \to \mathcal{O}_{Z,p}$. Denote by $\eta \colon \mathcal{O}_{Z,p} \to k$ the pre-image of 1.

Then, η defines a non-degenerate symmetric bilinear form by $\beta(x, y) = \eta(xy)$. The **index of σ at p**, denoted $\mathrm{ind}_p(\sigma)$, is the Grothendieck–Witt class of β.

Definition 2.47. The $\mathbb{A}^1$–Euler number, $e^{\mathbb{A}^1}(V, \sigma)$ is $\sum_{p\in Z} \mathrm{ind}_p(\sigma)$ where Z is the vanishing sub-scheme of σ.

Remark 2.48. It is unclear a priori that the $\mathbb{A}^1$–Euler number is well defined. In the following section, we will prove several well-definedness results, after we have introduced some computational results.

Example 2.49. Fix $X = \mathbb{A}^n_k = \mathrm{Spec}(k[x_1, \ldots, x_n])$. Then we set $V = \mathcal{O}_X^{\oplus n}$. We remark that this vector bundle is globally trivial. Furthermore, we have a basis of TX corresponding to $\left\{\frac{d}{dx_i}\right\}_{i=1}^n$ coming from our coordinate basis $x_1, \ldots, x_n$. Then, we have $\mathrm{Hom}(\det TX, \det V) \cong \mathcal{O}_X^{\otimes 2}$, with the tensor factor comparing the scale factor of where the top form is sent. This canonical isomorphism yields our relative orientation.

Now let $\sigma = (f_1, \ldots, f_n)$ and let us suppose the origin is an isolated zero of σ. Then, as in the construction of the η of Scheja Storch, we'll be able to find $a_{ij} \in k[x_1, \ldots, x_n]$ such that $f_i = \sum_{j=1}^n a_{ij} x_j$. Observe that expressing the g_j in terms of the x_i's instead of the tensors gives an equivalent choice of a_{ij}. Then, quite remarkably it turns out that we can always pick a presentation such that η can be any k-linear homomorphism on the stalk $\mathcal{O}_{X,p}$ that sends $\det(a_{ij})$ to 1.

So for a more specific example, we could take $n = 2$ and $\sigma = (y - x, y + x)$. Then we have a single isolated zero of σ located at $(0, 0)$. Expressing the two coordinate functions as above we obtain $-1 \cdot x + 1 \cdot y$ and $1 \cdot y + 1 \cdot x$. Hence, we get:

$$(a_{ij}) = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

So $\det(a_{ij}) = -2$. Furthermore, we note we have an isomorphism $\frac{k[x,y]}{\langle x-y, x+y\rangle} \cong k$, since $x - y = 0$ and $x + y = 0$ implies $2x = 0$, so $x = 0$. Meanwhile, we then obtain $y = 0$ from back substitution. So, we get $\eta(-2) = 1$. Thus, $\eta(1) = -\frac{1}{2}$. So we conclude $\beta(a, b) = -\frac{1}{2}ab$, thus $\mathrm{Ind}_{(0,0)} \sigma = \langle -\frac{1}{2}\rangle = \langle -2\rangle$.

For a second example, consider $n = 2$ but $\sigma = (y - x^2, y + x^2)$. We once more have an isolated zero at $(0, 0)$. Now we get $-x \cdot x + 1 \cdot y$ and $1 \cdot y + x \cdot x$. So our matrix (a_{ij}) now takes the form;

$$(a_{ij}) = \begin{bmatrix} -x & 1 \\ x & 1 \end{bmatrix}$$

So $\det(a_{ij}) = -2x$. Meanwhile, $\frac{k[x,y]_{(0,0)}}{\langle y-x^2, y+x^2\rangle} \cong \frac{k[x]_{(0,0)}}{x^2} \cong k \oplus k \cdot x$ since we again obtain $2y = 0$ implying $y = 0$, and $2x^2 = 0$ so $x^2 = 0$. Then, $\{1, x\}$ provides our basis. We require that $\eta(-2x) = 1$, so $\eta(x) = -\frac{1}{2}$. We can then set $\eta(1) = 0$ to

get in our basis that the form β has the following matrix representation

$$\beta = \begin{bmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{bmatrix}$$

We note that by taking the basis $\{1+x, 1-x\}$, we get that this form is equivalent to $\langle -2\rangle + \langle 2\rangle = \langle -1\rangle + \langle 1\rangle$.

Remark 2.50. We note that in the first example, we got a 1-dimensional form as our $\mathbb{A}^1$–Euler number. However, in the second example we got a 2-dimensional form. Ergo, the choice of section impacted the resultant $\mathbb{A}^1$–Euler number. This is a general feature. The remarkable fact is when our base scheme is further assumed to be *proper*, the $\mathbb{A}^1$–Euler number is independent of the section.

Now say we have $X = \mathbb{A}^1_{\mathbb{R}}$ with $\mathcal{O}_X$ as our vector bundle. Consider say the section $\sigma_f = x^2 - 1$. This has an isolated zero corresponding to the complex point $\langle x^2 + 1\rangle$. This has a length two stalk, which we could compute by hand, however computing simple zeros is much easier. Luckily, for a $\kappa(p)$-point where $\kappa(p)/k$ is separable, there is a useful formula for computing the index.

Definition 2.51 ([Neu13, Proposition I.2.6.])**.** Let L/k be a separable field extension of k. The **field trace** $\mathrm{Tr}_{L/k}\colon L \to k$ is a k-linear map defined a follows. Let $\sigma\colon L \to \bar{k}$ vary over all k-embeddings of L into an algebraic closure of k. Then,

$$\mathrm{Tr}_{L/k}(x) = \sum_{\sigma} \sigma(x)$$

Definition 2.52 ([Lam05, p. 189])**.** Suppose $\alpha \in L^\times$. The **Scharlau transfer**, denoted $\mathrm{Tr}_{L/k}(\langle\alpha\rangle)$, is the k-valued bilinear form on L, viewed as a k-vector space, given by the composition:

$$L \otimes L \xrightarrow{\langle\alpha\rangle} L \xrightarrow{\mathrm{Tr}_{L/k}} k$$

What this transfer does is allow us to make precise the intuition that a closed point with non-trivial residue field should separate upon base change into a Galois orbit of conjugate points, whose indices are related by the action of the Galois group. For example, a complex line on a real cubic is a pair of Galois conjugate geometric lines.

Then, suppose X is a smooth k-scheme with a point p whose residue field is a separable extension of k. Let $X_{\kappa(p)}$ be the base change of X to $\kappa(p)$. The map $\mathrm{Spec}(\kappa(p)) \to X$ induces by the universal property of the fibred product a canonical $\kappa(p)$ point of $X_{\kappa(p)}$, denoted $p_{\kappa(p)}$. Let $\sigma_{\kappa(p)}$ also be the base change of σ.

Proposition 2.53 ([KW21, Proposition 34])**.** *Let p be a point with residue field $\kappa(p)$ separable over k. There is an equality in the Grothendieck–Witt ring:*

$$\operatorname{ind}_p(\sigma) = \operatorname{Tr}_{\kappa(p)/k} \operatorname{ind}_{p_{\kappa(p)}} \sigma_{\kappa(p)}$$

Proof. We proceed by Galois descent. Let $L/\kappa(p)/k$ be a tower of finite extensions such that L is Galois over k. Denote by β the form representing $\operatorname{ind}_p(\sigma)$ and by $\beta_{\kappa(p)}$ the form representing $\operatorname{ind}_{p_{\kappa(p)}} \sigma_{\kappa(p)}$. Tensoring their underlying vector spaces $\mathcal{O}_{Z,p}$ and $\mathcal{O}_{Z_{\kappa(p)},p_{\kappa(p)}}$ by L produces two L-vector spaces, with induced bilinear forms $L \otimes_k \beta$ and $L \otimes_k \operatorname{Tr}_{\kappa(p)/k} \beta_{\kappa(p)}$, respectively.

First, $L \otimes_k \beta$ is defined on $L \otimes_k \mathcal{O}_{Z,p}$. Each coset $g \operatorname{Gal}(L/\kappa(p))$ of $\operatorname{Gal}(L/k)$ corresponds to an embedding $\sigma_g \colon \kappa(p) \to L$. Hence, by composing this embedding with the morphism $p \colon \operatorname{Spec} \kappa(p) \to X$, each coset determines an L-point of X, denoted gp. Since $\kappa(p)/k$ is separable and $\mathcal{O}_{Z,p}$ is an Artinian local k-algebra with residue field $\kappa(p)$, we obtain a decomposition

$$L \otimes_k \mathcal{O}_{Z,p} \cong \prod_{\sigma_g \colon \kappa(p) \to L} \mathcal{O}_{Z_L,gp}.$$

We claim that the restriction of $L \otimes_k \eta_{k,p}$ to each factor is $\eta_{L,gp}$. To see this, note that since $\mathcal{O}_{Z,p}$ is a finite-dimensional k-algebra, we have

$$L \otimes_k \operatorname{Hom}_k(\mathcal{O}_{Z,p}, k) \cong \operatorname{Hom}_L(L \otimes_k \mathcal{O}_{Z,p}, L).$$

Likewise, the commuting square in the Scheja–Storch construction gives the corresponding isomorphism for a presentation of $L \otimes_k \mathcal{O}_{Z,p}$, since the construction commutes with extension of scalars.

Hence, $L \otimes_k \eta_{k,p}$ corresponds to $1 \in L \otimes_k \mathcal{O}_{Z,p}$ under the Scheja–Storch isomorphism. Under the above product decomposition, this element is $(1, \ldots, 1)$. Consequently, its restriction to each factor corresponds to 1 in that factor, giving the desired result.

On the other side, $L \otimes_k \operatorname{Tr}_{\kappa(p)/k} \beta_{\kappa(p)}$ is defined on $L \otimes_k \mathcal{O}_{Z_{\kappa(p)},p_{\kappa(p)}}$. Each embedding $\sigma_g \colon \kappa(p) \to L$ induces an isomorphism

$$g \colon \mathcal{O}_{Z_{\kappa(p)},p_{\kappa(p)}} \to \mathcal{O}_{Z_{g\kappa(p)},gp}.$$

These give us quotient maps

$$L \otimes_k \mathcal{O}_{Z_{\kappa(p)},p_{\kappa(p)}} \to L \otimes_{g\kappa(p)} \mathcal{O}_{Z_{g\kappa(p)},gp},$$

defined by $a \otimes x \mapsto a \otimes g(x)$. Indeed, these maps simply impose the additional relations required to tensor over $g\kappa(p)$ rather than k.

Since $\kappa(p)/k$ is separable, we have

$$L \otimes_k \kappa(p) \cong \prod_{\sigma_g \colon \kappa(p) \to L} L.$$

Hence, the quotient maps induce an isomorphism

$$L \otimes_k \mathcal{O}_{Z_{\kappa(p)}, p_{\kappa(p)}} \cong \prod_{\sigma_g \colon \kappa(p) \to L} L \otimes_{g\kappa(p)} \mathcal{O}_{Z_{g\kappa(p)}, gp}.$$

But it follows from the construction of the fibre product via tensor products that

$$L \otimes_{g\kappa(p)} \mathcal{O}_{Z_{g\kappa(p)}, gp} \cong \mathcal{O}_{Z_L, gp}.$$

Thus, we have

$$L \otimes_k \mathcal{O}_{Z_{\kappa(p)}, p_{\kappa(p)}} \cong \prod_{\sigma_g \colon \kappa(p) \to L} \mathcal{O}_{Z_L, gp}.$$

Next, we claim that $\eta_{L,gp}(g \cdot x) = g \cdot \eta_{L,p}(x)$. This follows from the functoriality of the Scheja–Storch construction, since the isomorphism $\mathcal{O}_{Z_L,p} \to \mathcal{O}_{Z_L,gp}$ is compatible with the commuting square defining the Scheja–Storch isomorphism.

Finally, we claim that under the preceding identification, the form $L \otimes_k \operatorname{Tr}_{\kappa(p)/k} \beta_{\kappa(p)}$ corresponds to the orthogonal direct sum

$$\bigoplus_{\sigma_g \colon \kappa(p) \to L} \beta_{L,gp}.$$

This follows from the proof of [Lam05, Theorem VII.6.1], since the same computations follow through, when sums are taken over cosets rather than the whole Galois group, together with the functoriality of η established above.

Combining these identifications, we have now constructed an isometry

$$L \otimes_k \mathcal{O}_{Z,p} \cong L \otimes_k \mathcal{O}_{Z_{\kappa(p)}, p_{\kappa(p)}},$$

taking $L \otimes_k \beta$ to $L \otimes_k \operatorname{Tr}_{\kappa(p)/k} \beta_{\kappa(p)}$.

We now observe that this isometry is Galois equivariant. Indeed, the canonical action of $\operatorname{Gal}(L/k)$ on either tensor product acts on the L factor. Under our product decompositions, an element $g \in \operatorname{Gal}(L/k)$ permutes the factors by sending the factor corresponding to an embedding σ_h to that corresponding to $g \circ \sigma_h$, while applying g to its elements. Both decompositions are compatible with this action, and hence the resulting isometry is Galois equivariant.

Therefore, by Galois descent, this isometry descends to a k-linear isometry

$$\beta \cong \operatorname{Tr}_{\kappa(p)/k} \beta_{\kappa(p)},$$

proving the desired equality in the Grothendieck–Witt ring. □

These examples allow us to present the following algorithm.

1. Using the preceding proposition, we may reduce to the case where the zeroes of a section are k-points.

2. We choose for η any k-linear homomorphism $\eta\colon \mathcal{O}_{Z,p} \to k$ which takes the distinguished socle element to 1.

3. We compute the isometry class of the resultant β. This is the index.

The fact we can use any choice of η is a corollary of either [KW19], or it can be shown with some work from Theorem 2.80.

In particular, this computational method is particularly amenable to implementation via computer algebra systems. We present a computation using Macaulay2 in Appendix A. Indeed, there's recently been a dedicated Macaulay2 library released for the purpose of computing these indices. Likewise, [KW19] presents a computation via Mathematica.

2.5 Well-Definedness and Invariance

Finally, we conclude this chapter by discussing the well-definedness and invariance of the $\mathbb{A}^1$–Euler number. In order to define $e^{\mathbb{A}^1}(V, \sigma)$, we had to choose compatible relations g_i on each stalk of Z, a compatible choice of Nisnevich coordiantes on X and trivialisation of V around p, and a section σ. In this section, we will prove that nothing except the relative orientation affects the final value of the $\mathbb{A}^1$–Euler number.

In Example 2.49, we saw that using different sections of relatively oriented vector bundles may result in different $\mathbb{A}^1$–Euler numbers. However, when X is further assumed to be proper, we obtain $e^{\mathbb{A}^1}(V, \sigma)$ is independent of the section.

The proof of this sectional invariance utilises coherent duality. As a result it will only work on proper schemes. But, as a corollary of the proof, we further obtain independence of the indices from the coordinates, trivialisations and choice of generators (g_i).

Remark 2.54. This cohomological approach is the not the approach taken in [KW21]. Instead, the authors prove in their Lemma 29 the independence of the indices via computing change of basis matrices comparing different possible indices. This version of the result holds for non-proper base schemes, but we're only working over proper base schemes so we do not need the lengthy computation

involved. Likewise, they do not prove general invariance of the $\mathbb{A}^1$–Euler number. Rather, as Lemma 36 they prove invariance on a proper scheme, over sections that lie in families over $\mathbb{A}^1$.

While the argument is less technical, it obscures the applicability of the general framework to other enumerative problems, as one now needs to prove the existence of suitable families. The proof they give is very specific to cubic surfaces.

Coherent duality, also knows as Grothendieck duality, is a generalisation of Serre duality to the derived categories of a quasi-compact quasi-separated scheme. Specifically, for a map $f\colon X \to Y$ of such schemes, as explained in [Stacks, Tag 0A9E], it says there is a quasi-isomorphism

$$Rf_* R\mathcal{H}om(F^\bullet, f^! G) \to R\mathcal{H}om(Rf_* F^\bullet, G^\bullet).$$

Taking global sections recovers an adjunction $Rf_* \vdash f^!$.

To use this framework, recall by Lemma 2.4, the relative orientation of V induces an isomorphism $\rho\colon \det V \otimes \omega_{X/k} \to \mathcal{L}^{\otimes 2}$. Then we can define pairings:

$$\beta_{i,j}\colon H^i(X, \wedge^j V^* \otimes \mathcal{L}) \otimes H^{n-i}(X, \wedge^{n-i}(X, \wedge^{n-j} V^* \otimes \mathcal{L}) \to k$$

given by the composition

$$\begin{aligned} H^i(X, \textstyle\bigwedge^j V^* \otimes \mathcal{L}) \otimes H^{n-i}(X, \bigwedge^{n-j} V^* \otimes \mathcal{L}) & \\ \xrightarrow{\cup} H^n(X, \textstyle\bigwedge^j V^* \otimes \bigwedge^{n-j} V^* \otimes \mathcal{L}^{\otimes 2}) & \\ \to H^n(X, \det V^* \otimes \mathcal{L}^{\otimes 2}) \xrightarrow{\rho} H^n(X, \omega_{X/k}) \xrightarrow{\eta_f} k. & \end{aligned}$$

Here η_f is the co-unit map of the adjunction. It is otherwise known as the Serre duality trace map. The third map is induced by the canonical surjection $\wedge^j V^* \otimes \wedge^{n-j} V^* \to \wedge^n V^*$. This map is a perfect pairing. So for $i = n - i$ and $j = n - j$, we obtain that $\beta_{i,j}$ is a non-degenerate symmetric bilinear form on $H^i(X, \wedge^j V^* \otimes \mathcal{L})$. For other values of i and j, we have $\beta_{i,j} \perp \beta_{n-i,n-j}$ is a bilinear form on $H^i(X, \bigwedge^j V^* \otimes \mathcal{L}) \oplus H^{n-i}(X, \bigwedge^{n-j} V^* \otimes \mathcal{L})$. In this case, we acquire a hyperbolic form, as either of the subspaces $H^{n-i}(X, \bigwedge^{n-j} V^* \otimes \mathcal{L})$ or $H^n(X, \bigwedge^j V^* \otimes \mathcal{L})$ are, by non-degeneracy, half-dimension isotropic subspaces. Then we make, via a small abuse of notation, the following definition.

Definition 2.55 ([BW23, p. 683])**.** The **coherent duality $\mathbb{A}^1$–Euler number** of a relatively oriented vector bundle V is defined as the Grothendieck–Witt class

$$e^{\mathbb{A}^1}_{CD}(V) = \sum_{i,j} (-1)^{i+j} \beta_{i.j}$$

Remark 2.56. To explain our abuse of notation, note that as $(-1)^{n-i+n-j} = (-1)^{i+j}$, we can group the terms as $(-1)^{i+j}[\beta_{i,j} + \beta_{n-i,n-j}]$, which is a well defined Grothendieck–Witt class, making the sum well defined. We commit to this style of abuse of notation moving forward.

Remark 2.57. The coherent duality $\mathbb{A}^1$–Euler number depends only on the choice of relative orientation of V. There is no other input data.

Theorem 2.58 ([BW23, Theorem 1.1])**.** *Suppose a relatively oriented rank n vector bundle V on a smooth proper k-scheme X of dimension n has a section with only isolated zeroes. Then*

$$e^{\mathbb{A}^1}(V, \sigma) = e^{\mathbb{A}^1}_{CD}(V).$$

In particular, $e^{\mathbb{A}^1}(V, \sigma)$ is independent of the choice of σ, Nisnevich coordinates, and local trivialisation.

Notation 2.59. When X is a proper smooth k-scheme, we simply denote the $\mathbb{A}^1$–Euler number of a relatively oriented vector bundle V as $e^{\mathbb{A}^1}(V)$.

The proof of this theorem is involved, as it relies on extensive cohomological machinery. The strategy is summarised by this chain of currently undefined equalities:

$$e^{\mathbb{A}^1}_{CD}(V) = e^{\mathbb{A}^1}_{CD}(V, \sigma) = \sum_{p \in Z} \operatorname{ind}^{CD}_p(\sigma) = \sum_{o \in Z} \operatorname{ind}_p(\sigma) = e^{\mathbb{A}^1}(V, \sigma).$$

First, we will define a coherent duality $\mathbb{A}^1$–Euler number *with respect to a section*, $e^{\mathbb{A}^1}_{CD}(V, \sigma)$. Then in Proposition 2.66 we will use a hyper-cohomology spectral sequence to show the first equality. Next, we'll define using push forwards of a cohomology theory a coherent duality index $\operatorname{ind}^{CD}_p \sigma$ and show the second equality with Proposition 2.78. Finally, we conclude by showing $\operatorname{ind}_p(\sigma) = \operatorname{ind}^{CD}_p \sigma$, yielding the third equality. Since the coherent duality index won't depend on the choice of generators, coordinates or trivialisation, it will also give us well-definedness of the indices.

Say we have *any* section of a relatively oriented vector bundle V on a proper scheme. To define $e^{\mathbb{A}^1}_{CD}(V, \sigma)$, consider the Koszul complex, denoted $K(\sigma)^\bullet$. which is given by the complex:

$$0 \to \wedge^n V^* \to \wedge^{n-1} V^* \to \cdots \to V^* \to \mathcal{O} \to 0.$$

So this Koszul complex corresponds to the map $V^* \to \mathcal{O}$ given by evaluation at σ. It is endowed with a multiplication map $m\colon K(\sigma)^\bullet \otimes K(\sigma)^\bullet \to K(\sigma)^\bullet$ induced by the wedge product; in degree p', $m = \bigoplus_{i+j=p'} \mathrm{Id}_{\wedge^i V^*} \wedge \mathrm{Id}_{\wedge^j V^*}$. Furthermore, we have a projection $p\colon K(\sigma)^\bullet \to \det V^*[n]$. We define $\beta_{(V,\sigma)}\colon K(\sigma) \otimes K(\sigma) \to \det V^*[n]$ by $\beta_{(V,\sigma)} = pm$. Then tensoring by $\mathcal{L}^{\otimes 2}$ and reordering the tensor factors, we set $\beta_{(V,\sigma,\rho)}$ as the composition:

$$(K(\sigma)^\bullet \otimes \mathcal{L})^{\otimes 2} \xrightarrow{\beta_{(V,\sigma)} \otimes \mathrm{Id}_{\mathcal{L}}^{\otimes 2}} \det V^* \otimes \mathcal{L}^{\otimes 2} \xrightarrow{\rho'} \omega_{X/k}$$

The map ρ' is induced by the relative orientation map via tensor manipulation. Then if we apply Rf_* we get, by lax symmetric monoidality of the derived direct image, the composition in $D(k)$:

$$Rf_*\beta_{(V,\sigma,\rho)}\colon [Rf_*(K(\sigma)^\bullet \otimes \mathcal{L})]^{\otimes 2} \to Rf_*\omega_{X/k} \to \mathcal{O}_k.$$

The final map arises from Serre duality. However, $D(k)$ is equivalent to the category of graded vector spaces, via the cohomology functor. A symmetric bilinear form β on a graded vector space V is said to be graded of degree p' if $\beta(V_i, V_j) = 0$ if $i + j \neq p'$. We take all graded forms to be degree 0 by default. Then under this equivalence $Rf_*\beta_{(V,\sigma,\rho)}$ may be viewed as a graded symmetric bilinear form on a k-vector space.

Notation 2.60. For a graded bilinear form β, denote by $\beta^{(i)}$ the restriction of β to $V^{(i)} \coloneqq V_i \oplus V_{-i}$.

Definition 2.61 ([BW23][p. 683])**.** Say V is a relatively oriented vector bundle on a proper scheme X. For any section σ, the coherent duality $\mathbb{A}^1$–Euler number **with respect to** σ is defined by

$$e^{\mathbb{A}^1}_{CD}(V, \sigma) \coloneqq \sum_{i=0}^{n} (-1)^i [(Rf_*\beta_{(V,\sigma,\rho)})^{(i)}] \in \mathrm{GW}_0(k).$$

Our goal now is to prove these two $\mathbb{A}^1$–Euler numbers are equal. To do so, we'll realise them as forms on the E_1 and E_∞ pages of a spectral sequence. Specifically, we'll use the hyper-cohomology spectral sequence $E_r^{i,j}(K^\bullet)$ associated to a bounded complex $K^\bullet$ of locally free sheaves on X, with first page given by:

$$E_1^{i,j}(K^\bullet) = H^j(X, K^i) \implies R^{i+j} f_* K^\bullet.$$

The convergence of this sequence induces a decreasing filtration of the abutment $R^* f_* K^\bullet$, which is given by:

$$F^i R^m f_* K^\bullet \coloneqq \mathrm{Im}(R^m f_* K^{\bullet \geq i} \to R^m f_* K^\bullet).$$

Then, the E_∞ page is identified with the associated graded object of this filtration.

Let β be a symmetric pairing of complexes $\beta\colon K^\bullet \otimes K^\bullet \to \omega_{X/k}[n]$, such that the induced pairings $K^i \otimes K^{-n-i} \to \omega_{X/k}$ are perfect. The trace map $Rf_*\omega_{X/k}[n] \to k$, together with the cup product, induces a pairing β' on $R^*f_*K^\bullet$. Likewise, lax symmetric monoidality of R^*f_* induces a pairing β_1 on the E_1 page. Then, k is in bi-degree $(-n, n)$. So this pairing is zero unless the components have bi-degrees (i, j) and (i', j') satisfying $i + i' = -n$ and $j + j' = n$.

Definition 2.62. For an object C with decreasing filtration F^i, its r-th **filtration shift** $C(r)$ is defined by $F^i(C(r)) = F^{i-r}C$.

The following proposition is asserted without proof in [BW23]. However, much of it is proved in [Hel17], albeit with homological notation and increasing filtrations. We dualise appropriately.

Proposition 2.63 ([BW23, Discussion on p. 689])**.** *The pairings β' and β_1 satisfy the following properties:*

1. *After placing k in the codomain of β_1 in bidegree $(-n, n)$, β_1 is a map of bigraded vector spaces and satisfies the Leibniz rule with respect to d_1. Hence, it induces a pairing β_2 on the E_2 page, which also satisfies the Leibniz rule, inducing β_3 and so on.*

2. *All the pairings β_r, including β_∞, are perfect.*

3. *The pairing β' satisfies $\beta'(F^i, F^k) = 0$ for $i + k > -n$.*

4. *This induces a pairing on $\operatorname{gr}_F R^*f_*K^\bullet$. Under the isomorphism $\operatorname{gr}_F R^*f_*K^\bullet \cong E_\infty$, this pairing agrees with β_∞.*

5. *For the pairing β', the induced pairing $F^iR^*f_*K^\bullet \otimes R^*f_*K^\bullet/F^{-n-i+1} \to k$ is perfect.*

Proof. To use the results of [Hel17] first we translate this to a filtered homological algebra problem. Equip $K^\bullet$ with the stupid filtration. This induces on $R^*f_*K^\bullet$ the filtration we defined. Then, we give k the filtration $F^ik = k$ for $i \leq 0$, else it is zero. Hence, $F^ik(-n) = k$ for exactly $i \leq -n$. Since $k(-n)$ is concentrated in degree zero, the spectral sequence with it correctly filtered in the abutment is the one placing it in bi-degree $(-n, n)$.

Notice that the pairing β vanishes on $K^p \otimes K^q$ unless $p + q = -n$. Hence, it is compatible with the stupid filtrations when the codomain $\omega_{X/k}[n]$ is placed in

filtration degree $-n$. So the cup product and the trace map induce the desired filtered pairings. We're now ready to prove the claims.

1. By [Hel17, Lemma 3.5.2. and Proposition 3.4.2.], the filtered pairing induces pairings β_r on every page which satisfy the Leibniz rule. The Leibniz rule takes the form $0 = \beta_r(d_r u, v) + (-1)^{|u|}\beta_r(u, d_r v)$ as the differential is trivial on the right hand side. So we observe if u, v are in the kernel of the differential

$$\beta_r(u + dx, v) = \beta(u, v) + \beta(dx, v) = \beta(u, v) \pm \beta(x, dv) = \beta(u, v).$$

   So by symmetry we conclude β_r descends to a well defined form β_{r+1} on the E_{r+1} page. The same proposition gives the Leibniz rule here as well, so this continues indefinitely.

2. We have β_1 is perfect by assumption. Now suppose inductively that β_r is perfect. So its adjoint will provide an isomorphism

$$E_r^{p,q} \xrightarrow{\sim} (E_r^{-n-p,n-q})^\vee$$

   By the first property, these isomorphisms are compatible with differentials, up to the usual sign rules. Then, since dualisation over k is exact, taking cohomology gives an isomorphism on the $r+1$ page. This is precisely the adjunction of β_{r+1} since it was defined by taking cohomology. Hence β_{r+1} is perfect. As the filtration is finite, working over a proper scheme, β_∞ is perfect too.

3. The filtered pairing $R^* f_* K^\bullet \otimes R^* f_* K^\bullet \to k(-n)$ induces an extended pairing of the associated Cartan–Eilenberg systems, as in [Hel17, Lemma 3.5.2]. Since the filtration is finite, the corresponding spectral sequences converge strongly. Thus [Hel17, Proposition 3.4.4] gives a commutative diagram

$$\begin{array}{ccc} \mathrm{gr}^p R^m f_* K^\bullet \otimes \mathrm{gr}^q R^{m'} f_* K^\bullet & \longrightarrow & \mathrm{gr}^{p+q} R^{m+m'} f_*(k(-n)) \\ \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} \\ E_\infty^{p,m-p} \otimes E_\infty^{q,m'-q} & \xrightarrow{\beta_\infty} & E_\infty^{p+q,m+m'-p-q}(k(-n)). \end{array}$$

   The upper horizontal map is induced by β' on the associated graded of the abutment. Hence this pairing agrees with β_∞ under the identification $\mathrm{gr}_F R^* f_* K^\bullet \cong E_\infty$.

4. Since the pairing $K^p \otimes K^q \to \omega_{X/k}[n]$ vanishes unless $p+q=-n$, its restriction to $K^{\geq i} \otimes K^{\geq k}$ vanishes whenever $i+k > -n$. Hence $\beta'(F^i, F^k) = 0$ whenever $i+k > -n$.

5. By three, the pairing β' induces a pairing

$$\phi_i \colon F^i R^* f_* K^\bullet \to (R^* f_* K^\bullet / F^{-n-i+1} R^* f_* K^\bullet)^\vee.$$

Our first claim is this is injective. Consider a non-zero x in $F^i R^* f_* K^\bullet$. Then choose a $p \geq i$ such that x is in the degree p part of the filtration but not the $p+1$ part. So x has a non-zero class $\overline{x}$ in the degree p part of the associated graded. Applying perfection of β_∞ and 2 and the identification of 4 we have $\mathrm{gr}^p R^* f_* K^\bullet \otimes \mathrm{gr}^{-n-p} R^* f_* K^\bullet \to k$ is perfect. So there must be some $\bar{y}$ such that $\beta_\infty(\bar{x}, \bar{y}) \neq 0$.

Then, we choose a lift of $\overline{y}$ to the degree $-n-p$ part. By property 4's identification we know $\beta'(x,y) \neq 0$. Since β' vanishes on $F^i R^* f_* K^\bullet \otimes F^{-n-i+1} R^* f_* K^\bullet$, we have $\phi_i(x) \neq 0$. So ϕ_i is injective.

Then, perfection of β_∞ tells us that $\mathrm{gr}^p R^* f_* K^\bullet$ and $\mathrm{gr}^{-n-p} R^* f_* k^\bullet$ have the same dimension. Finally, since the filtration is finite

$$\begin{aligned} \dim F^i H &= \sum_{p \geq i} \dim \mathrm{gr}_F^p H \\ &= \sum_{p \geq i} \dim \mathrm{gr}_F^{-n-p} H \\ &= \dim(H / F^{-n-i+1} H). \end{aligned}$$

Therefore ϕ_i is an isomorphism, so the induced pairing is perfect. □

Lemma 2.64 ([BW23, Lemma 2.5])**.** *Let X be a graded k vector space with a finite decreasing filtration $X \supset \cdots \supset X_j \supset X_{j+1} \supset \ldots$. Suppose $X \otimes X \to k$ is a perfect symmetric bilinear pairing satisfying properties 3 and 5 of Proposition 2.63. Let X^i denote the grade i subspace of X, and X^i_j the grade i subspace of X_j. Then in $\mathrm{GW}_0(k)$ we have*

$$\sum_i (-1)^i [X^i] = \sum_i (-1)^i [\mathrm{gr}_\bullet X^i].$$

Proof. Property 5 implies that the induced pairing on $\mathrm{gr}_\bullet X$ is perfect. For $i \neq 0$, the degree i and $-i$ parts on either side form a metabolic space, since either part is an isotropic subspace of half dimension. Both sides have the same rank,

as passing to the associated graded preserves dimension. Their contributions in nonzero degrees therefore agree, so it suffices to consider the case $X = X^0$. We may now omit the grading.

Let N be maximal such that $X_N \neq 0$, and put $m = -n - N$. Property 5 gives a perfect pairing $X_{N+1} \otimes X/X_m \to k$. Since $X_{N+1} = 0$, we have $X_m = X$. If $m \geq N$, then $X = X_N$ and the filtration has only one nonzero graded piece, so we are done.

Suppose instead that $m < N$. Property 5 also gives a perfect pairing $X_N \otimes X/X_{m+1} \to k$, so $X_N^\perp = X_{m+1}$. Similarly, $X_{m+1}^\perp = X_N$. Choose a complement X' to X_N in X_{m+1}, with filtration $X'_j = X_j \cap X'$. The restriction of the pairing to X' is non-degenerate, since X_N is the orthogonal complement under the pairing on X_{m+1}. The induced filtration on X_{m+1}/X_N is perfect in the sense of Property 5, and hence so is the filtration on X'. Moreover, $\mathrm{gr}_j X'$ agrees with $\mathrm{gr}_j X$ for $m < j < N$. As X' has fewer nonzero filtration pieces, induction gives $[X'] = [\mathrm{gr}_\bullet X']$. Since X' is non-degenerate, we also have $X = X' \oplus (X')^\perp$.

It remains to show that $[(X')^\perp] = [\mathrm{gr}_m X \oplus \mathrm{gr}_N X]$. The subspace $X_N \subseteq (X')^\perp$ is isotropic by Property 3. It has half the dimension of $(X')^\perp$, since the perfect pairing $X_N \otimes X/X_{m+1} \to k$ gives $\dim X/X_{m+1} = \dim X_N$. Likewise, $\mathrm{gr}_N X = X_N$ is an isotropic subspace of half dimension in $\mathrm{gr}_m X \oplus \mathrm{gr}_N X$. Both spaces are therefore metabolic of the same rank, and so have the same class in $\mathrm{GW}_0(k)$. □

Lemma 2.65 ([BW23, Lemma 2.6])**.** *Let $E^\bullet$ be a bounded complex of finite-dimensional k vector spaces with a non-degenerate, symmetric bilinear form $E^\bullet \otimes E^\bullet \to k[0]$. Then in the Grothendieck–Witt ring*

$$\sum_i (-1)^i [H^i(E)] = \sum_i (-1)^i [E^i].$$

Proof. Since dualising finite-dimensional vector spaces is exact, the form induces a perfect pairing on $H^\bullet(E)$. Both sides have the same rank: for each i, $\dim H^i(E) = \dim E^i - \mathrm{rk}(d^i) - \mathrm{rk}(d^{i-1})$, and the terms involving the differentials cancel in the alternating sum.

For $i \neq 0$, the degree i and $-i$ parts of either graded form combine into a metabolic space. It therefore suffices to compare the degree 0 forms in the Witt group.

Choose a subspace $H \subseteq \ker(d^0)$ which maps isomorphically onto $H^0(E)$. The restriction of the pairing to H is perfect, since it agrees with the pairing on $H^0(E)$. Thus $E^0 = H \oplus H^\perp$, and it remains to show that $H^\perp$ is metabolic.

Compatibility with the differential gives $\beta(dv, w) = (-1)^{|v|+1}\beta(v, dw)$. In particular, if $B = d(E^{-1})$, then $\beta(dv, dw) = \beta(v, d^2w) = 0$, so B is isotropic. Also $\beta(dv, h) = \beta(v, dh) = 0$ for $h \in H$, giving $B \subseteq H^\perp$. Since $\ker(d^0) = H \oplus B$, we have $\ker(d^0|_{H^\perp}) = B$.

Finally, perfection identifies d^{-1} with the dual of d^0, up to sign, so $\mathrm{rk}(d^{-1}) = \mathrm{rk}(d^0)$. Rank-nullity now gives $\dim H^\perp = \dim B + \mathrm{rk}(d^0) = 2\dim B$. Hence B is an isotropic subspace of half dimension in $H^\perp$, making $H^\perp$ metabolic. The degree 0 forms have the same Witt class and we're done. □

Theorem 2.66 ([BW23, Proposition 2.3.])**.** *Let σ be a section of any relatively oriented vector bundle V on a proper scheme X. There is an equality in the Grothendieck–Witt ring*

$$e^{\mathbb{A}^1}_{CD}(V) = e^{\mathbb{A}^1}_{CD}(V, \sigma).$$

Proof. Take $K^\bullet = K(\sigma)^\bullet \otimes \mathcal{L}$. The pairing β_1 agrees component wise with $e^{\mathbb{A}^1}_{CD}(V)$. So:

$$e^{\mathbb{A}^1}_{CD}(V) = \sum_{p+q}(-1)^{p+q}[E_1^{p,q}].$$

Then, since each β_r is perfect, we can apply Lemma 2.65 to increase the page number. This gives for every r:

$$e^{\mathbb{A}^1}_{CD}(V) = \sum_{p+q}(-1)^{p+q}[E_r^{p,q}.]$$

Then, since X is proper and of finite dimension, the spectral sequence is bounded, and hence the pages stabilise. So we can go to $r = \infty$. Then, under the isomorphism $E_\infty^{i,k} \cong \mathrm{gr}^i R^{i+j} f_* K^\bullet$ we know β_∞ agrees with the pairing on the associated graded. Applying Lemma 2.64, we get:

$$e^{\mathbb{A}^1}_{CD}(V) = \sum_{p,q}(-1)^{p+q}[R^{p+q} f_* K(\sigma)^\bullet] = e^{\mathbb{A}^1}_{CD}(V, \sigma).$$ □

Now let's assume σ has isolated zeroes. Our next step is to define appropriate local indices $\mathrm{ind}^{CD}_p(\sigma)$, which we will do so as a push-forward of a cohomology theory. Fix $i\colon Z \to X$ for the closed regular immersion defining the vanishing locus of σ. Recall that an immersion is regular if locally its ideal is generated by a regular sequence. Further, recall a perfect complex is a complex quasi-isomorphic to a bounded complex of finite locally free $\mathcal{O}_X$-modules. They form a subcategory of $D(X)$ denoted $D^{perf}(X)$.

Notation 2.67. The set $\mathrm{BL}_{naive}(D(X), \mathcal{L}[n])$ is the set of isomorphism classes of non-degenerate symmetric bilinear forms on $D^{perf}(X)$, with respect to the duality function $\mathrm{Hom}(-, \mathcal{L})$, in the sense of Definition 1.50.

Example 2.68. Under the equivalence $D(k) \cong \mathrm{GrVect}_k$ given by taking cohomology, we can observe that perfect complexes correspond to finite dimensional graded vector spaces. So $\mathrm{BL}_{naive}(D(k), k[0])$ is the set of isomorphism classes of graded forms on graded k-vector spaces.

For a proper morphism $f\colon X' \to X$ the coherent duality adjunction supplies a trace map $\eta_{f,\mathcal{L}}\colon Rf_* f^!(\mathcal{L}) \to \mathcal{L}$ given by the co-unit map.

Remark 2.69. For coherent sheaves, the formula $f^!\mathcal{L} \cong f^*\mathcal{L} \otimes f^!\mathcal{O}_X$ from [Nee21, Lemma 2.4(ii)] computes $\eta_{f,\mathcal{L}}$ in terms of the standard trace map

$$f_*(f^!\mathcal{L}) \cong f_*(f^!\mathcal{O}_X \otimes f^*\mathcal{L}) \cong f_* f^!\mathcal{O}_X \otimes \mathcal{L} \xrightarrow{\eta_f \otimes \mathrm{id}_{\mathcal{L}}} \mathcal{L}.$$

We can use $\eta_{f,\mathcal{L}}$ to define a push-forward of cohomology theories.

Definition 2.70 ([BW23, p. 692])**.** For $f\colon X' \to X$ a proper locally complete intersection morphism, define the pushforward

$$f_*\colon \mathrm{BL}_{naive}(D(X'), f^!\mathcal{L}) \to \mathrm{BL}_{naive}(D(X), \mathcal{L})$$

via

$$[E \otimes E \xrightarrow{\phi} f^!\mathcal{L}] \mapsto [Rf_*E \otimes f_*E \to Rf_*(E \otimes E) \xrightarrow{Rf_*\varphi} Rf_* f^!\mathcal{L} \xrightarrow{\eta_{f,\mathcal{L}}} \mathcal{L}].$$

The following example is essential to the proof of the theorem.

Example 2.71. Suppose that $f\colon X \to k$ is a proper smooth scheme over k with a relatively oriented vector bundle V. Then, $f_*\beta_{(V,\sigma,\rho)}$ is observed to be $e^{\mathbb{A}^1}_{CD}(V, \sigma)$, with the $(-1)^i$ factor in the latter's definition arising from the Koszul signs on $E \otimes E$.

From here, we will assume that σ is locally given by only a Koszul-regular sequence.

Notation 2.72. We denote by L_f the cotangent complex of a morphism of schemes. We refer the reader to [Stacks, Tag 08P5] for the definition and properties.

Definition 2.73 ([KM76, p. 31])**.** For a bounded complex of vector bundles $E^\bullet$, the **graded determinant** is defined by

$$\widetilde{\det}(E^\bullet) = \left(\bigotimes_i (\det E^i)^{\otimes(-1)^i}, \sum_i (-1)^i \operatorname{rk}(E^i) \right) \in \operatorname{Pic}(X) \times H^0(X, \mathbb{Z})$$

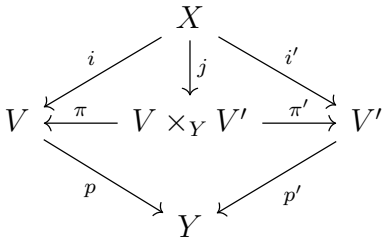
Proposition 2.74 ([BW23, Proposition B.1.])**.** *Suppose $f\colon X \to Y$ is an lci morphism. Then there is a canonical isomorphism $f^!\mathcal{O}_Y \cong \widetilde{\det} L_f$.*

Proof. Locally on X, by definition f factors as $p \circ i\colon X \to V \to Y$ for p smooth and i a regular immersion. We will initially construct our isomorphisms locally. For i, [Stacks, Tag 0BR0] and [Stacks, Tag 08SJ] yield $\widetilde{\det} L_i \cong i^!\mathcal{O}_V$. Likewise for p, [Stacks, Tag 0BRT] provides $\widetilde{\det} L_p \cong p^!\mathcal{O}_Y$.

Now, we'd like to use this to prove the result for the composition. First, [Nee21, Lemma 2.4(ii)] yields $(pi)^!\mathcal{O}_Y \cong i^*p^!\mathcal{O}_Y \otimes i^!\mathcal{O}_V$. Likewise, additivity of the determinant over distinguished triangles, combined with the standard distinguished triangle $i^*L_p \to L_{pi} \to L_i$ gives $\widetilde{\det} L_{pi} \cong i^*\widetilde{\det} L_p \otimes \widetilde{\det} L_i$. So by functoriality of the tensor product, we have the desired isomorphism locally.

Now, we argue that provided we have the required identifications, gluing is possible. Since both $f^!\mathcal{O}_Y$ and $\widetilde{\det} L_f$ are locally line bundles, we can invert them. This gives us the bundle $A = f^!\mathcal{O}_Y \otimes (\widetilde{\det} L_f)^{-1}$. Then the local isomorphism of the two components gives this is an $\mathcal{O}_X$-module constructed in degree 0. So an isomorphism can be constructed as a global section of a, so we can glue.

To have gluing data, it suffices to prove the isomorphism constructed before is independent of choice of factorisation. Suppose we have an alternative factorisation $X \xrightarrow{i'} V \xrightarrow{p'} Y$. Consider the induced map $j\colon X \to V \times_Y V'$:



The map j is a regular immersion. This holds since it factors as a section of the smooth map $X \times_Y V' \to X$, followed by the base change of i along π. Being a regular immersion is stable under base change, and a section of a smooth morphism is a regular immersion by [Stacks, Tag 067R]. Since π and π' are smooth, by symmetry we may compare each factorisation with the one through $V \times_Y V'$.

This reduces the problem to comparing factorisations which fit in a commutative diagram of the form

$$\begin{array}{ccccc} X & \xrightarrow{i'} & V' & \xrightarrow{p'} & Y \\ \| & & \downarrow{\scriptstyle q} & & \| \\ X & \xrightarrow[i]{} & V & \xrightarrow[p]{} & Y, \end{array}$$

where q is smooth. As explained in the proof of [Har66, Proposition III.2.2] the decomposition of $(pi)^!$ is constructed using the first fundamental sequence of Kähler differentials. So the two isomorphisms agree if we're composing two smooth things. So we can assume p is the identity.

Then the isomorphism $i'^! q^! \cong i^!$ can be reduced using the proof of [Har66, Proposition III.8.2.] to the case of a smooth morphism with a section. I.e., let $p\colon V \to X$ be smooth and $i\colon X \to V$ a regular immersion which is a section of p. Then we have an isomorphism $\omega X/V \otimes i^*\omega_{V/X} \cong \mathcal{O}_X$. All that is left to do is show the isomorphism $\widetilde{\det} L_i \otimes i^* \widetilde{\det} L_p \cong \widetilde{\det} L_{id} = \mathcal{O}_X$ is this isomorphism. But checking [Har66, Lemma III.8.1], we see the first proof arises from the second fundamental exact sequence of Kähler differentials and we're done. □

Notation 2.75. Denote by 1_X the symmetric bilinear form $\mathcal{O}_X \otimes \mathcal{O}_X \to \mathcal{O}_X$.

Proposition 2.76 ([BW23, Proposition 2.11.])**.** *Say V is a relatively oriented vector bundle on a smooth proper k-scheme X, and σ is a section with isolated zeroes. There is a canonical equivalence $i^! \det(V^*)[n] \cong \mathcal{O}_Z[0]$. Under the push-forward on bilinear forms, we have $i_*(1_Z) = \beta_{V,\sigma}$.*

Proof. Recall that we have a projection $p\colon K(\sigma)^\bullet \to \det V^*[n]$. Since σ is locally given by a Koszul-regular sequence, the map $r\colon K(\sigma)^\bullet \to i_*\mathcal{O}_Z$ is a quasi-isomorphism. Thus in the derived category we have $p \circ r^{-1}\colon i_*\mathcal{O}_Z \to \det V^*[n]$. It determines by adjunction a map $\eta\colon \mathcal{O}_Z \to i^! \det V^*[n]$. We claim that η is an equivalence.

For a closed immersion, we have $i_* i^! \mathcal{F} \cong R\mathcal{H}om_X(i_*\mathcal{O}_Z, \mathcal{F})$. Since $K(\sigma)^\bullet$ is a bounded locally free resolution of $i_*\mathcal{O}_Z$, it follows that $i_* i^! \det V^*[n] \cong \mathcal{H}om^\bullet_{\mathcal{O}_X}(K(\sigma)^\bullet, \det V^*[n])$.

The pairing $\beta_{(V,\sigma)} = p \circ m$ induces, by self-duality of the Koszul complex, an isomorphism of complexes $K(\sigma)^\bullet \cong \mathcal{H}om^\bullet_{\mathcal{O}_X}(K(\sigma)^\bullet, \det V^*[n])$, since exterior multiplication gives perfect pairings between complementary exterior powers. By Koszul regularity, this Hom complex therefore has cohomology $i_*\mathcal{O}_Z$ in degree zero and zero in every other degree.

To identify the map induced by η, we work locally on $\operatorname{Spec} A$. Then we can trivialise V, and write $\sigma = (f_1, \ldots, f_n)$ and $I = (f_1, \ldots, f_n)$. The degree-zero term of the Hom complex is $\operatorname{Hom}_A(\det V^*, \det V^*) \cong A$ since the determinant bundle is rank 1.

Further, the image of the differential into the degree 0 term is I. Since p is the identity on the top exterior power, it represents $1 \in A/I$. Under adjunction, the map on degree-zero cohomology induced by $i_*\eta$ sends 1 to this class, and hence is the identity of A/I. Both sides have cohomology concentrated in degree zero, so $i_*\eta$ is an equivalence. Since i_* is conservative, it detects quasi-isomorphisms, and therefore η is an equivalence.

Finally, under the identification of $i^*V^* \cong \mathcal{I}/\mathcal{I}^2$ with the conormal bundle, the class of p corresponds to the identity in $i^* \det(V^*) \otimes \det(\mathcal{I}/\mathcal{I}^2)^\vee$, which maps to 1 by evaluation. Thus η is the inverse of the equivalence supplied by Proposition 2.74.

Next, to show that $i_*(1_Z) = \beta_{(V,\sigma)}$, consider the diagram:

$$\begin{array}{ccccccc}
K(\sigma)^\bullet \otimes^{\mathrm{L}} K(\sigma)^\bullet & \xrightarrow{r\otimes r} & i_*\mathcal{O}_Z \otimes^{\mathrm{L}} i_*\mathcal{O}_Z & & & & \\
\downarrow{\scriptstyle m_K} & & \downarrow{\scriptstyle m_Z} & & & & \\
K(\sigma)^\bullet & \xrightarrow[r]{} & i_*\mathcal{O}_Z & \xrightarrow{i_*\eta} & i_*i^! \det V^*[n] & \xrightarrow{\mathrm{tr}} & \det V^*[n]
\end{array}$$

Here m_K is multiplication in the Koszul complex, m_Z is the lax monoidal map for i_* followed by multiplication in $\mathcal{O}_Z$, and tr is the counit of the adjunction $i_* \dashv i^!$. The square commutes because r respects multiplication. Then, the composite from the top right to $\det V^*[n]$ is the push-forward pairing $i_*(1_Z)$.

By the triangle identity for the adjunction $i_* \dashv i^!$, we have $\mathrm{tr} \circ i_*\eta = pr^{-1}$. Hence, the bottom row composes to p. After pre-composing $i_*(1_Z)$ with $r \otimes r$, the diagram gives $i_*(1_Z) \circ (r \otimes r) = p \circ m_K = \beta_{(V,\sigma)}$. Since $r \otimes r$ is an equivalence, this identifies $i_*(1_Z)$ with $\beta_{(V,\sigma)}$ in $\mathrm{BL}_{\mathrm{naive}}(D(X), \det V^*[n])$. □

Lemma 2.77 ([BW23, Lemma 2.12.])**.** *Let $g\colon Z \to Y$ and $f\colon Y \to X$ be proper maps. Given equivalences $f^!\mathcal{L} \cong \mathcal{M}[n]$ and $g^!\mathcal{M}[n] \cong \mathcal{N}$, the equivalence $(fg)^! \cong g^!f^!$ gives $(fg)^!\mathcal{L} \cong \mathcal{N}$. These equivalences induce push-forwards*

$$\begin{aligned}
g_*\colon \mathrm{BL}_{naive}(D(Z), \mathcal{N}) &\to \mathrm{BL}_{naive}(D(Y), \mathcal{M}[n]),\\
f_*\colon \mathrm{BL}_{naive}(D(Y), \mathcal{M}[n]) &\to \mathrm{BL}_{naive}(D(X), \mathcal{L}),\\
(fg)_*\colon \mathrm{BL}_{naive}(D(Z), \mathcal{N}) &\to \mathrm{BL}_{naive}(D(X), \mathcal{L}).
\end{aligned}$$

There is a canonical equivalence $(fg)_* \cong f_*g_*$.

Proof. Let $E \otimes E \xrightarrow{\phi} \mathcal{N}$ be a symmetric form. We suppress the chosen equivalences between $\mathcal{N}$, $g^!\mathcal{M}[n]$, and $(fg)^!\mathcal{L}$ in the following composites. Pushing the form forward along fg gives

$$(fg)_*E \otimes (fg)_*E \to (fg)_*(E \otimes E) \xrightarrow{(fg)_*\phi} (fg)_*\mathcal{N} \xrightarrow{\eta_{fg,\mathcal{L}}} \mathcal{L}.$$

Pushing it forward first along g and then along f gives

$$f_*g_*E \otimes f_*g_*E \to f_*(g_*E \otimes g_*E) \to f_*g_*(E \otimes E) \xrightarrow{f_*g_*\phi} f_*g_*\mathcal{N} \xrightarrow{f_*(\eta_{g,\mathcal{M}[n]})} f_*\mathcal{M}[n] \xrightarrow{\eta_{f,\mathcal{L}}} \mathcal{L}.$$

The first two arrows in the second composite agree with the lax monoidal map for $(fg)_*$, so it remains to compare the traces.

By Remark 2.69, $\eta_{f,\mathcal{L}}$ is $\eta_{f,\mathcal{O}_X} \otimes \mathrm{id}_{\mathcal{L}}$ under the projection formula. Since $f^!\mathcal{L} \cong f^!\mathcal{O}_X \otimes f^*\mathcal{L}$, applying the same formula to g identifies $\eta_{f,\mathcal{L}} \circ f_*(\eta_{g,f^!\mathcal{L}})$ with $(\eta_{f,\mathcal{O}_X} \circ f_*(\eta_{g,\omega_f})) \otimes \mathrm{id}_{\mathcal{L}}$.

The counit of the composite adjunction $f_*g_* \dashv g^!f^!$ satisfies

$$\eta_{f,\mathcal{O}_X} \circ f_*(\eta_{g,\omega_f}) = \eta_{fg,\mathcal{O}_X}.$$

Applying Remark 2.69 once more identifies the resulting map with $\eta_{fg,\mathcal{L}}$. Thus the two composites define the same pairing, as desired. □

Now, factor the structure map of the vanishing sub-scheme Z/k of σ as $\varpi = f \circ i$ where f is the structure map of X. The equivalence $i^! \det V^*[n] \cong \mathcal{O}_Z[0]$, and the equivalence $i^!\mathcal{L} \cong i^!\mathcal{O}_X \otimes i^*\mathcal{L}$, when combined with the projection formula, yield the equivalence

$$i^!(\det V^*[n] \otimes \mathcal{L}^{\otimes 2}) \cong i^! \det V^*[n] \otimes i^*\mathcal{L}^{\otimes 2}.$$

Combining this with the isomorphism $\det V^*[n] \otimes \mathcal{L}^{\otimes 2} \cong \omega_{X/k}[n]$ induced by the orientation, we obtain $i^!(\omega_{X/k}[n]) \cong i^*\mathcal{L}^{\otimes 2}$. Then, as f is a smooth map, using Proposition 2.74, we have an equivalent $f^!\mathcal{O}_k \cong \omega_{X/k}[n]$. So $\varpi^!\mathcal{O}_K \cong i^*\mathcal{L}^{\otimes 2}$. Hence, we end up with a derived push-forward.

Proposition 2.78. *When σ is a section with isolated zeroes of a relatively oriented vector bundle V, there is an equality*

$$e^{\mathbb{A}^1}_{CD}(V, \sigma) = \varpi_*(i^*\mathcal{L} \otimes i^*\mathcal{L} \to i^*\mathcal{L}^{\otimes 2}).$$

Proof. The orientation ρ identifies $\det(V^*)[n] \otimes \mathcal{L}^{\otimes 2}$ with $\omega_{X/k}[n]$. By Proposition 2.76, $i^! \det(V^*)[n] \cong \mathcal{O}_Z$ and $i_*(1_Z) = \beta_{(V,\sigma)}$. The projection formula therefore gives $i^!\omega_{X/k}[n] \cong i^*\mathcal{L}^{\otimes 2}$ and $i_*i^*\mathcal{L} \cong i_*\mathcal{O}_Z \otimes \mathcal{L} \cong K(\sigma)^\bullet \otimes \mathcal{L}$. Under these identifications, pushing forward the multiplication pairing on $i^*\mathcal{L}$ gives $\beta_{(V,\sigma)}$ tensored with $\mathcal{L}^{\otimes 2}$ and composed with ρ. This is $\beta_{(V,\sigma,\rho)}$.

By Lemma 2.77 and Example 2.71, we conclude that

$$\varpi_*(i^*\mathcal{L} \otimes i^*\mathcal{L} \to i^*\mathcal{L}^{\otimes 2}) = f_*\beta_{(V,\sigma,\rho)} = e^{\mathbb{A}^1}_{CD}(V,\sigma). \qquad \square$$

As σ has isolated zeros, we can consider the decomposition of Z into the connected components Z_z containing each isolated zero z. Denote by i_Z the closed immersion $Z_z \to X$ and by ϖ_z the structure map of Z_z. This then gives us a decomposition:

$$e^{\mathbb{A}^1}_{CD}(V,\sigma) = \sum_{z \in Z} \varpi_{z*}(i_z^*\mathcal{L} \otimes i_z^*\mathcal{L} \to i_z^*\mathcal{L}^{\otimes 2}).$$

Definition 2.79 ([BW23, Definition 2.29.])**.** Let z be an isolated zero of a section σ of a relatively oriented vector bundle V on a proper scheme. The **coherent duality index** of z is defined as

$$\mathrm{ind}^{CD}_z(\sigma) \coloneqq \varpi_{z*}(i_z^*\mathcal{L} \otimes i_z^*\mathcal{L} \to i_z^*\mathcal{L}^{\otimes 2})$$

To conclude the proof of Theorem 2.58, all that remains to show the Coherent Duality indices agree with the Scheja-Storch indices via a local computation.

Theorem 2.80 ([BW23, Theorem 2.18])**.** *Let us suppose that Z is realised as a relative global complete intersection in X. The map* $\mathrm{Hom}_{\mathcal{O}_k}(\varpi_*\mathcal{O}_Z, \mathcal{O}_k) \to \varpi_*\mathcal{O}_Z$ *induced by the $\chi(\Delta)$ map of Scheja-Storch duality is a symmetric isomorphism and induces a symmetric bilinear structure on $\varpi_*\mathcal{O}_Z$. This is the same structure as $\varpi_*(1_Z)$.*

Proof. The claim is local, so we may assume $Z = \mathrm{Spec}(B)$. Then since ϖ is finite, there is by [Har66, Theorem III.8.7.iii] a canonical isomorphism $\varpi^! \cong \mathcal{H}om_k(B, -)$. We want to make explicit the isomorphism:

$$\mathcal{H}om_k(B,k) \cong \varphi^!(\mathcal{O}_k) \cong i^!\pi^!(\mathcal{O}_A) \cong \omega_{Z/X} \otimes \omega_X \cong \mathcal{O}_k$$

It can be shown the isomorphism arises by computing $\mathrm{Ext}^n_X(B, \mathcal{O}_X)$ in two ways. First, the Koszul regular sequence $f_1, \ldots, f_n$ gives a resolution $K_k(f) \to B$ over X.

Meanwhile, there is the embedding of Z into $X \times_k Z$. The image's ideal is generated by the regular sequence $T_i - t_i$, where $T_i \otimes 1 - 1 \otimes T_i$ generate the diagonal of X, and t_i is the image of T_i in B. This gives a resolution $K_B(T-t)^\bullet \to B$ over $X \times Z$. Since $p\colon X \times Z \to X$ is finite, being the base change of a finite map, $p_*(K_B(T-t)^\bullet) \to p_*B = B$ is a resolution.

Applying $\operatorname{Hom}_X(-, \mathcal{O}_X)$ to the second resolution computes $\operatorname{Ext}^n_X(B, \mathcal{O}_X)$ as the cokernel of the map into $\operatorname{Hom}_X(K_B(T-t)_n, \mathcal{O}_X)$. The top term is $B \otimes_k \mathcal{O}_X$. Since B finite and projective as a complete intersection, we have $\operatorname{Hom}_X(K_B(T-t)_n, \mathcal{O}_X) \cong B^\vee \otimes_A \mathcal{O}_X$. In particular, an element $\alpha \in B^\vee$ gives a map $\xi(\alpha)$ in this top dual term, defined by $\xi(\alpha)(b \otimes r) = \alpha(b)r$. The relations imposed by the preceding Koszul differential identify the resulting cokernel with $B^\vee$. Thus ξ induces $B^\vee \cong \operatorname{Ext}^n_X(B, \mathcal{O}_X)$. Computing the same Ext group with $K_A(f)$ gives $\operatorname{Ext}^n_X(B, \mathcal{O}_X) \cong B$.

To compare these identifications, restrict the a_{ij} to $X \times_k Z$ and write their images as $\bar{a}_{ij}$. The relation $f_i = \sum_j \bar{a}_{ij}(T_j - t_j)$ defines a map of resolutions $\zeta\colon K_A(f) \to K_B(T-t)$ which sends the ith degree-one generator e_i to $\sum_j \bar{a}_{ij} e'_j$. In top degree, ζ is therefore multiplication by $\det(\bar{a}_{ij})$. Dualising ζ gives the middle map in

$$B^\vee \cong H^n \operatorname{Hom}_X(K_B(T-t), \mathcal{O}_X) \xrightarrow{H^n(\zeta^\vee)} H^n \operatorname{Hom}_X(K_A(f), \mathcal{O}_X) \cong B.$$

This composite is the isomorphism.

Write the image of $\det(a_{ij})$ in $B \otimes_k B$ as $\Delta = \sum_\ell b_\ell \otimes b'_\ell$. Evaluating the top-degree map on $\xi(\alpha)$ and then reducing modulo $(f_1, \dots, f_n)$ gives

$$\theta(\alpha) = \sum_\ell \alpha(b_\ell) b'_\ell = \chi(\Delta)(\alpha).$$

Hence the Scheja–Storch isomorphism $\chi(\Delta)$ is the isomorphism $B^\vee \cong \varpi^!(k) \cong B$ supplied by coherent duality. The trace is evaluation at the identity, so this identifies the bilinear form defined by Δ with $\varpi_*(1_Z)$. □

With this theorem, we are now placed to prove Theorem 2.58.

Proof of Theorem 2.58. All that remains is to show $\operatorname{ind}^{CD}_z(\sigma) = \operatorname{ind}_z(\sigma)$ for each zero z.

Fix z, and choose Nisnevich coordinates $x_1, \dots, x_n$ and a compatible trivialisation of V near z. Write $\sigma = (f_1, \dots, f_n)$ in this trivialisation, and put $I = \langle f_1, \dots, f_n \rangle \subseteq \mathcal{O}_{X,z}$.

Recall that the generators g_i used to present $\mathcal{O}_{Z,z}$ satisfy $g_i - f_i \in \mathfrak{m}^{2m}$, where $\mathfrak{m}^m \subseteq I$. Hence $g_i - f_i \in I^2$, so $\overline{g}_i = \overline{f}_i$ in I/I^2. Under the identification

$$\varpi_z^! k \cong \det(I/I^2)^\vee \otimes i_z^* \omega_{X/k},$$

the coordinates and equations therefore give the same generator

$$(\overline{g}_1 \wedge \cdots \wedge \overline{g}_n)^\vee \otimes (dx_1 \wedge \cdots \wedge dx_n) = (\overline{f}_1 \wedge \cdots \wedge \overline{f}_n)^\vee \otimes (dx_1 \wedge \cdots \wedge dx_n).$$

By the definition of compatibility with the relative orientation, this generator corresponds to the $\mathbf{r}_U^{\otimes 2}$ of Definition 2.16 under $\varpi_z^! k \cong i_z^* \mathcal{L}^{\otimes 2}$. Let $\eta = \chi(\Delta)^{-1}(1)$ be the functional defining $\mathrm{ind}_z(\sigma)$ for the chosen presentation. Theorem 2.80 identifies the generator, and hence $\mathbf{r}_U^{\otimes 2}$, with $\eta \in \mathrm{Hom}_k(\mathcal{O}_{Z,z}, k)$.

Since the trace is evaluation at the identity, the push-forward pairing defining $\mathrm{ind}_z^{CD}(\sigma)$ sends

$$(a\mathbf{r}_U, b\mathbf{r}_U) \mapsto ab\mathbf{r}_U^{\otimes 2} \mapsto ab\eta \mapsto (ab\eta)(1) = \eta(ab).$$

Thus trivialising $i_z^*\mathcal{L}$ by $\mathbf{r}_U$ identifies this pairing with the Scheja–Storch pairing on $\mathcal{O}_{Z,z}$. Hence $\mathrm{ind}_z^{CD}(\sigma) = \mathrm{ind}_z(\sigma)$, as required. □

Chapter 3

Lines on Cubics

In this chapter, we apply the machinery developed over the previous two chapters to produce an arithmetic count of the lines on a cubic surface. We begin by choosing Nisnevich coordinates and a trivialisation of the relevant vector bundle, and describing a *canonical* relative orientation. We then interpret the index associated to a line on a smooth cubic surface via the Morel degree of an involution associated to that line. This gives a geometric interpretation of the local terms in the count. Combined with showing that in the smooth case the section has simple zeros, this explains in what sense the $\mathbb{A}^1$–Euler number is a *signed* count.

We next compute that

$$e^{\mathbb{A}^1}(V) = 15\langle 1\rangle + 12\langle -1\rangle.$$

Combining this with the interpretation of the local indices, we obtain as our arithmetic count:

$$\sum (\#\text{lines of type } \beta)\,\mathrm{Tr}_{k(\beta)/k}\langle\beta\rangle = 15\langle 1\rangle + 12\langle -1\rangle.$$

We use this equality to derive restrictions on the numbers of lines of different types on a smooth cubic surface.

Finally, we use Macaulay2 to compute the local contributions for a singular cubic and verify that their sum agrees with the $\mathbb{A}^1$–Euler number in the smooth case, as predicted by Theorem 2.58.

3.1 Set-Up

First, let us make precise what we mean by a line.

Definition 3.1 ([KW21, Definition 4])**.** A line in $\mathbb{P}^3_k$, with field of definition L, is a closed point $\mathrm{Spec}(L) \to \mathrm{Gr}_k(4,2)$.

Recall that the Grassmannian's functor of points is given by:

$$\mathrm{Gr}_k(4,2)(\mathrm{Spec}(A)) = \{A^4 \twoheadrightarrow Q : Q \text{ is a projective } A\text{-module of rank } 2\}/\sim .$$

In the case A is the field extension L of k, lines are equivalent to L-linear surjections $L^4 \twoheadrightarrow L^2$. Then, we can consider the kernel of this surjection, S. The annihilator of S generates a homogeneous ideal of $\mathrm{Sym}((L^4)^\vee)$. To see this, say $\{e_1, e_2, e_3, e_4\}$ is the standard basis of k^4. Dualising, we obtain the dual basis $\{e_1^\vee, e_2^\vee, e_3^\vee, e_4^\vee\}$. We can then identify the elements of the annihilator with linear polynomials in x_1, x_2, x_3, x_4 that vanish on S by mapping $e_i^\vee \mapsto x_i$. This ideal then defines a closed subscheme of $\mathbb{P}^3_L$. Often, this subscheme will by abuse of notation also be denoted by ℓ.

Example 3.2. Consider the line in $\mathbb{P}^3_L$ corresponding to the map $\mathrm{Spec}(\mathbb{R}) \hookrightarrow \mathrm{Gr}_{\mathbb{R}}(4,2)$ defined by $\mathbb{R}^4 \to \mathbb{R}^2$, $(a_1, a_2, a_3, a_4) \mapsto (a_3, a_4)$. The kernel is given by $\mathrm{Span}(e_1, e_2)$. Then the annihilator is $\mathrm{Span}(x_3, x_4)$, so taking the homogeneous ideal, we get the subscheme generated by $\langle x_3, x_4\rangle$ of $\mathbb{P}^3_{\mathbb{R}}$, or $V(x_3, x_4)$ is the corresponding subscheme of $\mathbb{P}^3_{\mathbb{R}}$. This contains the closed real points $[a : b : 0 : 0]$ for $a, b \in \mathbb{R}$ not both equal to 0. If we take an affine neighbourhood, we get $[1 : t : 0 : 0]$ or $[t : 1 : 0 : 0]$ which both match our intuition for what lines are, 1 dimensional affine subspaces. This line has the field of definition $\mathbb{R}$.

Example 3.3. Consider the line $\mathrm{Spec}(\mathbb{C}) \hookrightarrow \mathrm{Gr}_{\mathbb{R}}(4,2)$ defined by the map $\mathbb{C}^4 \to \mathbb{C}^2$ given by $(a_1, a_2, a_3, a_4) \mapsto (a_1 + a_2 i, a_3 + a_4 i)$. The kernel of this map is given by $\mathrm{Span}_{\mathbb{C}}(e_1 + ie_2, e_3 + e_4 i)$. So the annihilator is $\mathrm{Span}_{\mathbb{C}}(x_2 - ix_1, x_4 - ix_3)$ so the homogeneous ideal is generated by $\langle x_2 - ix_1, x_4 - ix_3\rangle$. This is then the subspace $\{[s : si : t : ti]\}$ of $\mathbb{P}^3_{\mathbb{C}}$, as we'd expect.

Then, conjugating the line yields $[s : -is : t : -it]$ with homogeneous ideal $\langle x_2 + ix_1, x_4 + ix_3\rangle$. While the two lines are distinct over $\mathbb{C}$, they describe the same closed point of the real Grassmannian, since the corresponding maps from $\mathrm{Spec}\,\mathbb{C}$ differ by pre-composition with the automorphism induced by complex conjugation, and hence have the same image. So, this closed point of the real Grassmannian describes a Galois orbit of geometric lines.

Set $X = \mathrm{Gr}_k(4,2)$, the Grassmannian of 2-planes in $\mathbb{A}^4_k$. Equivalently, it is the space of lines, or copies of $\mathbb{P}^1$, in $\mathbb{P}^3$. Then denote by $\mathcal{S}$ and $\mathcal{Q}$ the tautological sub-bundle and quotient bundle on X. We set $V \coloneqq \mathrm{Sym}^3(\mathcal{S}^\vee)$.

We use this bundle since V is the vector bundle whose fibre at a point $[S]$, corresponding to the 2-plane S in $\mathbb{A}^4_k$, is precisely $\mathrm{Sym}^3(S^\vee)$. Upon selecting a basis e_1, e_2 of S, labelling the dual basis x_1, x_2 induces an identification of this fibre with the space of homogeneous cubics on S.

Then, the choice of a homogeneous cubic polynomial f on $\mathbb{P}^3_k$ induces a global section of V. Formally, we view f as an element of $\mathrm{Sym}^3((k^4)^\vee)$, with x_i corresponding to the dual of the coordinate basis vector e_i. Then we consider the homomorphism $\mathrm{Sym}^3((\mathcal{O}_X^4)^\vee) \to \mathrm{Sym}^3(\mathcal{S}^\vee)$ induced by dualising the inclusion $\mathcal{S} \subseteq \mathcal{O}_X^4$. The section σ_f is the image of f under this map. Intuitively, this is the section sending each point $[S]$ to the point $f|_S$ in the fibre V_S.

Hence, the zeros of σ_f correspond to the planes S in $\mathbb{A}^4_k$ on which $f|_S$ is the zero polynomial. Upon quotienting by scaling, these planes become lines in $\mathbb{P}^3$ on which f is identically zero. Hence, the zeros of this section are exactly the lines on the cubic surface defined by the vanishing locus of f.

Part of our input data into the $\mathbb{A}^1$–Euler number is a choice of Nisnevich coordinates, a local trivialisation and a relative orientation. In this section we will provide this data.

To relatively orient V, we first note that for the Grassmannian, as explained in [Vak25, Section 21.3.10.], there is a canonical isomorphism:

$$TX \cong \mathcal{H}om(\mathcal{S}, \mathcal{Q}) \cong \mathcal{S}^\vee \otimes \mathcal{Q}.$$

Then, we recall the standard affine coordinates on the Grassmannian:

Definition 3.4 ([KW21, Definition 42])**.** If $\underline{e} = \{e_1, ..., e_4\}$ is a basis for k^4, we define the elements of $(k[x, x', y, y'])^4$:

$$\tilde{e}_1 \coloneqq e_1 \quad \tilde{e}_2 \coloneqq e_2$$

$$\tilde{e}_3 \coloneqq xe_1 + ye_2 + e_3 \text{ and } \tilde{e}_4 \coloneqq x'e_1 + y'e_2 + e_4.$$

These form a basis of the $k[x, x', y, y']$-module $(k[x, x', y, y'])^4$. Denote by $\tilde{\phi}_i$ the corresponding dual basis.

These coordinates describe the planes that can be parameterised as:

$$\begin{bmatrix} x & y & 1 & 0 \\ x' & y' & 0 & 1 \end{bmatrix}.$$

Lemma 3.5 ([KW21, Lemma 43])**.** *If $\{e_1, ..., e_4\}$ is a basis for k^4, then the morphism*

$$\mathrm{Spec}(k[x, x', y, y']) \to \mathrm{Gr}_k(4, 2),$$

with the property that $\mathcal{S}$ pulls back to the subspace

$$k[x,x',y,y'] \cdot \tilde{e}_3 + k[x,x',y,y'] \cdot \tilde{e}_4 \subseteq (k[x,x',y,y'])^4,$$

is an open immersion.

Proof. Due to our pullback condition, we pick a distinguished map $\mathcal{O}_X^2 \to \mathcal{O}_X^4$ given by $(a,b) \to ae_1 + be_2$. Then given a surjection $q\colon \mathcal{O}_X^4 \to \mathcal{Q}$, we can form the composition $\mathcal{O}_X^2 \to \mathcal{O}_X^4 \xrightarrow{q} \mathcal{Q}$.

Next, note this composition is an isomorphism when the kernel of q is complementary to the span of e_1 and e_2. However, this condition is equivalent to the kernel of q being spanned by some $\tilde{e_3}$ and $\tilde{e_4}$. Now our morphism $\mathrm{Spec}(k[x,x',y,y']) \to X$ represents the sub-functor of the Grassmannian parameterising choices of q such that the composition is an isomorphism. This is an open immersion, as the composition is an isomorphism when the determinant of the composition is invertible, which is a Zariski open condition. □

Definition 3.6. For an ordered basis $\underline{e}$ of k^4, denote by $U(\underline{e})$ the open image of the map in Lemma 3.5.

These provide a covering of each point that is also locally an isomorphism, yielding Nisnevich coordinates. By construction, Lemma 3.5 gives us that $\mathcal{S}^\vee$ is trivialised by $\tilde{\phi_3}$ and $\tilde{\phi_4}$ over $U(\underline{e})$. Hence, it follows the quotient bundle $\mathcal{Q}$ is trivialised over $U(\underline{e})$ by $[\tilde{e}_1]$ and $[\tilde{e}_2]$, where the square brackets denote the classes of $\tilde{e}_i$ in the quotient bundle.

Appealing to our earlier description of the tangent bundle as $\mathcal{S}^\vee \otimes \mathcal{Q}$, we can formulate the following proposition.

Proposition 3.7 ([KW21, Proposition 45])**.** *As $\{e_1,\dots,e_4\}$ varies over all bases of k^4, the local trivialisations given by $\{\tilde{\phi_3} \otimes [\tilde{e}_1], \tilde{\phi_3} \otimes [\tilde{e}_2], \tilde{\phi_4} \otimes [\tilde{e}_1], \tilde{\phi_4} \otimes [\tilde{e}_2]\}$ are compatible with an orientation of TX.*

Proof. Let $\underline{b} = \{b_1,\dots,b_4\}$ be another choice of basis with dual basis $\theta_1,\dots,\theta_4$. Define $\tilde{b_i}$ as above, but replacing x,y with w,z.

Let M_{be} be the change of basis matrix relating the trivialisations $\{\tilde{\phi_3} \otimes [\tilde{e}_1], \tilde{\phi_3} \otimes [\tilde{e}_2], \tilde{\phi_4} \otimes [\tilde{e}_1], \tilde{\phi_4} \otimes [\tilde{e}_2]\}$ and $\{\tilde{\theta_3} \otimes [\tilde{b}_1], \tilde{\theta_3} \otimes [\tilde{b}_2], \tilde{\theta_4} \otimes [\tilde{b}_1], \tilde{\theta_4} \otimes [\tilde{b}_2]\}$. We claim that its determinant is a square in $\mathcal{O}^*(U(\underline{e}) \cap U(\underline{b}))$.

To see this, note that $M_{be} = A \otimes B$, where

$$A = \begin{bmatrix} \tilde{\theta_3}(\tilde{e_3}) & \tilde{\theta_4}(\tilde{e_3}) \\ \tilde{\theta_3}(\tilde{e_4}) & \tilde{\theta_4}(\tilde{e_4}) \end{bmatrix} \qquad B = \begin{bmatrix} \tilde{\phi_1}(\tilde{b_1}) & \tilde{\phi_1}(\tilde{b_2}) \\ \tilde{\phi_2}(\tilde{b_1}) & \tilde{\phi_2}(\tilde{b_2}) \end{bmatrix}.$$

These are the change of basis matrices relating $\{\tilde{\phi}_3, \tilde{\phi}_4\}$ and $\{\tilde{\theta}_3, \tilde{\theta}_4\}$, and $\{[\tilde{b}_1], [\tilde{b}_2]\}$ and $\{[\tilde{e}_1], [\tilde{e}_2]\}$ over $\mathcal{S}^\vee(U(\underline{e}) \cap U(\underline{b}))$ and $\mathcal{Q}(U(\underline{e}) \cap U(\underline{b}))$, respectively. In particular, their determinants are invertible regular functions. By the standard identity for determinants of tensor products, we have

$$\det M_{be} = (\det A)^2 (\det B)^2.$$

Hence, $\det M_{be}$ is the square of an invertible regular function.

We therefore obtain a line bundle $\mathcal{L}$ with transition functions $\det A \det B$ on $U(\underline{e}) \cap U(\underline{b})$. Then, on each trivialisation, sending the wedge product of the four sections to $1 \otimes 1$ gives the desired isomorphism $\det TX \cong \mathcal{L}^{\otimes 2}$. □

Likewise, we have:

Proposition 3.8 ([KW21, Proposition 47])**.** *As $\{e_1, \dots, e_4\}$ varies over all bases of k^4, the local trivialisations given by the sections $\{\tilde{\phi}_3^3, \tilde{\phi}_3^2\tilde{\phi}_4, \tilde{\phi}_3\tilde{\phi}_4^2, \tilde{\phi}_4^3\}$ are compatible with an orientation of V.*

Proof. In the proof of Proposition 3.7, we computed a change of basis matrix A relating $\{\tilde{\phi}_3, \tilde{\phi}_4\}$ and $\{\tilde{\theta}_3, \tilde{\theta}_4\}$ for two different choices of bases $\underline{e}$ and $\underline{b}$. The point here is to realise that our transition matrix M_{be} is now $\mathrm{Sym}^3(A)$, as we have taken the third symmetric power of both bases. Then we note that

$$\det(\mathrm{Sym}^3(A)) = (\det A)^6.$$

This fact follows from computing the product of the eigenvalues of $\mathrm{Sym}^3(A)$ over an algebraic closure.

Hence, $\det M_{be} = ((\det A)^3)^2$. Applying the same strategy as in the Proposition 3.7, we obtain a line bundle $\mathcal{L}$ with transition functions $(\det A)^3$, together with an isomorphism $\det V \cong \mathcal{L}^{\otimes 2}$. □

Let $v(\underline{e})$ be the section of $\mathcal{H}om(\wedge^4 TX, \wedge^4 V)|_{U(\underline{e})}$ that sends the wedge product of $\tilde{\phi}_3 \otimes [\tilde{e}_1], \tilde{\phi}_4 \otimes [\tilde{e}_1], \tilde{\phi}_3 \otimes [\tilde{e}_2], \tilde{\phi}_4 \otimes [\tilde{e}_2]$ to the wedge product of $\tilde{\phi}_3^3, \tilde{\phi}_3^2\tilde{\phi}_4, \tilde{\phi}_3\tilde{\phi}_4^2, \tilde{\phi}_4^3$.

Corollary 3.9 ([KW21, Corollary 48])**.** *Let $\mathcal{L} = \det \mathcal{S}^\vee$. There is a distinguished relative orientation*

$$j \colon \mathcal{H}om(\det TX, \det V) \xrightarrow{\sim} \mathcal{L}^{\otimes 2}$$

satisfying the following properties:

1. *For every basis $\underline{e}$, the section $j(v(\underline{e}))$ is a square.*

2. The isomorphism j is defined over $\mathrm{Spec}(\mathbb{Z})$.

Additionally, the last property determine j up to a sign when required to hold over all fields.

Proof. We begin by computing, using $TX = \mathcal{S}^\vee \otimes \mathcal{Q}$, that

$$\begin{aligned}\mathcal{H}om(\det TX, \det V) &\cong \det(\mathcal{S}^\vee \otimes \mathcal{Q})^\vee \otimes \det \mathrm{Sym}^3(\mathcal{S}^\vee) \\ &\cong (\det \mathcal{S})^{\otimes 2} \otimes (\det \mathcal{Q}^\vee)^{\otimes 2} \otimes (\det \mathcal{S}^\vee)^{\otimes 6} \\ &\cong (\det \mathcal{S}^\vee)^{\otimes 2},\end{aligned}$$

where the last identification follows from additivity of the determinant after trivialising $\det \mathcal{O}_X^4$ in the tautological exact sequence.

Then, by [Dol03, Lemma 11.1.] $\mathrm{Pic}(X) \cong \mathbb{Z}$, so it is torsion-free. Hence $\mathcal{L} = \det \mathcal{S}^\vee$ is the unique square root of $\mathcal{H}om(\det TX, \det V)$ up to isomorphism. Further, the determinant computation is valid over $\mathbb{Z}$. Hence it gives an isomorphism j defined over $\mathrm{Spec}(\mathbb{Z})$. We choose its sign so that $j(v(\underline{e}))$ is the square of the corresponding local generator, the $\mathbf{r}_U$ of Definition 2.16, on the standard affine chart. By the transition function computations in Proposition 3.7 and Proposition 3.8, then $j(v(\underline{e}))$ is a square for every basis $\underline{e}$.

Suppose that j' is another such isomorphism defined over $\mathbb{Z}$. Since any two isomorphisms of line bundles differ by a unit, and the only units over $\mathbb{Z}$ are ± 1, we must have $j' = \pm j$. □

Remark 3.10. As we will see in Proposition 3.27, there is a third property we can require which determine the sign as well.

3.2 Interpreting the Indices

Recall the section σ_f which is the image of f under the induced map $\mathrm{Sym}^3((\mathcal{O}_X^3)^\vee) \to \mathcal{S}\dagger\Updownarrow^3(\mathcal{S}^\vee)$. When f is a smooth cubic, the indices $\mathrm{ind}_\ell(\sigma_f)$ have a geometric interpretation.

Let us suppose that a line ℓ contained in $W = V(f)$ is defined by the closed immersion $\mathrm{Spec}(L) \to \mathrm{Gr}_k(4,2)$. Then we can view ℓ as being the corresponding line on $W \otimes_k L$.

Definition 3.11 ([KW21, Definition 5.])**.** Set $T \subseteq (L^4)^\vee = H^0(\mathbb{P}^3_L, \mathcal{O}(1))$ as the vector space of linear polynomials that vanish on ℓ.

Remark 3.12. We can view T in two ways. First, it is a subspace of $H^0(W \otimes L, \mathcal{O}_{W\otimes L}(1))$, the linear polynomials on $W \otimes_k L$. Likewise, T is a subspace of $H^0(\mathcal{I}_\ell(1))$ where $\mathcal{I}_\ell(1) \coloneqq \mathcal{I}_\ell \otimes \mathcal{O}_W(1)$. This is the sheaf of linear polynomials on W that vanish on ℓ.

Remark 3.13. Note that T is the annihilator of the subspace $S \subseteq L^4$ that quotients under scaling to yield ℓ.

Since ℓ is a co-dimension 1 subscheme of $W \otimes_k L$, the Weil-Cartier divisor correspondence yields the ideal sheaf $\mathcal{I}_\ell$ is a line bundle. Hence, we've defined two linear systems on $W \otimes_k L$. First, we have $(T, \mathcal{O}_W(1))$. The elements of T each cut out planes containing ℓ. Since we're here viewing them as functions on W, we're then looking at the intersection of these planes with W.

Meanwhile, the elements of $(T, I_\ell(1))$ are the *residual* intersections with these planes. In effect, the tensor with $\mathcal{I}_\ell$ allows us to factor out multiples of the line from $\mathcal{O}_W(1)$, or remove the divisor $[\ell]$. This means for a plane $S' \subseteq \mathbb{P}^3$, we're left with $[S' \cap W \otimes_k L] - [\ell]$, or the non trivial part of the intersection.

Lemma 3.14 ([KW21][Lemma 6.])**.** *The linear system $(T, \mathcal{I}_\ell(1))$ is base-point free. I.e. there exists a selection of sections in $T \subseteq \Gamma(\mathcal{I}_\ell(1))$ that do not simultaneously vanish.*

Proof. The sheaf $\mathcal{I}_\ell(1)$ is a quotient of the sheaf $\mathcal{I}_\ell \otimes \mathcal{O}(1)$ on $\mathbb{P}^3_L$. Then, the sheaf on $\mathbb{P}^3_L$ is generated by T by definition of ℓ. Then generation by T is equivalent to base-point-freeness, as a basis of T descends to a collection of non-simultaneously vanishing sections. □

Then, a base-point-free linear system $(T, \mathcal{L})$ induces a map $\pi\colon W \to \mathbb{P}(T^\vee)$ via the functor of points of projective space.

Lemma 3.15 ([KW21, Lemma 10])**.** *Suppose that W contains the line ℓ defined by the subspace* $\mathrm{Span}_L(e_3, e_4)$*. We can then set $f = x_1 \cdot P_1 + x_2 \cdot P_2$ where P_1 and P_2 are homogeneous quadratic polynomials.*

Then, the fibre of $\pi\colon \ell \to \mathbb{P}(T^\vee)$ over the k point corresponding to subspace $(a, b, 0, 0) \in T^\vee$ is given by:

$$\{a \cdot P_1(0, 0, x_3, x_4) + b \cdot P_2(0, 0, x_3, x_4) = 0\} \subseteq \ell.$$

The proof below differs from that of [KW21] and was communicated to the author by his co-adviser. We present it as it is a more direct style of argument.

Proof. Observe that since we're taking residual intersections, $\pi|_\ell^{-1}[a : b : 0 : 0] = V(bx_1 - ax_2) \cap \ell - \ell$ as divisors. Then, $\pi^{-1}[a : b : 0 : 0]$ is going to be the points subject to the equation $bx_1 - ax_2 = 0$, and $x_1P_1(x_1, x_2, x_3, x_4) + x_2P_2(x_1, x_2, x_3, x_4) = 0$. Setting $bx_1 - ax_2 = 0$ and assuming without loss of generality that $a \neq 0$, we get that $x_2 = \frac{b}{a}x_1$. So the equation cutting out our intersection becomes $x_1P_1(x_1, \frac{b}{a}x_1, x_3, x_4) + \frac{b}{a}x_1P_2(x_1, \frac{b}{a}x_1, x_3, x_4) = 0$. In this chart, our line ℓ is cut out by x_1, so we can factor out a copy of ℓ's equation to get $x_1(P_1(x_1, \frac{b}{a}x_1, x_3, x_4) + \frac{b}{a}P_2(x_1, \frac{b}{a}x_1, x_3, x_4))$. Thus, on ℓ the residual intersection will be cut out by $P_1(0, 0, x_3, x_4) + \frac{b}{a}P_2(0, 0, x_3, x_4) = 0$. Multiplying by a gives us the desired result. □

Since f is smooth, $P_1|_\ell$ and $P_2|_\ell$ have no shared zeros. Hence we have computed that the map $\pi|_\ell$ is finite of degree 2. So, when $\mathrm{Char}(k) \neq 2$, the map is Galois.

Definition 3.16 ([KW21, Definition 7.])**.** Set $i\colon \ell \to \ell$ as the non-trivial automorphism of ℓ in the Galois group of $\pi|_\ell$.

We note that i is an involution as the Galois group is of cardinality 2.

This involution admits a geometric description. We can view $\mathbb{P}(T^\vee)$ as the space of planes in $\mathbb{P}^3_L$ containing ℓ. This holds since $T^\vee$ is the dual of the annihilator of a plane S. Hence $T^\vee$ is canonically isomorphic to L^4/S. Then we note any non-zero vector in L^4/S defines a 3 dimensional subspace containing S. Scalar multiples of a vector corresponding to the same subspace. Then upon quotienting by scaling, this becomes a 2-plane in $\mathbb{P}^3_L$ containing ℓ.

Hence π admits a description as the map sending a point on W not on ℓ to the plane containing it and ℓ. On ℓ, we have π is the extension of the projection, sending each point to the *tangent plane* to W containing each point. Since f is constant on ℓ, that line lies in the tangent plane. Likewise, the residual intersection of the plane with W is by Bezout's theorem a conic, since the plane can't intersect the line with multiplicity by smoothness of f. Hence, we have there are at most 2 points on the line with the same tangent plane.

So i swaps p and q if $T_pW = T_qW$. This is equivalent to swapping the two points of intersection of the residual conic with ℓ, or if the residual conic ramifies, we get a fixed point.

Theorem 3.17 ([Mor07, p. 1037])**.** *There is a degree map*

$$\deg^{\mathbb{A}^1}\colon [\mathbb{P}^n/\mathbb{P}^{n-1}, \mathbb{P}^n/\mathbb{P}^{n-1}]^{\mathbb{A}^1} \to \mathrm{GW}_0(k),$$

where the left hand side is the homotopy classes of maps between the two simplicial sheaves, in the sense of Morel-Voevodsky's $\mathbb{A}^1$–homotopy theory. This map is always surjective, and is an isomorphism for $n \geq 2$.

Then we observe $\mathbb{P}^1/\mathbb{P}^0 \cong \mathbb{P}^1$. So the theorem yields for each involution $i\colon \mathbb{P}^1 \to \mathbb{P}^1$ a choice of element in the Grothendieck–Witt ring. This $--A^1$ degree turns out to be relatively easy to compute.

Theorem 3.18 ([WW20, p. 933])**.** *Let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be an endomorphism of the projective line. Consider a k-point p. Set $f^{-1}(p) = \{q_1, ..., q_n\}$. Suppose $\mathrm{Jac}(f)|_{q_i} \neq 0$ for all i and all the q_i are k-points. Then:*

$$\deg^{\mathbb{A}^1}(f) = \sum_{i=1}^{n} \langle \mathrm{Jac}(f)|_{q_i} \rangle \in \mathrm{GW}_0(k),$$

Definition 3.19 ([KW21, Definition 9.])**.** The **type** of a line on W is $\langle -1 \rangle \cdot \deg^{\mathbb{A}^1}(i)$.

Definition 3.20 ([KW21, Definition 9.])**.** We say a line is **hyperbolic** if it has type $\langle 1 \rangle$. Else it is **elliptic**.

Then, we have the following result:

Theorem 3.21 ([KW21, Lemma 51])**.** *Let ℓ be a line on a smooth cubic surface W. Then the type equals the index of σ_f.*

To prove this formula, we will provide a formula for both the type and the index, which proves the claim.

The automorphism group of $\mathbb{P}^1_L$ is the group $\mathrm{PGL}_2(L)$. This acts on the point $[z : 1]$ via:

$$\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} [z : 1] = \frac{\alpha z + \beta}{\gamma z + \delta}.$$

Then using either of our formulae above, we obtain the $\mathbb{A}^1$–degree of i is given by $\langle \alpha\delta - \beta\gamma \rangle$.

We have the following result from group theory:

Lemma 3.22 ([Bea10, Theorem 4.2])**.** *Every non-trivial involution $i\colon \mathbb{P}^1_L \to \mathbb{P}^1_L$ is conjugate to the involution $z \mapsto -\frac{\alpha}{z}$ for some $\alpha \in L$.*

The $\mathbb{A}^1$ degree of $-\frac{\alpha}{z}$ is then readily computed as $\langle \alpha \rangle$.

Lemma 3.23 ([KW21, Corollary 13.]). *Let i be a non-trivial involution on $\mathbb{P}^1_k$ and $\mathcal{D}$ be the discriminant of the polynomial defining the fixed point locus of i. Then:*

$$\langle -1 \rangle \cdot \deg^{\mathbb{A}^1}(i) = \langle \mathcal{D} \rangle.$$

Proof. As outlined, the $\mathbb{A}^1$–degree is invariant under conjugation. Likewise, the square class of the discriminant is invariant under conjugation, as it agrees with that of the discriminant of the characteristic polynomial of a lift of i to $\mathrm{GL}_2(k)$. So we reduce to $i(z) = -\frac{\alpha}{z}$. Then note

$$-\frac{\alpha}{z} = z \iff z^2 + \alpha = 0$$

This polynomial has discriminant -4α so:

$$\langle \mathcal{D} \rangle = \langle -4\alpha \rangle = \langle -\alpha \rangle = \langle -1 \rangle \cdot \deg^{\mathbb{A}^1}(i).$$

□

Definition 3.24 ([Eis95][p. 312]). Let $f(x)$ and $g(x)$ be univariate polynomials of degrees m and n. Their **Sylvester matrix** is the $(m+n) \times (m+n)$ matrix whose first n columns consist of the coefficients of f, shifted down by one position in each successive column and padded with zeros. The remaining m columns are constructed analogously from the coefficients of g.

The determinant of this matrix is the **resultant** of the polynomials.

Proposition 3.25 ([KW21, Proposition 14]). *Let $e_1, ..., e_4$ be a basis of k^4 such that the subspace $S = \mathrm{Span}_k(e_3, e_4)$ defines a line in W, with dual basis $x_1, ..., x_4$. Then we have:*

$$\langle -1 \rangle \cdot \deg^{\mathbb{A}^1}(i) = \left\langle \mathrm{Res}\left(\left.\frac{\partial f}{\partial x_1}\right|_S, \left.\frac{\partial f}{\partial x_2}\right|_S \right) \right\rangle.$$

Proof. We assume that f is then of the form $x_1 P_1 + x_2 P_2$.

By Lemma 3.23 it suffices to show that the resultant is the class of the discriminant. The locus of points fixed by i is observed to be the ramification locus of π, via the description of π as the extension of the projection. So now we just need to compute the ramification locus.

We found that the fibre of π over $[a : b : 0 : 0]$ was given by $a \cdot P_1(0, 0, x_3, x_4) + b \cdot P_2(0, 0, x_3, x_4)$. So π ramifies when this polynomial has a double zero. So the ramification locus is the vanishing set of the discriminant in x_3 and x_4 of this polynomial. So the discriminant of the ramification locus is given by

$$\mathrm{Disc}_{a,b}(\mathrm{Disc}_{x_3,x_4}(a \cdot P_1(0, 0, x_3, x_4) + b \cdot P_2(0, 0, x_3, x_4)))$$

To conclude the proof, it suffices to compute both sides explicitly. Differentiating f we get:

$$\frac{\partial f}{\partial e_1}|_S = P_1(0,0,x_3,x_4),$$
$$\frac{\partial f}{\partial e_2}|_S = P_2(0,0,x_3,x_4).$$

Further, assume that

$$P_1(0,0,x_3,x_4) = \sum_{i+j=2} a_{ij}x_3^i x_4^j,$$
$$P_2(0,0,x_3,x_4) = \sum_{i+j=2} b_{ij}x_3^i x_4^j.$$

Then we get for the Sylvester matrix:

$$\begin{bmatrix} a_{20} & 0 & b_{20} & 0 \\ a_{11} & a_{20} & b_{11} & b_{20} \\ a_{02} & a_{11} & b_{02} & b_{11} \\ 0 & a_{02} & 0 & b_{02} \end{bmatrix}$$

It has determinant given by:

$$a_{02}^2b_{20}^2 + a_{11}^2b_{02}b_{20} + a_{20}^2b_{02}^2 + a_{02}a_{20}b_{11}^2 - a_{02}a_{11}b_{11}b_{20} - a_{11}a_{20}b_{02}b_{11} - 2a_{02}a_{20}b_{02}b_{20}$$

A fairly quick computation yields that this is one-sixteenth of the discriminants, and hence has the same square class. □

Lemma 3.26 ([KW21, Lemma 50])**.** *Let $f \in k[x_1,...,x_4]$ be a cubic homogeneous polynomial. If a plane $S \subseteq k^4$ satisfies $f|_S = 0$, the differential of σ_f at the corresponding k-point of X is the map*

$$\mathcal{S}^\vee \otimes \mathcal{Q} \to \mathrm{Sym}^3(\mathcal{S}^\vee)$$

given by

$$\phi \otimes (v+S) \mapsto \left(\frac{\partial f}{\partial v}\right)|_S \cdot \phi.$$

Proof. Extend a basis of S to a basis $\underline{e}$ of k^4 such that $S = \mathrm{span}_k(e_3,e_4)$. Then, we pick coefficients such that

$$f = \sum_{\substack{i,j,k,l\geq 0 \\ i+j+k+l=3}} a_{i,j,k,l}\phi_1^i\phi_2^j\phi_3^k\phi_4^l$$

Then, considering our basis of $\mathcal{S}^\vee \otimes \mathcal{Q}$ from Proposition 3.7, the linearity of the map we're comparing the differential to tells us it is completely defined by:

$$\begin{aligned}
\phi_3 \otimes [e_1] &\mapsto (a_{1,0,2,0}\phi_3^2 + a_{1,0,1,1}\phi_3\phi_4 + a_{1,0,0,2}\phi_4^2)\cdot\phi_3,\\
\phi_4 \otimes [e_1] &\mapsto (a_{1,0,2,0}\phi_3^2 + a_{1,0,1,1}\phi_3\phi_4 + a_{1,0,0,2}\phi_4^2)\cdot\phi_4,\\
\phi_3 \otimes [e_2] &\mapsto (a_{0,1,2,0}\phi_3^2 + a_{0,1,1,1}\phi_3\phi_4 + a_{0,1,0,2}\phi_4^2)\cdot\phi_3,\\
\phi_4 \otimes [e_2] &\mapsto (a_{0,1,2,0}\phi_3^2 + a_{0,1,1,1}\phi_3\phi_4 + a_{0,1,0,2}\phi_4^2)\cdot\phi_4.
\end{aligned}$$

Then, we know V is trivialised on $U(\underline{e})$ by $\tilde{\phi}_3^3, \tilde{\phi}_3^2\tilde{\phi}_4, \tilde{\phi}_3\tilde{\phi}_4^2, \tilde{\phi}_4^3$. On this chart, σ_f is the function $\mathbb{A}_k^4 \to \mathbb{A}_k^4$ defined by taking the coefficients of the u, v power terms in

$$f|_S = \sum_{\substack{i,j,k,l\geq 0\\ i+j+k+l=3}} a_{i,j,k,l}(x\tilde{\phi}_3 + x'\tilde{\phi}_4)^i(y\tilde{\phi}_3 + y'\tilde{\phi}_4)^j\tilde{\phi}_3^k\tilde{\phi}_4^l$$

and differentiating with respect to x, x', y and y'. Then taking partial derivatives at 0 returns the same map as we just computed and we're done. □

Proposition 3.27 ([KW21, Lemma 51])**.** *Suppose the surface defined by a smooth homogeneous cubic polynomial $f \in k[x_1, ..., x_4]$ contains the line defined by $S = \mathrm{span}_k(e_3, e_4)$ for a basis $\underline{e} = \{e_1, e_2, e_3, e_4\}$ of k^4.*

Then we may choose the sign of the relative orientation of V such that

$$\mathrm{Ind}_S(\sigma_f) = \left\langle \mathrm{Res}\left(\left.\frac{\partial f}{\partial x_1}\right|_S, \left.\frac{\partial f}{\partial x_2}\right|_S \right) \right\rangle.$$

Proof. This is effectively a corollary of the Lemma 3.26. We note that under such a basis, we identified the differential with the map whose matrix is with respect to the $\phi_3 \otimes e_1, \ldots, \phi_4 \otimes e_2$ and $\phi_3^3, \ldots, \phi_4^3$ bases was

$$\begin{bmatrix}
a_{1,0,2,0} & 0 & a_{0,1,2,0} & 0\\
a_{1,0,1,1} & a_{1,0,2,0} & a_{0,1,1,1} & a_{0,1,2,0}\\
a_{1,0,0,2} & a_{1,0,1,1} & a_{0,1,0,2} & a_{0,1,1,1}\\
0 & a_{1,0,0,2} & 0 & a_{0,1,0,2}.
\end{bmatrix}$$

This is the Sylvester matrix of $\left.\frac{\partial f}{\partial x_1}\right|_S$ and $\left.\frac{\partial f}{\partial x_2}\right|_S$. Hence the determinant of this matrix is the resultant.

Recall that by Corollary 3.9, our relative orientation was determined up to a choice of sign. Since the ordered bases used to define the matrix are those defining $v(\underline{e})$, the two possible choices give local indices equal to either the class of the resultant or its negation, in the sense of multiplication by $\langle -1\rangle$.

The sign is independent of the chosen line and basis, since the transition determinants between the corresponding local trivialisations are squares.

By Proposition 3.25, the former is the type of the corresponding line. We therefore pick a distinguished relative orientation by requiring that the index of a line agrees with its type. This fixes the sign of the orientation and yields the desired equality. □

Corollary 3.28 ([KW21, Corollary 52])**.** *The type of a line on a smooth cubic is, after base change, the index of σ_f at the corresponding zero.*

Proof. This is immediate by combining the proposition with Proposition 3.25. □

Corollary 3.29 ([KW21, Corollary 53])**.** *When f is smooth the section σ_f has only simple zeros. More generally, a line that is disjoint from the singular locus of f corresponds to a simple zero.*

Proof. We note that, as discussed in the proof of [Vak25, Theorem 27.2.1], when f is smooth the two quadratics defining $\frac{\partial f}{\partial e_1}$ and $\frac{\partial f}{\partial e_2}$ in the proposition's assumptions cannot simultaneously vanish without forcing the Jacobian to vanish. Hence, the resultant won't vanish as its vanishing implies the polynomials share a common root in $\bar{k}$. □

Corollary 3.30 ([KW21, Corollary 54])**.** *The field of definition of a line contained in a smooth cubic surface is a separable extension of k.*

Proof. As σ_f has only simple zeros, its vanishing subscheme Z is geometrically reduced. However, $\mathrm{Spec}(L) \to Z$ identifies $\mathrm{Spec}(L)$ with a connected component of Z, as σ_f has isolated zeroes by smoothness of f. But since we've identified $\mathrm{Spec}(L)$ with a geometrically reduced component, we conclude it is geometrically reduced as well, which is equivalent to L being separable. □

Remark 3.31. Thanks to this corollary, we can use Proposition 2.53 to compute the indices of lines not defined over our base field.

Corollary 3.32. *Suppose that f is smooth. Let $A\colon \mathbb{P}^3_L \to \mathbb{P}^3_L$ be a projective automorphism, that fixes W. Then*

$$\mathrm{ind}_\ell(\sigma_f) = \mathrm{ind}_{A.\ell}(\sigma_f).$$

Proof. This follows from the assertion that $\deg^{\mathbb{A}^1}$ is a group homomorphism, and hence is multiplicative with respect to composition. Note as A fixes W, and

automorphisms of $\mathbb{P}^3_L$ are smooth, we have $A(T_pW) = T_{A\cdot p}W$. Hence, we obtain by the description of the involution:

$$i_\ell = A|^{-1}_{A\cdot\ell} \cdot i_{A\cdot\ell} \cdot A|_\ell.$$

Viewing A as a map $\mathbb{P}^1 \to \mathbb{P}^1$, multiplicativity gives us the desired results. □

Remark 3.33. The fact that the section has only simple zeros is what allows us to interpret the $\mathbb{A}^1$–Euler number as a signed count. Indeed, if $p \in Z$ is a k-point corresponding to a line on $V(f)$, then simplicity of the zero implies that the stalk $\mathcal{O}_{Z,p}$ is one-dimensional over k. Hence the local index at p is represented by a rank 1 form $\langle\alpha\rangle$.

More generally, suppose that a line is defined over a separable finite extension L/k. By the preceding remark, after tensoring with the separable extension L, its Galois conjugates give distinct simple zeros, and hence distinct rank 1 forms. The transfer from L to k then combines these conjugate contributions into a single class over k.

Thus, the $\mathbb{A}^1$–Euler number may be viewed as assigning a square class to each geometric line, with the contribution of the lines over the extension accounting for conjugacy.

3.3 The Arithmetic Count

In this section, we will actually compute the arithmetic count, and then prove a result restricting the number of lines of different types.

Theorem 3.34 ([KW21, Main Theorem])**.** *Let* $\mathrm{Sym}^3(\mathcal{S}^\vee)$ *be relatively oriented with the canonical relative orientation of Corollary 3.9. Then, over any field not of characteristic 2, one has:*

$$e^{\mathbb{A}^1}(\mathrm{Sym}^3(\mathcal{S}^\vee)) = 15\langle 1\rangle + 12\langle -1\rangle.$$

In Theorem 2.58, we already established the $\mathbb{A}^1$–Euler number is independent of the choice of a section with isolated zeroes. Hence, it suffices to compute it for nice cubics with isolated lines. Due to characteristics issues, we will need two cubic surfaces: the Fermat Cubic and the Clebsch Cubic. These are smooth when the characteristic is not equal to 3, and not equal to 5 respectively.

Definition 3.35. The **Fermat Cubic** is the cubic given by:

$$f(x_0, x_1, x_2, x_3) = x_0^3 + x_1^3 + x_2^3 + x_3^3.$$

We first note that when the characteristic is not 3, a Jacobian computation shows this cubic is clearly smooth. In characteristic 3, its vanishing locus $V(f)$ is a multiplicity 3 hyperplane.

Lemma 3.36 ([Vak25, Ex. 27.1.C])**.** *When k contains a non-trivial cubic root of unit, denoted ζ_3, then $V(f)$ contains 27 lines of the form:*

$$\mathrm{Span}_k((-1, \zeta_3^{i_1}, 0, 0), (0, 0, -1, \zeta_3^{i_2}))$$

for $i_1, i_2 = 0, 1, 2$, and up to permutation of the non-zero coordinates.

Hence, we split into two cases. First, when $k = k(\zeta_3)$, we have 27 lines on $V(f)$. The automorphism group of the Fermat cubic acts transitively on them, since by inspection the symmetry of the Fermat cubic gives us we are free to permute coordinates using automorphisms. Likewise, we can scale them by factors of ζ_3.

Hence, using Corollary 3.32 we obtain:

$$e^{\mathbb{A}^1} = 27 \cdot \mathrm{Type}(\mathrm{Span}_k((-1, 1, 0, 0), (0, 0, -1, 1))).$$

Using Proposition 3.25, we simply need to find the resultant. Extending $(-1, 1, 0, 0)$ and $(0, 0, -1, 1)$ to an ordered basis by pre-pending $(1, 0, 0, 0)$ and $(0, 0, 1, 0)$ we can compute the partial derivatives are:

$$\frac{\partial f}{\partial e_1} = 3x_0^2 \qquad \frac{\partial f}{\partial e_2} = 3x_2^2.$$

To restrict the partial derivatives to S, we just apply the relation $x_0 = -x_1$ and $x_2 = -x_3$, which returns the same partial derivatives. Then, the resultant is clearly 81 by computing the Sylvester matrix as $3 \cdot \mathrm{Id}_4$. Since the resultant is $81 = 9^2$, we conclude the type of ℓ is $\langle 1 \rangle$. Then, note that if $\zeta_3 \in k$, we have:

$$-1 = \zeta_3^2 + (\zeta_3^2)^2.$$

Hence by Remark 1.48 we obtain:

$$e^{\mathbb{A}^1}(V) = 27 \cdot \langle 1 \rangle = 15\langle 1 \rangle + 12\langle -1 \rangle.$$

Next, we have the case where $k(\zeta_3) \neq k$. By inspection, we have exactly 3 lines with field of definition k, given by taking $i_1 = i_2 = 0$ and permuting coefficients. By our previous computation, these 3 lines have type $\langle 1 \rangle$. Then, there are 12 lines with field of definition $k(\zeta_3)$, corresponding to the Galois orbits of the geometric lines over $k(\zeta_3)$. Applying Proposition 2.53 we can base change to $k(\zeta_3)$, and note

by the previous case that all 24 lines have type $\langle 1\rangle$. By Proposition 2.53, taking the Scharlau transfer of Definition 2.52 for a single representative of each Galois orbit, we get:

$$e^{\mathbb{A}^1}(V) = 3 \cdot \langle 1\rangle + 12 \cdot \mathrm{Tr}_{k(\zeta_3)/k}\langle 1\rangle.$$

Then we compute $12\,\mathrm{Tr}_{k(\zeta_3)/k}\langle 1\rangle = 12\mathbb{H}$, since taking $1, \zeta_3$ as a basis we can quickly obtain:

$$\begin{bmatrix} 2 & -1 \\ -1 & -1 \end{bmatrix}.$$

As this is defined over the prime field, if we have 12 copies of it, taking the rank and determinant tells us over $\mathbb{F}_p$ it is isometric to the form with rank 24 and determinant 1, which is equivalent to $12\mathbb{H}$. Meanwhile, over $\mathbb{Q}$ we can observe it is of the form $\langle a\rangle + \langle -\beta\rangle$ since the form is neither positive nor negative-definite. Hence, we can apply Proposition 1.47, to get $12\,\mathrm{Tr}_{k(\zeta_3)/k} = 12\mathbb{H}$.

Either way, we conclude:

$$e^{\mathbb{A}^1}(V) = 15\langle 1\rangle + 12\langle -1\rangle.$$

To resolve the characteristic 3 case, we instead use the Clebsch cubic.

Definition 3.37. The **Clebsch cubic** is the cubic given by:

$$f(x_0, x_1, x_2, x_3) = \sum_{i,j=0 i\neq j}^{3} x_i^2 x_j + 2\sum_{i=0}^{3} x_0 x_1 x_2 x_3 x_i^{-1}$$

In $\mathrm{char}(k) \neq 3$ it is equivalent to:

$$V(x_0 + \cdots + x_4) \cap V(x_0^3 + \cdots + x_3^3) \subseteq \mathbb{P}^4_k$$

This becomes a variety in $\mathbb{P}^3_k$ by setting $x_4 = -x_0 - \cdots - x_3$

It can be verified that this cubic is smooth when the characteristic is not 5. A quick computation shows:

Lemma 3.38. *When $k = k(\sqrt{5})$, the Clebsch cubic contains the following lines:*

$$\ell_1 = \mathrm{Span}_k((1,-1,0,0),(0,0,1,-1)),$$
$$\ell_2 = \mathrm{Span}_k((2,\varphi^{-1},-\varphi,-\varphi),(\varphi^{-1},-\varphi,-\varphi,\varphi^{-1})).$$

Here $\varphi = \frac{1+\sqrt{5}}{2}$.

The Clebsch cubic has a natural S_5 action on it. This can be seen in some cases from the second definition we gave of it.

$$\sigma(x_i) = \begin{cases} x_{\sigma(i)} & \text{if } \sigma(i) \neq 4 \\ -x_0 - \cdots - x_3 & \text{if } \sigma(i) = 4 \end{cases}$$

We claim that all 27 geometric lines are attained via the orbits of these two examples. First, we note that the coordinates on ℓ_1 satisfy $x_0 = -x_1$ and $x_2 = -x_3$. So any stabiliser of ℓ_1 can either interchange 0 and 1, interchange 2 and 3 or swap the pairs $0, 1$ and $2, 3$. This gives $2 \cdot 2 \cdot 2 = 8$ choices, so by the Orbit-Stabiliser theorem, there are 15 lines in the orbit of ℓ_1.

Meanwhile, it can be shown (e.g. [CR15, Proposition 7.8.]) that the stabiliser of ℓ_2 is isomorphic to D_5 which has order 10. So it has 12 lines in its orbit. Then, since S_5 acts by linear automorphisms we can apply Corollary 3.32.

When $k = k(\sqrt{5})$, we can apply the resultant lemma. We can complete $(1, -1, 0, 0)$ and $(0, 0, 1, -1)$ to a basis of k^4 by adjoining $(1, 0, 0, 0)$ and $(0, 0, 1, 0)$ as our first two basis vectors. Differentiating with respect to their duals, we get:

$$\frac{\partial f}{\partial x_0} = (x_0 + x_1 + x_2 + x_3)^2 - x_0^2,$$
$$\frac{\partial f}{\partial x_2} = (x_0 + x_1 + x_2 + x_3)^2 - x_2^2.$$

Then we apply the formulas $x_0 = -x_1$ and $x_2 = -x_3$ to restrict to ℓ_1, giving:

$$\left.\frac{\partial f}{\partial x_0}\right|_{\ell_1} = -x_1^2, \qquad \left.\frac{\partial f}{\partial x_2}\right|_{\ell_1} = -x_3^2.$$

Their resultant is easily computed as 1.

Similarly, for ℓ_2 we adjoin $(0, 1, 0, 0)$ and $(0, 0, 0, 1)$. Differentiating with respect to x_1 and x_3, we obtain

$$\frac{\partial f}{\partial x_1} = (x_0 + x_1 + x_2 + x_3)^2 - x_1^2,$$
$$\frac{\partial f}{\partial x_3} = (x_0 + x_1 + x_2 + x_3)^2 - x_3^2.$$

The coordinates of points on ℓ_2 satisfy $x_1 = \varphi(x_0 + x_2)$ and $x_3 = -\varphi x_0 - x_2$. Substituting these expressions gives

$$\left.\frac{\partial f}{\partial x_1}\right|_{\ell_2} = -\varphi x_0^2 - 2x_0x_2,$$
$$\left.\frac{\partial f}{\partial x_3}\right|_{\ell_2} = -\varphi(x_0^2 - x_2^2).$$

The Sylvester matrix of these two polynomials is

$$\begin{bmatrix} -\varphi & 0 & -\varphi & 0 \\ -2 & -\varphi & 0 & -\varphi \\ 0 & -2 & \varphi & 0 \\ 0 & 0 & 0 & \varphi \end{bmatrix}$$

A quick computation shows the determinant is $-\sqrt{5}\varphi$, or $-\frac{5+\sqrt{5}}{2}$. Hence

$$\text{type}(\ell_2) = \left\langle -\frac{5+\sqrt{5}}{2} \right\rangle.$$

When $k = k(\sqrt{5})$ the $\mathbb{A}^1$–Euler number is just the sum of these forms over the two S_5–orbits. Then we note $\frac{5+\sqrt{5}}{2} = 1 + \varphi^2$. So applying Remark 1.48 we conclude:

$$\begin{aligned} e^{\mathbb{A}^1}(V) &= 15\langle 1\rangle + 12\left\langle -\frac{5+\sqrt{5}}{2} \right\rangle \\ &= 15\langle 1\rangle + 12\langle -1\rangle. \end{aligned}$$

For our final case, we have when $\sqrt{5} \notin k$, lines in the orbit of ℓ_1 remain defined over k. Then base changing and taking a single representative of each Galois orbit, since the Scharlau transfer is a group homomorphism, we have that:

$$e^{\mathbb{A}^1}(V) = 15\langle 1\rangle + 6\,\mathrm{Tr}_{k(\sqrt{5})/k}\,\langle -1\rangle\,.$$

Computing the Scharlau transfer of $\langle -1\rangle$ we get $\langle -2\rangle + \langle 10\rangle$. Since $2 = 1^2 + 1^2$ and $10 = 1^2 + 3^2$, we can apply Remark 1.48 and we're done. We have now proved Theorem 3.34.

Corollary 3.39 ([KW21, p 678])**.** *When all* 27 *lines on a smooth cubic over* $\mathbb{F}_q$ *for* q *odd are defined on* $\mathbb{F}_q$*, then there is always an even number of elliptic lines.*

Proof. Recall that a line is elliptic if it isn't of type $\langle 1\rangle$. As $\mathbb{F}_q^\times$ is finite, and squaring is a 2 to 1 map, we know lines on $\mathbb{F}_q$ are either of type $\langle 1\rangle$ or type $\langle \alpha\rangle$ for α not a square. Then, taking the determinant of $15\langle 1\rangle + 12\langle -1\rangle$, we always get a square, as the number of possibly non square factors is even. So if a cubic f has m elliptic lines, and n hyperbolic, since the determinant of $e^{\mathbb{A}^1}(V)$ will be the determinant of $n \cdot \langle 1\rangle + m \cdot \langle \alpha\rangle$, or the class of α^m, we conclude m must be even. □

3.4 Singular Cubics

Finally, we examine how the theory generalises to the singular case. Theorem 2.58 tells us that the $\mathbb{A}^1$–Euler number is independent of the cubic used to define the section, provided its scheme of lines is finite. Thus, even for a singular cubic, we obtain the same overall count of $15\langle 1\rangle + 12\langle -1\rangle$.

However, the zeros of σ_f need no longer have multiplicity one. Consequently, the indices of lines can have higher rank, allowing them to encode arithmetic information beyond their classical multiplicities. This gives further motivation for studying the individual indices in the singular case.

Computing these indices is significantly more difficult than in the smooth case. For a line passing through a singularity, the resultant in Proposition 3.27 vanishes, so we cannot use that result to compute its index. Likewise, our construction of the distinguished involution i no longer applies to such a line. Consequently, we cannot use the multiplicativity of $\mathbb{A}^1$–degree to prove invariance of type under projective automorphism.

Nevertheless, one advantage of the local construction of the $\mathbb{A}^1$–Euler number in [KW21] is that it is amenable to systematic computation. In particular, the algorithm given in Section 2.4 can be implemented in Macaulay2, as demonstrated in [Pau23]. There is also a Macaulay2 package, `A1BrouwerDegrees`, dedicated to computing local and global $\mathbb{A}^1$-Brouwer degrees.

For the remainder of this section, assume that k has characteristic zero. As explained in [Dol12, Section 9.2.], singular cubic surfaces with finitely many lines are classified by Dynkin diagrams. These diagrams describe the intersection patterns of the exceptional curves that arise upon resolving the singularities.

By [Dol12, Table 9.1], the number of geometric lines on such a singular cubic ranges from 21 for a single singularity of type A_1 to just 1 for a singularity of type E_6. In the latter case, the unique line has multiplicity 27.

We focus on the case of two singularities of type A_1. Consider the cubic defined by

$$f(x_0, x_1, x_2, x_3) = x_3(x_0x_2 - x_1^2) - (x_0 - 2x_1 + x_2)(x_0 - ax_1)(x_1 - bx_2).$$

Here $a, b \in k$ are generic parameters satisfying $a, b \neq 0, 1$, are not equal to each other and $a - b - 1 \neq 0$.

This cubic has two singularities, located at $[0 : 0 : 0 : 1]$ and $[1 : 1 : 1 : (1 - a)(1 - b)]$. It contains 16 geometric lines, as recorded in *loc. cit.* These fall

into four families. There are seven lines that do not intersect the singular locus. There are also four lines passing through each singularity without passing through the other, each of multiplicity two. Finally, the line joining the two singularities has multiplicity four.

A lengthy computation gives the following equations for the 16 lines, grouped into the four families described in Table 3.1.

Family	Defining equations	
0	$x_0 = x_1$	$x_1 = x_2$
1	$x_0 = 0$	$x_1 = 0$
	$x_1 = 0$	$x_2 = 0$
	$x_0 = a^2 x_2$	$x_1 = a x_2$
	$x_0 = b^2 x_2$	$x_1 = b x_2$
2	$x_0 = x_1$	$x_3 = (1-a)(x_1 - b x_2)$
	$x_1 = x_2$	$x_3 = (1-b)(x_0 - a x_2)$
	$x_0 = (b+1)x_1 - b x_2$	$x_3 = (1-b)(x_0 - a x_1)$
	$x_0 = (a+1)x_1 - a x_2$	$x_3 = (1-a)(x_1 - b x_2)$
3	$x_1 = 0$	$x_3 = -b(x_0 + x_2)$
	$x_0 = (a+b)x_1 - ab x_2$	$x_3 = -b(x_0 - 2x_1 + x_2)$
	$x_0 = b x_1$	$x_3 = (a-b)(x_0 - 2x_1 + x_2)$
	$x_1 = a x_2$	$x_3 = (a-b)(x_0 - 2x_1 + x_2)$
	$x_3 = 0$	$x_0 = 2x_1 - x_2$
	$x_3 = 0$	$x_0 = a x_1$
	$x_3 = 0$	$x_1 = b x_2$

Table 3.1: Equations of lines on a cubic surface with singularities of type $2A_1$.

In particular, all 16 lines are defined over k.

The script in Appendix A computes matrices representing the indices of the nine lines in Families 0, 1, and 2. For each line, it first expresses the section σ_f in local affine coordinates on the Grassmannian. Classical geometry gives the corresponding stalk length 2 for lines through one singularity and length 4 for the line through both. Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{Z,p}$ for a point p in the vanishing sub-scheme $Z \subseteq \mathrm{Gr}_k(4,2)$ of σ_f. In these cases, as can be computationally verified, $\mathfrak{m}^3 = 0$. Hence, the stalk can be computed by truncating

the defining equations modulo $\mathfrak{m}^3$. The script then chooses a linear functional η sending the Jacobian determinant to the length of the stalk and computes the Gram matrix of the pairing $(u, v) \mapsto \eta(uv)$ with respect to a monomial basis.

Then, we can observe these matrices are equivalent to $\mathbb{H}$ for all except the rank 4 matrix which is equivalent to $2\mathbb{H}$, since they're of the form:

$$\begin{bmatrix} 0 & A \\ A^\intercal & 0 \end{bmatrix}$$

This form has a half dimensional isotropic subspace via restriction to the span of half of the basis vectors. So it is hyperbolic.

Finally, computing the resultant or the Jacobian gives the type of the other 7 lines. These are $\langle ab\rangle, \langle ab\rangle, \langle a(a-b)\rangle, \langle a(a-b)\rangle, \langle 1\rangle, \langle -b(a-b)\rangle, \langle -b(a-b)\rangle$. We then have

$$\langle a(a-b)\rangle + \langle -b(a-b)\rangle = \langle 1\rangle + \langle -ab\rangle$$

Hence

$$2(\langle ab\rangle + \langle a(a-b)\rangle + \langle -b(a-b)\rangle) = 2\mathbb{H} + 2\langle 1\rangle$$

So, the total $\mathbb{A}^1$–Euler number is

$$10\mathbb{H} + 5\langle 1\rangle + 2\langle -1\rangle = 15\langle 1\rangle + 12\langle -1\rangle,$$

as expected.

3.5 Future Directions

In the preceding computation, we observed that the index of the line joining the two singularities is $2\mathbb{H}$. This is equal to $\mathbb{H} \cdot \mathbb{H}$. Hence, its index is the product of the indices of lines passing through only the first or second singularity. This parallels the factorisation of the line joining the singularity's classical multiplicity as $4 = 2 \cdot 2$, with one factor corresponding to each singularity.

We also note that quadratically enriched intersection theories, such as Chow–Witt theory, provide a possible framework for explaining this factorisation. This motivates the following conjecture, which suggests an intersection-theoretic interpretation of the indices.

Conjecture A. *The index of a line passing through several singular points can be computed as the product of local contributions at each singularity, up to multiplication by a rank* 1 *form. These local contributions depend on both the type of each singularity and how the line passes through it.*

We allow multiplication by a rank 1 form because, even in the smooth case, where there are no singularities, the indices of lines can differ by multiplication by a rank 1 form.

As a first step towards investigating this conjecture, we propose computing the indices of lines on cubic surfaces with other configurations of singularities, such as a surface of type $2A_2$. The richer geometry of such surfaces may suggest a more precise formulation of the conjecture.

Another possible approach is to consider the minimal resolution of a singular cubic surface with isolated singularities. Over an algebraic closure, this can be realised as the blow-up of $\mathbb{P}^2$ at six points in a non-general configuration, possibly including infinitely near points. Studying deformations of these six points and their effect on the intersection theory of the resulting surface may provide an explanation for the observed factorisation.

Appendix A

Macaulay2 Code

Listing A.1: Computation of the local indices on the $2A_1$ cubic

```
k = frac(QQ[a,b]);
R = k[A,C,B,D];

-- Function to compute Index
eklAtOrigin = f -> (
    Q := R/(ideal f + (ideal vars R)^3);
    bQ := basis Q;
    mu := numColumns bQ;
    mons := flatten entries lift(bQ,R);
    j := lift(det(sub(jacobian matrix{f},Q)),R);

    pivot := 0;
    while coefficient(mons#pivot,j) == 0 do pivot = pivot+1;
    c := coefficient(mons#pivot,j);
    eta := q -> (mu/c)*coefficient(mons#pivot,lift(q,R));

    gram := matrix(k, for i from 0 to mu-1 list (
        for h from 0 to mu-1 list eta(bQ_(0,i)*bQ_(0,h))
    ));
    print gram;
    gram
);

-- Get polynomials with p,q on line and r,s are complementary.
formAtLine = (name,p,q,r,s) -> (
    print name;
    S := R[u,v];
    x := for i from 0 to 3 list (
        sub(p#i,S)*u + sub(q#i,S)*v
```

```
30          + sub(r#i,S)*(A*u+B*v)
31          + sub(s#i,S)*(C*u+D*v)
32      );
33      F := x#3*(x#0*x#2-(x#1)^2)
34          - (x#0-2*x#1+x#2)*(x#0-a*x#1)*(x#1-b*x#2);
35      f := {coefficient(u^3,F), coefficient(u^2*v,F),
36            coefficient(u*v^2,F), coefficient(v^3,F)};
37      eklAtOrigin(f)
38  );
39
40  -- List the points we're feeding in
41  e0 = {1,0,0,0};
42  e1 = {0,1,0,0};
43  e2 = {0,0,1,0};
44  P = {0,0,0,1};
45  Qnode = {1,1,1,(1-a)*(1-b)};
46
47  -- Compute things
48  GramLine0 = formAtLine("Line0",P,Qnode,e0,e1);
49  GramLine1 = formAtLine("Line1",P,{1,0,0,0},e1,e2);
50  GramLine2 = formAtLine("Line1",P,{0,0,1,0},e0,e1);
51  GramLine3 = formAtLine("Line3",P,{a^2,a,1,0},e0,e1);
52  GramLine4 = formAtLine("Line4",P,{b^2,b,1,0},e0,e1);
53  GramLine5 = formAtLine("Line5",Qnode,{1,0,0,1-b},e0,e1);
54  GramLine6 = formAtLine("Line6",Qnode,{1,1,0,1-a},e0,e1);
55  GramLine7 = formAtLine("Line7",Qnode,{a+1,1,0,1-a},e0,e1);
56  GramLine8 = formAtLine("Line8",Qnode,{b+1,1,0,-(b-1)*(b+1-a)},e0,e1);
```

Appendix B

Indices of $2A_1$ Lines

Line	Gram matrix
$x_0 = x_1,$ $x_1 = x_2$	$\begin{pmatrix} 0 & 0 & -2 & 0 \\ 0 & 0 & -\frac{2}{(a-1)(b-1)} & -\frac{1}{(a-1)(b-1)} \\ -2 & -\frac{2}{(a-1)(b-1)} & 0 & 0 \\ 0 & -\frac{1}{(a-1)(b-1)} & 0 & 0 \end{pmatrix}$
$x_1 = 0,$ $x_2 = 0$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$
$x_0 = 0,$ $x_1 = 0$	$\begin{pmatrix} 0 & -\frac{1}{a} \\ -\frac{1}{a} & 0 \end{pmatrix}$
$x_0 = a^2 x_2,$ $x_1 = a x_2$	$\begin{pmatrix} 0 & \frac{1}{a} \\ \frac{1}{a} & 0 \end{pmatrix}$
$x_0 = b^2 x_2,$ $x_1 = b x_2$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$
$x_1 = x_2,$ $x_3 = (1-b)(x_0 - a x_2)$	$\begin{pmatrix} 0 & -\frac{1}{(a-1)(b-1)^4} \\ -\frac{1}{(a-1)(b-1)^4} & 0 \end{pmatrix}$

Line	**Gram matrix**
$x_0 = x_1,$ $x_3 = (1-a)(x_1 - bx_2)$	$\begin{pmatrix} 0 & \frac{1}{b(a-1)^4(b-1)} \\ \frac{1}{b(a-1)^4(b-1)} & 0 \end{pmatrix}$
$x_0 = (a+1)x_1 - ax_2,$ $x_3 = (1-a)(x_1 - bx_2)$	$\begin{pmatrix} 0 & -\frac{1}{(a-1)^4(a-b)(b-1)} \\ -\frac{1}{(a-1)^4(a-b)(b-1)} & 0 \end{pmatrix}$
$x_0 = (b+1)x_1 - bx_2,$ $x_3 = (1-b)((b+1-a)x_1 - bx_2)$	$\begin{pmatrix} 0 & \frac{a-2b}{b(a-1)(a-b)(b-1)^4(a-b-1)^3} \\ \frac{a-2b}{b(a-1)(a-b)(b-1)^4(a-b-1)^3} & 0 \end{pmatrix}$

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